

Lightning Detector

Lightning Detection Safety and Tracking Device for Picnickers

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Chapter 1 - Executive Summary

1.1 Executive summary

Lightning poses a significant safety hazard for individuals participating in outdoor activities such as athletics, construction, recreation, and field operations. While commercial lightning detection systems exist, many are either expensive, lack portability, or provide limited user feedback. The objective of this project is to develop a portable lightning detection system capable of detecting nearby lightning activity and providing users with clear audible and visual warnings. A companion lightning emulator is also

being developed to simulate lightning events for system testing, demonstration, and validation without relying on naturally occurring storms.

The proposed system utilizes a Very Low Frequency (VLF) receiver to detect electromagnetic emissions generated by lightning strikes. An embedded microcontroller processes the received signals and determines when hazardous lightning activity is present. When a threat is detected, the system provides immediate user notification through visual indicators and prerecorded spoken voice alerts generated using a dedicated voice playback integrated circuit. Bluetooth Low Energy (BLE) connectivity is incorporated to enable wireless communication with a mobile application for displaying system information and enhancing the user experience.

This report summarizes the research, design decisions, and preliminary development completed during the first half of the Senior Design project. Topics include the scientific principles of lightning detection, evaluation of alternative hardware and software technologies, subsystem architecture, and component selection. Initial hardware and software designs have been established for the detector and emulator, providing the foundation for implementation, integration, and system testing during the remainder of the project. Future work will focus on hardware fabrication, firmware development, enclosure design, subsystem integration, and performance validation to produce a reliable and portable lightning detection system.

Chapter 2 - Project Description

2.1 Project Background and Motivation

It is our team's intention to provide a design effort for the implementation of a Lightning Detection Safety and Tracking Device for Picnickers and associated circuitry. System primary features will include VLF (very low frequency) radio receiver, lightning simulation transmitter, voice synthesizer for verbal warnings, additional audio and optical warning indicators. As a secondary design goal, we may design a Bluetooth radio device implementation and associated phone app to notify user of approaching lightning. The criteria used to formulate this proposal are primarily based on the information provided by the UCF Senior Design Advisors and Team Lightning.

Lighting is an existential safety concern in many areas of the world, Florida being no exception. The sooner individuals who are outdoors can receive notification of the presence of lightning or of its approach, the greater the reduction of exposure to lightning strikes on people. It is therefore that we propose the following solution.

Moreover, lighting is an AC electrical phenomenon. As such, it generates electromagnetic waves (EM waves) in the atmosphere. These EM waves are in fact radio waves at a very low frequency VLF. The frequency of lightning is circa 3-30 KHz. Radio

receivers can be designed to both receive and measure the signal strength of such EM energy. This can in turn be normalized and extrapolated to distance or range from the lightning to the VLF radio receiver and user.

Our specific application is targeted for picknickers who are in parks or camping. This device would detect the presence of lightning and possibly if it is approaching or going away. The way a user would be informed of this would be by the use of a voice synthesizer which would broadcast warning messages to those nearby the device. There would be additional optical and audio alarms which would get the attention of those further away from the device. As a secondary design goal, a BLE transceiver would transmit messages to the user's phone and inform them of the lightning status.

In order to test this system, it will be necessary to design and build a lightning simulator in hardware due to the impracticality of waiting for a lightning storm to occur nearby. This simulator would generate VLF EM energy in the range of 3-30 KHz to test the lightning detector radio.

2.2 Current Commercial Technologies

2.2.1 THOR GUARD Model TG 360

THOR GUARD has been manufacturing lightning prediction products since 1973. The TG 360 model is a commercial lightning prediction and warning system designed for outdoor facilities, athletic complexes, and other environments that may require weather hazard monitoring. According to THOR GUARD, the system monitors electrostatic changes in the atmosphere to detect potentially hazardous storm activity before lightning occurs. This model utilizes signal processing methods to analyze electric field variations and generates warning notifications [1]. One of the advantages of this product is its ability to continuously monitor the condition of the environment and provide automated warnings, improving the safety of individuals exposed to lightning hazards. Additionally, it also provides audible and visual alerts to notify nearby users.

Our senior design project shares some similarities with this THOR GUARD model through its objective of lightning safety and real-time user notification. Additionally, both of these systems are intended to reduce human exposure to lightning hazards by detecting electromagnetic activity. However, these two systems differ in their sensing methodology and implementation. Unlike the TG 360 THOR GUARD model, which analyzes atmospheric electrostatic field changes, our project detects the very low frequency electromagnetic radio waves that lightning discharges. In order to do so, our design utilizes a VLF radio receiver and antenna capable of measuring signal strength and the proximity and movement of lightning activity.

Moreover, our design is tailored for its portability, allowing users to utilize our project for recreational use, more specifically for picknickers and campers. On the other hand, the THOR GUARD TG 360 model is not designed for portability, and instead is meant for stationary use.

2.2.2 AcuRite Atlas Weather Station

The AcuRite Atlas Weather Station is a consumer weather monitoring product designed to provide warning notifications for lightning activity. According to AcuRite, this system detects lightning strikes within 25 miles by monitoring electromagnetic activity [2]. The sensor of this AcuRite Atlas Weather Station interacts with AcuRite's digital platform and provides users with estimated strike distance information, as well as lightning strike counting and wireless monitoring. One of the key advantages of this product is its ability to combine lightning detection with additional environmental signal detection, such as temperature, humidity, wind speed, rainfall, allowing users to also monitor weather conditions.

Our project shares some similarities with the AcuRite Atlas Weather Station through its focus on lightning detection and real-time user notification, as well as the use of a digital platform. Both products are design to improve outdoor safety by detecting lightning-related electromagnetic activity and warning users of approaching hazardous conditions. However, the systems differ significantly in their application. Unlike the AcuRite system, which functions as a stationary consumer weather station, our project focuses on portable lightning detection safety, targeting recreational use such as picnickers or campers.

Additionally, our system will incorporate a Bluetooth Low Energy (BLE) transceiver and mobile application in order to provide wireless smartphone notifications regarding lightning activity. Unlike the AcuRite design, which relies on a Wi-Fi connected display, our design is intended to support low power portable use, without the need for Wi-Fi connection.

2.2.3 StrikeAlert HD

The StrikeAler HD is a portable, personal lightning detection system that is designed to provide early warning notifications for nearby lightning activity. According to the manufacturer, this product utilizes lightning detection technology capable of detecting lightning strikes within a range of 40 miles. The device operates a graphical display that provides real-time lightning strike distance information, storm sensitivity, and trend visualization. One of the main advantages of this product is its lightning tracking capability, also allowing for trend visualization.

Our project shares some similarities with the StrikeAlert HD device, including its focus on portable lightning safety and real-time hazard detection. Both products emphasize the importance of low-power portable operation and multiple notification type methods, including audio and visual alerts. However, the designs differ significantly in system functionality as well as display features. Unlike the StrikeAlert HD, which functions as a standalone handheld device, our project aims to combine wireless connectivity and embedded signal processing into a comprehensive lightning safety and tracking platform. This platform, or app, would operate as a tool for users to seamlessly get notified of

hazardous conditions nearby, instead of manually checking a device, like the StrikeAlert HD provides.

2.2.4 Summary

Throughout the extensive research the team completed during the design and brainstorming phase of this project, we found a multitude of products designed to detect lightning and notify individuals of hazardous weather activity. However, we did not find any products that met all of our design specifications. For instance, the THOR GUARD TG 360 model has a similar objective to our project's focus on lightning safety and real-time user notification. However, the THOR GUARD detects hazardous weather activity measuring atmospheric electrostatic field changes [1]. In contrast of our design, which utilizes very low frequency electromagnetic radio waves that lightning discharges in order to detect lightning.

Furthermore, the AcuRite Atlas Weather Station is an additional design that has some similarities to our project. However, one of the key advantages of our design is the portability aspect of it, this is because we wanted users to be able to use this product recreationally. In contrast, the AcuRite Atlas Weather Station is a stationary device, mainly utilized in athletic field environments [2].

The research done by our team led us to design a device that had all of the advantages of the different lightning detector devices in the market, while also expanding on the aspects of accessibility, portability, and accuracy.

2.3 Objectives and Goals

2.3.1 Basic, Advanced and Stretch Goals

The primary objective of this project is to create a reliable, portable lightning detection system that provides a warning to users. To ensure a successful development lifecycle, the team has structured the project requirements into Basic, Advance, and Stretch goals across both hardware and software domains.

Overall Goals

- Design and build a system with a radio receiver and a microcontroller unit capable of detecting lightning and trigger peripheral alarms.
- Design and build a lightning emulator.
- Develop a functional mobile application interface that successfully pairs with the lightning detector's BLE transceiver.

Advanced Goals

Additionally, we also planned advanced goals for our project. These goals constitute planned additions to our main idea that are not necessarily “stretch goals”.

- Implement signal processing on the MCU to measure VLF signal strength, allowing the system to estimate the relative distance and directional trend (approaching vs. retreating) of the lightning
- Integrate and program a voice synthesizer to provide clear, verbal warnings (e.g., "Warning: Lightning approaching").
- Successfully establish a stable Bluetooth Low Energy (BLE) connection between the MCU and a mobile device.

Stretch goals

Finally, our stretch goals constitute planned additions to our project if we are able to complete the planned main and advanced goals in time.

- Implement a BLE radio with associated app.
- Optimize the power circuitry and MCU sleep states to achieve a minimal duty cycle, maximizing battery life for extended weekend camping or picnic trips
- Implement advanced analog or digital filtering to reject false positives from other AC electrical sources (e.g., power lines or car alternators).
- Enable background push notifications, ensuring the user receives a lightning alert on their phone even if the phone is locked or the app is closed.
- Integrate a visual "Danger Radius" map interface, mapping the user's GPS location against the estimated distance of the lightning strike.
- Allow the BLE transceiver to broadcast to multiple smartphones simultaneously, ensuring a whole group of picnickers receives the warning from a single hardware device

2.3.2 Objectives

These goals can be narrowed down into the objectives of the MCU and lightning simulator, as well as the mobile application.

MCU and lightning simulator objectives

- Successfully design and fabricate a VLF radio receiver capable of detecting electromagnetic signals in the 3–30 KHz range.
- Build a functional hardware lightning simulator/ transmitter to safely test the receiver without relying on actual weather events.
- Integrate a microcontroller unit (MCU) that triggers standard local audio (buzzer) and visual (LED) alarms upon detecting a VLF emission.

Mobile Application objectives

- Develop a functional mobile application interface that successfully pairs with the lightning detector's BLE transceiver.
- Display a simple, real-time boolean status on the app (e.g., "Safe" vs. "Lightning Detected") based on telemetry received from the hardware.

2.3.3 Hardware

The lightning detector's primary function is to detect potential lightning activity effectively and warn users using audio and visual alarms. The system will leverage a very low frequency (VLF) radio receiver capable of detecting electromagnetic signals in the 3 to 30kHz range. The team will design and fabricate the radio receiver accordingly. Additionally, our team will also build a functional lightning simulator in order to test the radio receiver without relying on weather events. Furthermore, the system will be integrated with a microcontroller unit in order to trigger audio and visual warnings, using a buzzer and an LED respectively, upon detecting a spike in the VLF.

Once our team successfully completes these primary goals for our project, we are expecting to include additional features to our system. Starting with some of the goals we deem the most viable include implementing advanced signal processing on the MCU that will measure the strength of VLF. This would allow our MCU to estimate the distance and directional trend of the lightning signals, such as whether the lightning is approaching or retreating. .

Moreover, an additional add-on to the system will be a programmed and integrated voice synthesizer to provide clear, verbal warnings such as a voice saying "Warning, lightning approaching." Our team also plans on (stretch goal) successfully establishing a stable Bluetooth Low Energy (BLE) connection in between the system and a mobile device. This would allow users to connect their phones to the system and seamlessly receive notifications and check status updates.

Ideally, we could also optimize the power circuitry and MCU states, which would allow the system to achieve a minimum duty cycle. This would in return provide the system with a maximized battery life, ideal for an extended weekend camping trip, or a long picnic.

Figure 2.1 - Lightning Detector Hardware Diagram

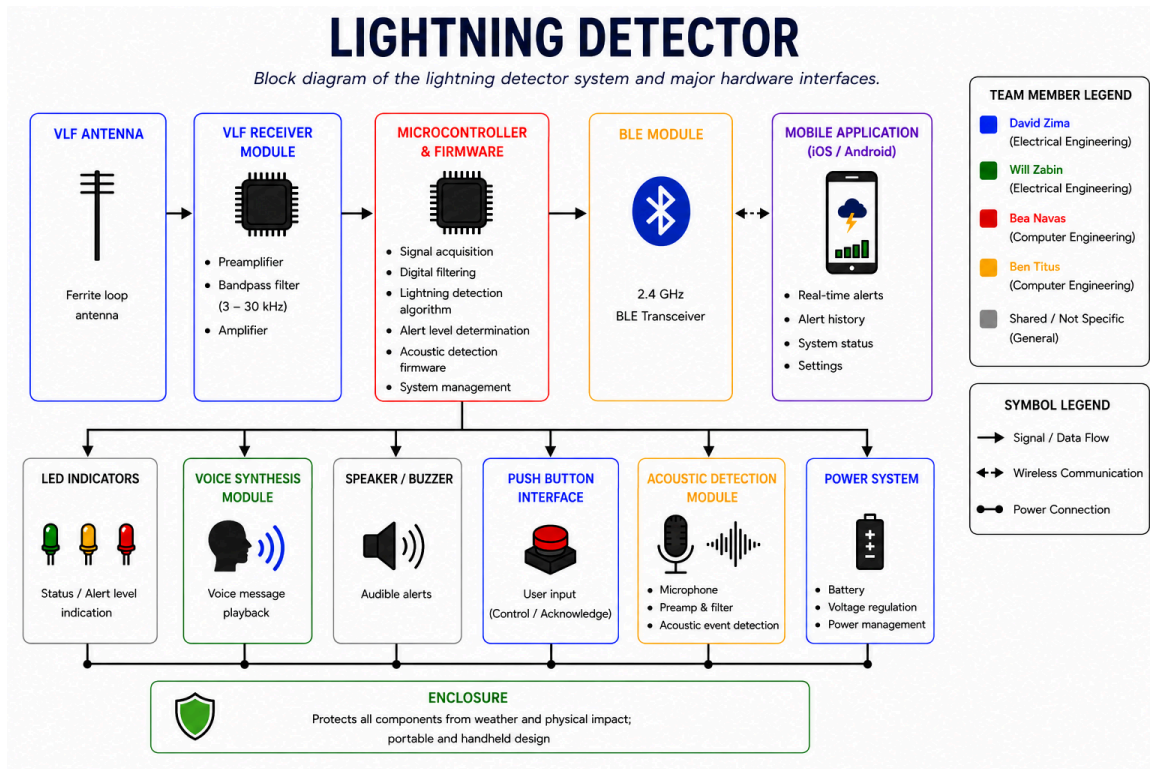
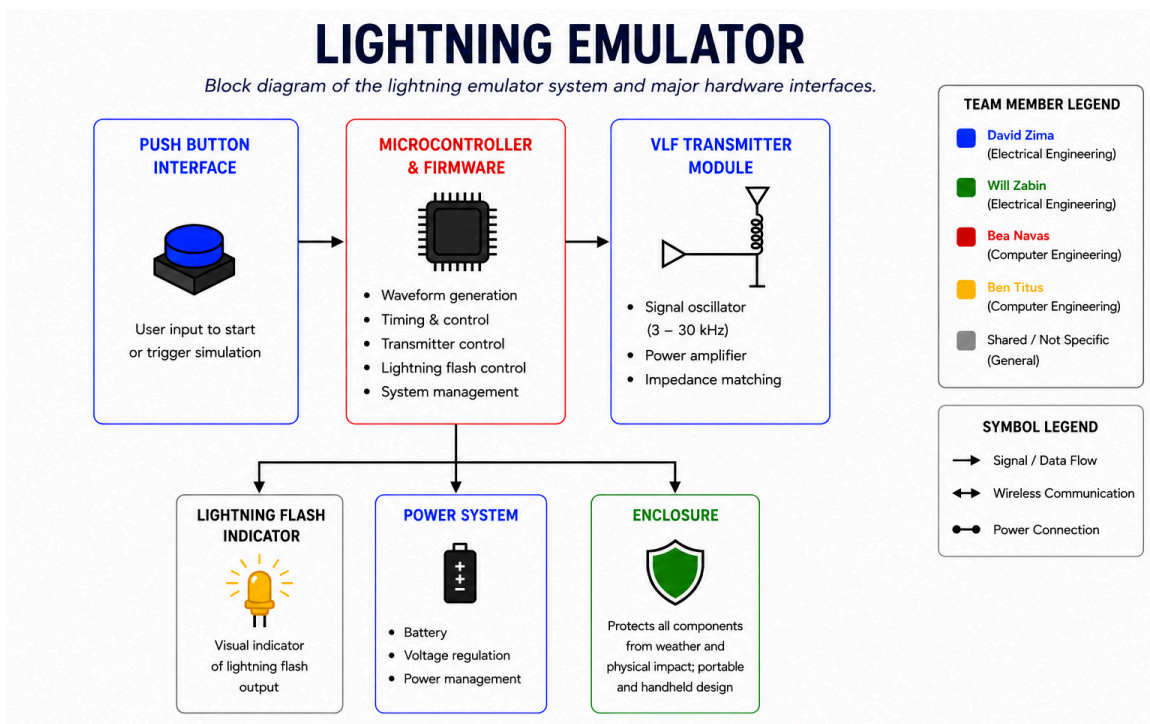


Figure 2.2 - Lightning Emulator Hardware Diagram



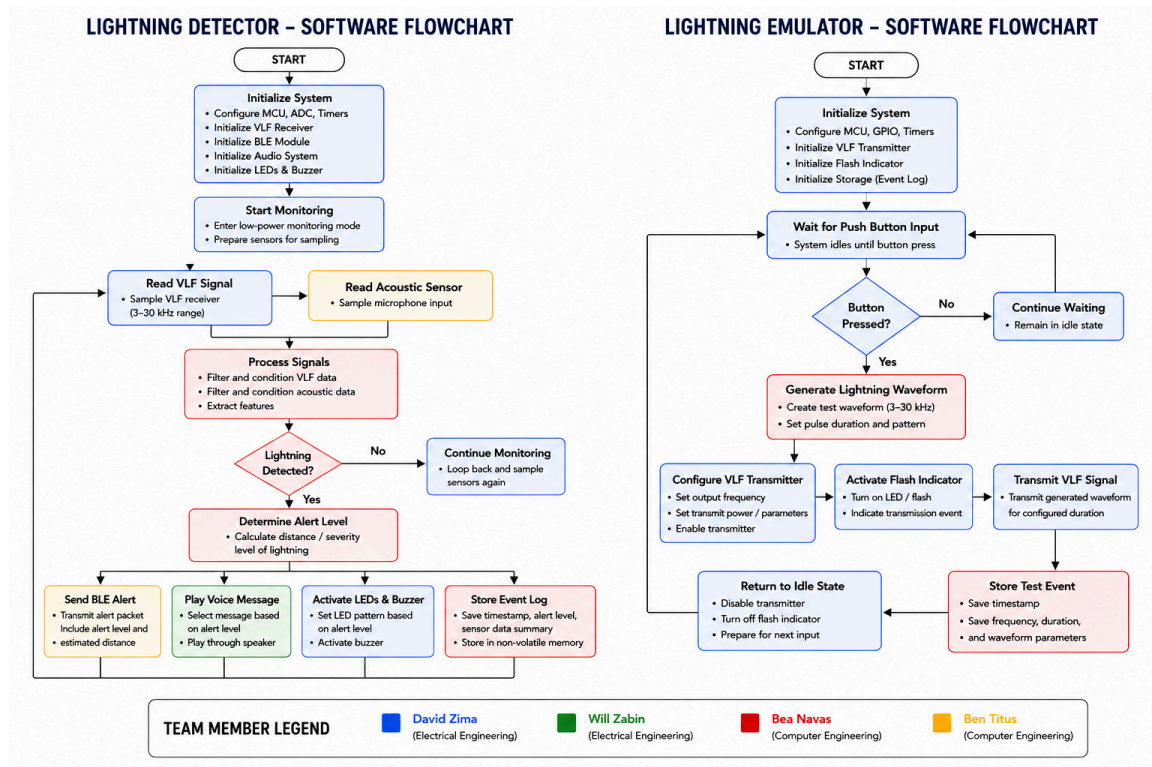
2.3.4 Software

The software features are essential for the portability and user mechanics of this project. For instance, the system needs a functional mobile application that successfully pairs with the lightning detector's BLE transceiver in order to display the results of the lightning detector. The mobile application will be a professional, user friendly environment for the user to check real-time updates on weather hazards nearby. For example, a real-time boolean status displaying the weather state, using "Safe" or "Lightning Detected" messages. For these real-time displays of weather hazards, the application will require telemetry received from the hardware. For the mobile app, we decided early on to build natively for iOS using SwiftUI and Apple's CoreBluetooth framework. The main reason for going native rather than a cross-platform tool like React Native or Flutter is that CoreBluetooth is the most reliable and well-documented way to communicate with BLE devices on an iPhone. It gives us direct access to the Bluetooth stack without any third-party library sitting between our code and the phone's hardware. Since BLE communication is the entire point of the app, that matters a lot.

Moreover, another aspect of the software side of the project is the firmware. For example, to display the estimated distance of the lightning strikes nearby, the MCU's firmware will process the signal strength data sent over BLE. Moreover, in order to estimate the direction of the lightning activity detected, the firmware will need to keep logs of the detected spikes in VLF, and calculate whether the lightning is nearing or retreating. On the BLE firmware side, our controller is configured as a GATT server, which is the standard Bluetooth Low Energy communication model. It advertises a custom service with a single characteristic that the app subscribes(Pub Sub architecture) to. When the controller's detection logic flags a lightning event, it writes a single byte to that characteristic — 0x01 for detected, 0x00 for clear — and the phone receives that update automatically because it has subscribed to notifications. The single-byte protocol is intentionally simple right now, which leaves room to add more data fields later, such as estimated distance or strike count, without redesigning the communication layer.

Moreover, our team has additional stretch goals that we plan to work on if time allows. For example, enabling background push notifications to ensure users receive the lightning alerts on their phone, even if their phone is locked. An advanced goal for the application is also to create a localized alert log or history within the app. This would allow for the app to visualize the frequency and timeline of recent strikes through the app.

Figure 2.3 - Software Flowcharts



2.4 Engineering Specifications

Table 2.1 Engineering specifications

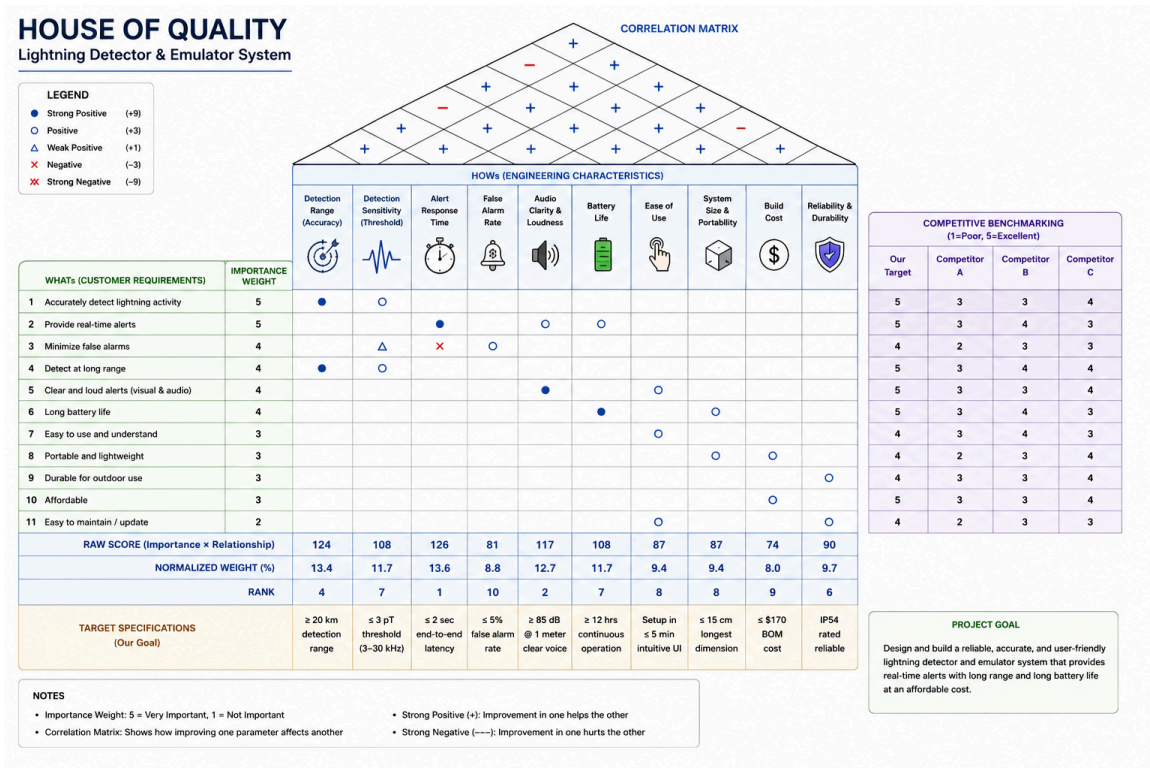
Emulator	
Emulator Frequency Range	500 KHz
Emulator Activation Time	Instantaneous
Detector	
Detection Frequency Range	500 KHz
Detection Distance Range	Ca. 30 miles
Alert Latency	100 mS
BLE Communication Range	100 ft
Visual Alert Visibility	250 ft
Alert Audio Volume	70 dB

App	
Mobile App Response Time	< 1 second
Enclosure	
Enclosure Weight	Ca. 8 oz.
Enclosure Dimensions	Ca. 10' round x 2" height

(Highlighted specifications are to be demonstrated during presentation)

2.5 House of Quality

Figure 2.4 - House of Quality



Chapter 3 - Research

We are grateful for the suggestions and guidance provided by Dr. Vladimir Rakov and Dr. Ziqin Ding from the University of Florida Lightning Research Group. The studies, research and data they shared was invaluable to our effort here.

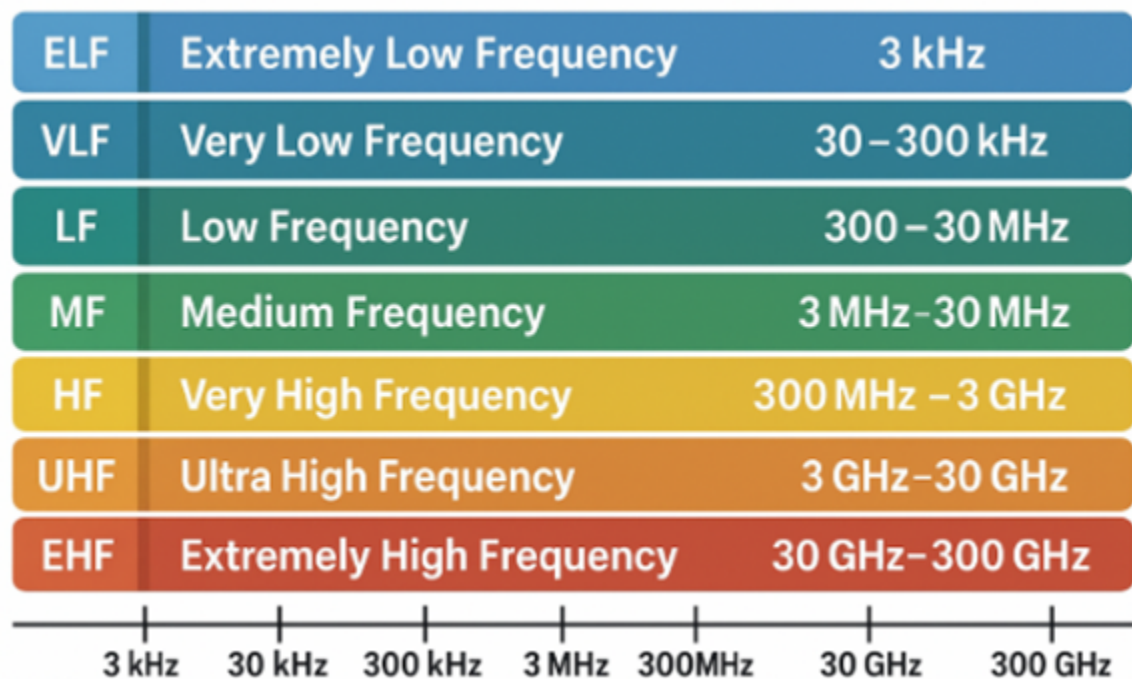


University of Florida Lightning Research Group

3.1 Lightning Theory Research

The premise of our design is that lightning generates radio emissions in the VLF or very low frequency and LF or low frequency part of the electromagnetic spectrum. Therefore, if we can build a VLF/LF radio with some filtering and processing, we should be able to detect the presence of lightning. Lightning is very high energy, so the signal strength does not require a radio of any significant sensitivity.

Figure 3.1 - Frequency Ranges



Further data from researchers at UF Gainesville [7] have provided other background information that will be helpful to our design effort. When an active thundercloud passes overhead, the charges within the cloud intensify the static electric field in the region below. This is not something we intend to measure as our approach is not detection of electrostatic energy but electromagnetic energy. We did, however, consider this approach. We found this to be encumbering technically as it would have required the development of a lightning leader and its path of propagation are largely determined by the shape and magnitude of the electric field between the storm-cloud and the surface of the earth (Petrov & Waters, 1995). It would have also required solving for the electric field using Laplace's equation in the space between buildings and other structures. The numerical techniques for the solution to this class of problems would have been much too cumbersome, inaccurate, or time-consuming for our design goals and scope of our Senior Design project.

Much of the following research and data are taken from the following two studies:

[8] **Location and analysis of acoustic infrasound pulses in lightning** by Arechiga, R., M. Stock, R. Thomas, H. Erives, W. Rison, H. Edens, and J. Lapierre (2014), Location and analysis of acoustic infrasound pulses in lightning, *Geophys. Res. Lett.*, 41, 4735–4744, doi:10.1002/2014GL060375.

[9] **Imaging thunder** by Johnson, J. B., R. O. Arechiga, R. J. Thomas, H. E. Edens, J. Anderson, and R. Johnson (2011), Imaging thunder, *Geophys. Res. Lett.*, 38, L19807, doi:10.1029/2011GL049162.

For lightning detection, [2] The New Mexico Lightning Mapping Array (LMA) uses very high frequency (VHF) radio emissions to pinpoint ionization events associated with the formation of lightning channels [[Rison et al., 1999](#)]. Stepped leaders often form dendritic channel structures [[Bazelyan and Raizer, 2000](#)], which can be mapped with accuracies of tens of meters [[Thomas et al., 2004](#)]. In central New Mexico typical negative cloud-to-ground and intracloud lightning produce hundreds to thousands of locatable VHF emissions (referred to hereon as rf sources) occurring during a fraction of a second.

[3] Though stepped leader progression does not generate the most energetic thunder, it is responsible for forming an ionized channel that subsequently carries the predominant current events such as return strokes. Rapid channel heating results in the generation of shock waves along the channel that quickly decay to linear acoustic waves with intense energy in the audible band [[Few, 1969](#); [Holmes et al., 1971](#)]. Simultaneously, it is predicted that the electrostatic relaxation within a charged source region will produce a volumetric contraction and a rarefactive lower frequency infrasound (below a few Hz) pulse [[Dessler, 1973](#); [Bohannon et al., 1977](#)]. Precursory charge accumulation [[Pasko, 2009](#)], or positive streamer formation [[Few, 1985](#)] may also contribute to infrasonic pulses.

[4] Acoustic reconstruction of lightning channel geometry has been attempted using microphone arrays and ray tracing back to the acoustic source [e.g., [Few, 1970](#); [Few and Teer, 1974](#); [Teer and Few, 1974](#); [MacGorman et al., 1981](#); [Arechiga et al., 2011](#)]. Infrasound emissions have not been conclusively located in the same manner, but they have been correlated to electrostatic sources

detected with electric field meters [[Balachandran, 1983](#)]. A study by [Assink et al. \[2008\]](#) related infrasound sources recorded at regional distances (up to 50 km) with rf sources from electrical storm activity, where the infrasound signals are probably low-frequency remnants of long-distance propagating broadband thunder. Another study by [Arechiga et al. \[2011\]](#) compared locations of rocket-triggered lightning sources (mapped with the New Mexico LMA) with broadband thunder sources occurring within a few km. This paper further explores techniques to jointly map and compare LMA and multiple thunder sources using array coherence mapping.

[5] In the summer of 2009 researchers recorded thunder using broadband microphones (AllSensors™ 1-inch MEMS transducers operated in absolute mode), which possess similar sensitivity, linear dynamic range to ± 125 Pa, and flat responses in the band 0.01 to 500 Hz. Microphones deployed in this study consisted of a three-station ‘network’, where one station (referred to as GTM) was a four-element infrasound array with sensor nodes located at the center and vertices of an approximate 45-m equilateral triangle ([Figure 1](#)). Two other infrasound stations (KVH and LAN) were located within 1.5 km of GTM. At all stations broadband sound was recorded with GPS time-synced Reftek RT-130 data loggers acquiring at a resolution of 24 bits and a sample rate of 1000 Hz during six weeks from late July through the end of August.

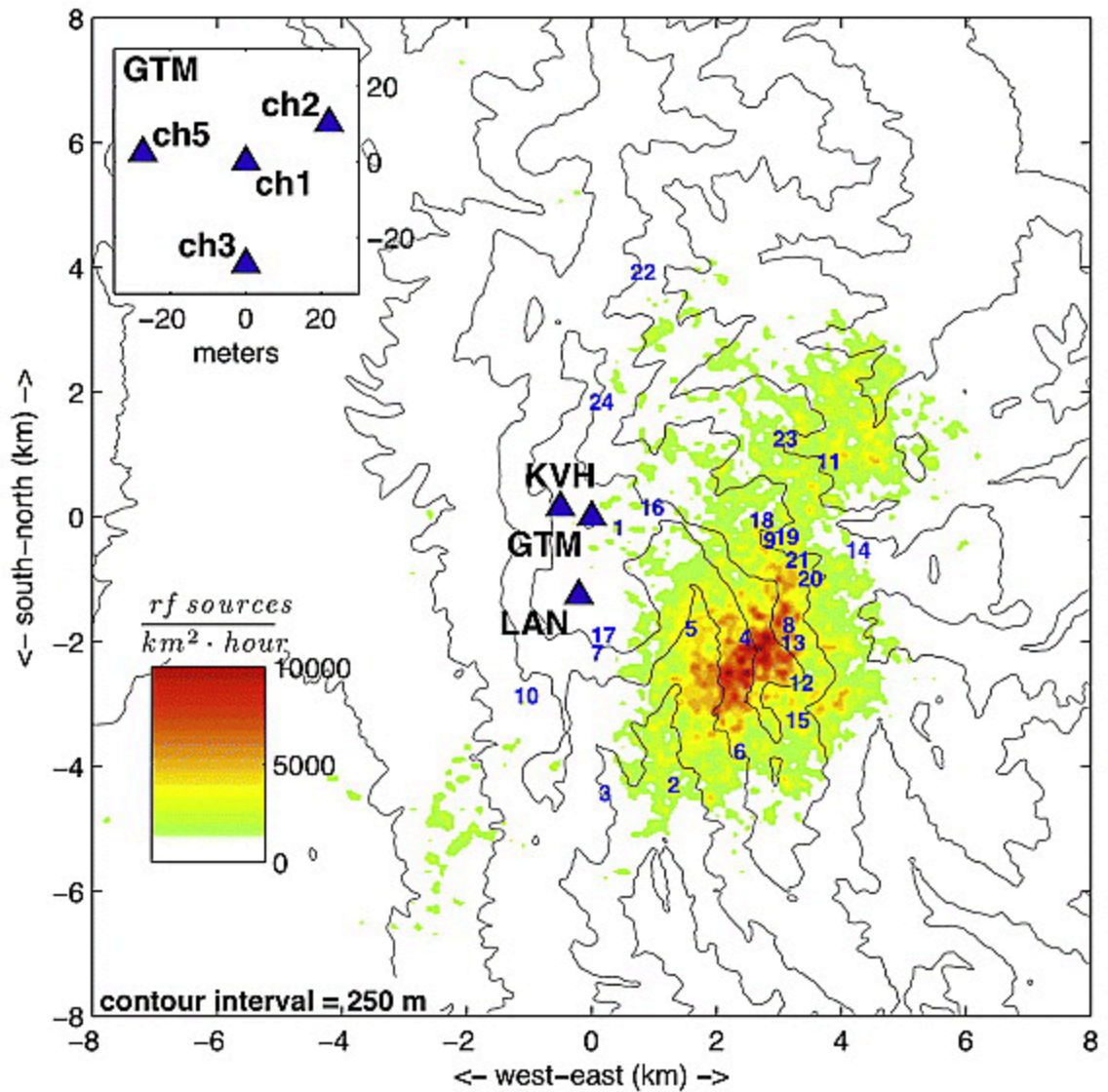


Figure 1

Topographic map showing location of acoustic monitoring arrays for the July–August 2009 thunder campaign. Origin of map is located at GTM (array detail shown in inset) with coordinates of 33.99012 N, 107.18237 W, and 3122 m above sea level. Color scale indicates LMA event density for 90 distinct lightning flashes occurring between 19 and 20 UTC (1:00 to 2:00 PM local time) on July 24th. Numeric indices correspond to the central locations of rf sources for events shown in [Figures 2](#) and [3](#).

[6] The experiment was conducted during central New Mexico's “monsoon” electrical storm activity. At least three separate storm systems (on July 24th, July 30th, and August 6th) traversed between the network stations providing a diversity of thunder signals that were recorded locally at ranges of a few hundred meters to more than ten kilometers ([Figure 1](#)). This study focuses on a one-hour period between 1 and 2 PM (19:00–20:00 UTC) on July 24th, during which 90 distinct lightning events were independently catalogued by the New Mexico LMA ([Figure 2](#)). Proximity of this lightning activity to the microphone network, time isolation of flashes, and generally low levels of ambient noise (owing to low winds), contributed to making this the best acoustic signal-to-noise dataset of all recorded thunderstorms.

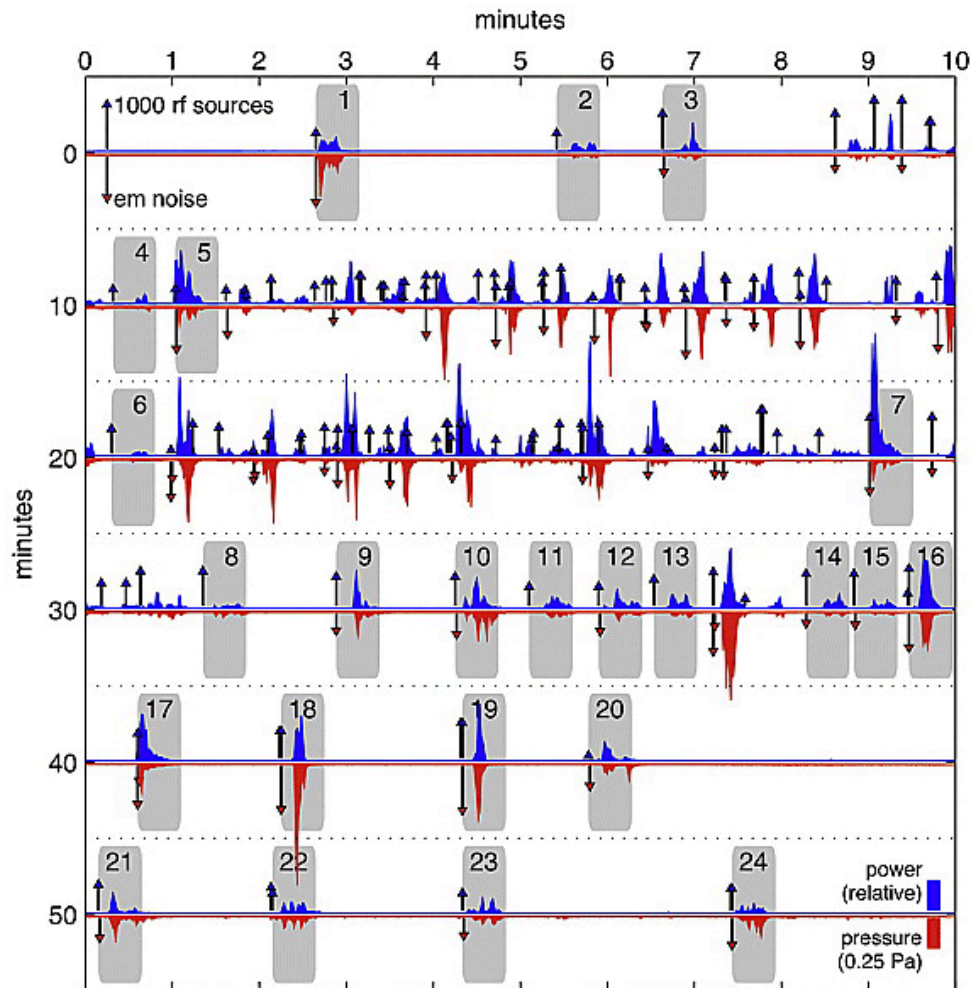


Figure 2

One hour of lightning/thunder activity occurring between 1 and 2 PM local time on July 24th, 2009. LMA event times are indicated by upward-pointed arrows whose vertical stem lengths are proportional to rf source counts per second (refer to scale bar). Those events with more than 10 rf sources, and which are isolated from preceding and following events by at least 30 s, are indicated in grey shading and subject to further analysis (e.g., [Figure 3](#) and [auxiliary material](#)). Normalized thunder power (shown as a blue envelope) is calculated from LMA event times and estimated acoustic propagation times to the GTM array modeled with a $1/r^2$ energy decay. The 1-s averaged absolute pressures recorded at GTM are shown as red envelopes (refer to scale bar). Downward pointed arrow stems have lengths that are proportional to the electromagnetic noise picked up by the GTM channel 2 cable.

[7] GTM was designated as the master array owing to its excellent signal-to-noise on all 4 array elements on July 24th. We use this array to locate thunder source regions through a combined slowness and distance ranging technique outlined below. We then compare thunder locations to LMA locations to assess the resolution capabilities for the linked processes and to elucidate the respective generation mechanisms of sound and VHF signals. Distinctive acoustic pulses are also located and used to predict corresponding pulse arrival times across the microphones network.

[8] Following seismic array data processing techniques for presumed incident plane waves [e.g., [Lacoss et al., 1969](#)], we apply a 2-D slowness search (or velocity filtering) to identify coherent acoustic backazimuths and then calculate vertical incidence angles for band filtered acoustic signals. Horizontal slowness components (s_x and s_y) are equivalent to the inverse of horizontal apparent velocity in east-west and north-south directions and have units of s/m. Horizontal slowness searches, using multi-element seismic arrays, permit determination of plane wave propagation directions from one or more concurrent sources, both transient and tremor-like [e.g., see [Almendros et al., 2002](#)]. However, because

seismic velocities are usually poorly constrained, there is inherent ambiguity in the vertical wave incidence angle of these sources.

[9] Acoustic waves in the atmosphere, on the other hand, have relatively unvarying speed (c_0) that is predictable as a function of temperature (equal to 338 m/s at 10°C with $\sim\pm 0.6$ m/s variation per $\pm^\circ\text{C}$). By assuming a temperature at the array, the vertical slowness (s_z) can be uniquely determined from the two horizontal slowness components, i.e., $s_z = \sqrt{1/c_0^2 - s_x^2 - s_y^2}$. Apparent source backazimuth and incidence are then calculated from the slowness unit vector ($[c_0 s_x, c_0 s_y, c_0 s_z]$) and this, combined with distance ranging, can provide an absolute 3-D spatial location [e.g., [Few, 1970](#)].

[10] Slowness searches are performed in multiple frequency bands and over time to locate frequency band and distance-limited sources of thunder. In this study results are calculated for overlapping bands of 1–10 Hz and 4–40 Hz sound for respective moving time windows (Δt) of 2 s and 0.5 s with $\Delta t/2$ window overlap. Relatively broad spectral bands are analyzed to minimize array response aliasing that can be problematic for 4-element arrays. Window length is fixed as twice the period associated with the low corner frequency.

[11] Distance (r) ranging is calculated as the product of the presumed ambient sound speed (c_0) and the elapsed time ($t - t_0$) since lightning flash, allowing for a thunder source location, $[x_{th}, y_{th}, z_{th}]$, to be mapped as:

$$[x_{th}, y_{th}, z_{th}] = c_0^2(t - t_0) \left[s_x, s_y, \sqrt{1/c_0^2 - s_x^2 - s_y^2} \right] + [x_0, y_0, z_0] \quad (1)$$

where $[x_0, y_0, z_0]$ are the coordinates of the acoustic array.

[12] Elapsed time since flash is calculated directly from the thunder time series relative to the median occurrence times of the LMA rf sources for the event (see upward-pointed arrows in [Figure 2](#)). New Mexico LMA locations were all within

the confines of the LMA network and were recorded continuously during the study interval. Event flash times and locations, accurate to a few tens of meters, were determined using 6 or more out of 9 active New Mexico LMA stations [Rison et al., 1999; Thomas et al., 2004].

[13] To avoid interference of overlapping thunder signals from multiple lightning flashes, we limit our analysis to time-isolated LMA events separated from previous and following events by at least 30 s. This reduces the July 24th dataset to 24 events (see Figure 2 shading) out of 90 total events identified in the LMA catalogue. We note that it is also possible to identify lightning event times (t_0) directly from the electromagnetic noise picked up on microphone channels that are connected by 30-m cables to the central datalogger. This noise appears as a 100–600 ms sequence of distinctive high-amplitude spikes (Figure 3). In the absence of LMA data this noise provides convenient lightning occurrence timing. The duration of the electromagnetic noise is concurrent with the duration of the LMA flashes, but its intensity is not well correlated with the number of rf sources.

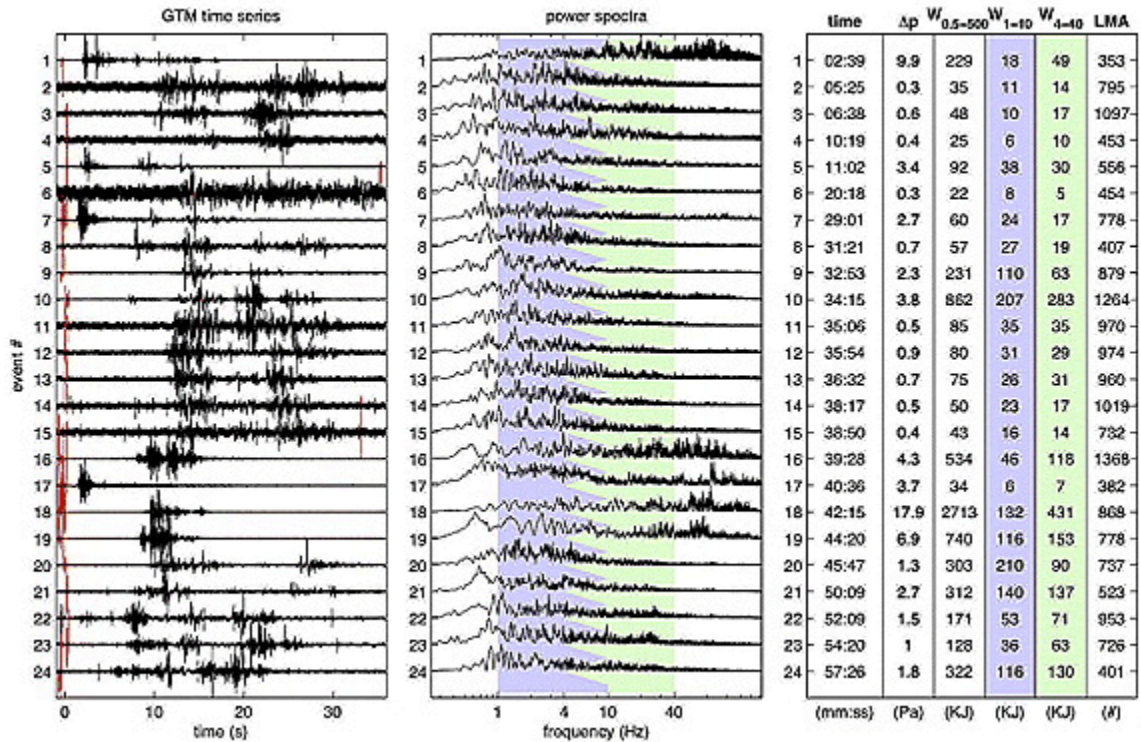


Figure 3

Normalized thunder waveforms for those signals identified in [Figure 2](#) are from GTM and have been filtered above 0.5 Hz. Channel 1 waveforms (black) overlay channel 2 waveforms (red), which are similar except that they also indicate em noise pick-up by the extended sensor cable. Times of lightning flashes are aligned at $t_0 = 0$ s. (middle) Logarithmic power spectra are shown along with indicated frequency bands (1–10 Hz and 4–40 Hz) that are used in the thunder coherence mapping and quantified in the data table. (right) Data table shows event timing (relative to 1:00 PM local time), peak-to-peak pressure recorded at GTM, energy (in KJ) contained in the first 35 s of waveform calculated according to [equation \(4\)](#) for three different frequency bands (0.5–500 Hz, 1–10 Hz, and 4–40 Hz), and the number of associated rf sources located by the LMA within 12 km of GTM.

[14] For a given time window Δt centered at time $t = t_0 + r/c_0$ the coherence may be mapped as a function of source-receiver distance (r) and the horizontal slownesses. The energy of a beam-shifted stack of N array elements calculated in the time domain is proportional to:

$$E(s_x, s_y, r) \propto \sum_{j=t_0+r/c_0-\Delta t/2}^{t_0+r/c_0+\Delta t/2} \left\{ \sum_{n=1}^N \frac{p_n \left(j + \Delta x_n s_x + \Delta y_n s_y + \Delta z_n \left(\sqrt{1/c_0^2 - s_x^2 - s_y^2} \right) \right)}{N} \right\}^2 \quad (2)$$

for horizontal slowness values $s_x^2 + s_y^2 < 1/c_0^2$. The variables Δx_n , Δy_n , and Δz_n are the relative Cartesian coordinates of array element n with respect to the center of the array, j and Δt are the center sample and number of samples in the analyzed window, and p_n is the band-filtered pressure waveform for channel n . A measure of uncorrelated energy (E_{unc}) can be obtained by calculating the average of [equation \(2\)](#) for all slowness values where $s_x^2 + s_y^2 > 1/c_0^2$. E_{unc} corresponds to impossibly large slowness values traversing the array and is

assumed to be equal to the stacked energy of uncorrelated noise. A normalized coherence is then defined as:

$$X(s_x, s_y, r) = E/E_{unc} - 1. \quad (3)$$

Coherence values, X , are converted to a Cartesian space by finding the maximum coherence in each $(0.5 \text{ km})^3$ volume bin. These maximum values are then displayed in both plan and profile views (e.g., [Figure 4](#)). Coherence maps have advantages over the techniques for thunder localization presented by [MacGorman et al. \[1981\]](#), [Teer and Few \[1974\]](#), and [Arechiga et al. \[2011\]](#) in that multiple coherent sources can be mapped for an individual distance range and peakedness of a slowness solution provides information about location uncertainty and/or frequency content of the source. Coherence maps calculated for synthetic pulses reveal that broader bull's-eyes are primarily an artifact of the frequency content of the source signal (see [auxiliary material](#)).

[equation \(3\)](#) using the GTM array. Map and profile axes are relative to the location of the GTM array, which is situated at 33.99012°N, 107.18237°W, and 3112 m above sea level. The mapped LMA rf sources (blue points) correspond to those rf emissions occurring within ± 1 s of the designated event onset and possessing a nearest neighbor within 100 m. Thunder and LMA locations are mapped out to ~12 km from the GTM array (dash-dot lines) and are also shown in (top right) West-East and (middle left) North-South profile views. (top left) Waveforms are provided for the 4 elements of the GTM array with time axis relative to the LMA flash time. (bottom) Waveforms show thunder signals recorded at three different network stations and filtered in the same band. Blue shading denotes the predicted time shift between select thunder hypocenter ellipses (refer to error analysis in the auxiliary material) and respective recording site. Width of shading indicates timing uncertainties associated with the error ellipses depicted in the coherence maps. Coherence maps for all 24 highlighted events and 2 different frequency bands are provided in Figures S2 and S3 in the auxiliary material.

[15] General validation of the thunder imaging technique is evident through comparison with the distribution of rf sources located by the LMA ([Figures 4 and S3](#)). For the featured hour on July 24th most lightning events, and mapped thunder, occurred to the SE of the GTM array although lightning branches for some events extended to the NE (#9, 11, 14, 18, and 21–24), SW (#7, 10, and 16) and W (#1) of GTM. These branches were reliably detected by both LMA and acoustic sensing techniques. In general, thunder coherence maps are displayed as diffuse regions, especially for infrasonic (1–10 Hz) thunder, which encompass the clouds of rf sources located by the LMA. Coherence for higher frequency (4–40 Hz) sound is better for delineating fine-scale structure of thunder source regions. Coherence for even higher frequency sound (e.g., >10 Hz) is less good and tends to be poorly correlated with LMA rf sources.

[16] For select lightning events, correlation of multiple discrete infrasonic thunder pulses is evident across the network. As seen in [Figure 4](#) (bottom) select thunder hypocenters, and error ellipses correspond to time-isolated, high-coherence pulses. In the absence of information about atmospheric structure the thunder location range, azimuth, and inclination errors (in meters) are respectively taken as $\Delta r = \pm 160 \text{ m} + 0.065 \times r$ (for 0.5 s time windows), $\Delta \theta = \pm 0.05 \times r$, and $\Delta \phi \approx \pm r \times (i_0 - \sin^{-1}(0.93 \times \sin i_0) + 0.03 \times \cos i_0)$, where i_0 is incidence angle measured from the vertical. The 3-D error ellipses associated with these thunder pulses are then used to estimate a range of sound propagation time-of-flight to stations KVV and LAN. Correlated pulses can be identified across the network including those occurring at ~18 s (A), 23 s (B), 26 s (C), and 27.5 s (D) in the GTM trace of event #12 ([Figure 4](#)). For other lightning events the thunder pulses are often less well correlated across the network. This may occur due to complex interference arising from the different azimuthal and distance perspectives of the microphones.

[17] For a single point source at a given distance range, a ‘bull's-eye’ coherence map may be expected where the radius of the bull's-eye is related to the frequency content of the sound source. Broader coherence peaks should result from a lower frequency signal that might suggest a more diffuse, or extensive, source, such as would be attributed to electrostatic relaxation [e.g., [Balachandran, 1979](#)]. Higher frequency sources appear more peaked in the slowness solution and may reflect thermal shocking of an ionized channel, which is the source of audible thunder [e.g., [Few, 1969](#)]. In general, the 4–40 Hz coherence map appears to delineate finer-scale structure of thunder generating phenomena and more closely tracks channels identified by the LMA. Coherence maps may be calculated for synthetic thunder pulses demonstrating how location resolution capabilities are worse for longer wavelength signals.

[18] Misfit between LMA and thunder locations is of particular interest. In some cases, thunder imaging identifies regions of sound production, which are not “seen” in the LMA record. One of many type examples can be seen as in event

#1 (source C in the 1–10 Hz band). Other examples include thunder that is mapped as cloud-to-ground, such as those of events #3, 6, 10, 16, and 23, that are not obviously connected from cloud-to-ground in the LMA data. In other instances, notably in events #1, 5, 7, 17, 18, both LMA and thunder distributions clearly indicate cloud-to-ground lightning.

[19] Conversely, LMA mapping is able to detect some lightning channels, which are not well identified through thunder imaging. Events #16, 17, and 18 show kilometer-long “tendrils”, which are not identified in the acoustic mapping even though there is suggestion that their endpoints are associated with coherent thunder sources. Notwithstanding the possibility that acoustic detection is hampered by atmospheric propagation phenomena, inefficient sound radiation geometry, or obfuscating noise at the sensor array, it is probable that certain lightning channels do not ensonify the atmosphere due to relatively low current.

[20] Thunder imaging has the potential to elucidate processes in the atmosphere including both current conveyance and volume charging/discharging that are not easily seen with the LMA. Estimating the radiated acoustic energy in the infrasound and high frequency bands offers a means to potentially quantify or distinguish these two processes. For a homogeneous atmosphere with impedance ρc_0 the energy contained in a band-limited acoustic thunder signal can be approximated by assuming spherical radiation of sound and integrating over the travel time $(t-t_0)$ since lightning flash (modified from [Holmes et al.](#)

[1971]):

$$W = \frac{4\pi}{\rho c_0} \int_{t_0}^{t_0+\tau} p(t)^2 c_0^2 (t - t_0)^2 dt \quad (4)$$

The table in [Figure 3](#) shows this integral calculated for broadband (0.5 Hz to 500 Hz) signal as well as for band-limited (1–10 Hz and 4–40 Hz) signal out to $\tau = 35$ s. It is noteworthy that there is more than two orders of magnitude variability for the 24 featured events (ranging from 22 KJ total acoustic energy for event #6 to 2713 KJ for event #18). It is also significant that the spectral character of the thunder is widely variable, with some events (such as #1, 16, and 18) containing

three times more energy in the 4–40 Hz band than in the 1–10 Hz band, whereas other events (such as #5–9) appear to radiate more energy at 1–10 Hz than at 4–40 Hz. The calculated acoustic energy in these three frequency bands does not appear correlated with the quantity of rf sources, as determined by the LMA. This last observation supports the conclusion that VHF and acoustic radiation represent inherently distinct processes; the LMA sees channel formation and fast current impulses, whereas the acoustic radiation reflects net charge transport.

3.2 Thunder Theory Research

After some independent research as well as conferring with the FU Lighting Research Lab. We learned that an acoustic approach to detecting thunder is uncommon such as the one we are planning.

When it is used, in some cases, [1] a network of broadband microphones, including a 4-element array was used for example to locate the sources of thunder occurring during an electrical storm in central New Mexico on July 24th, 2009. This approach is more expansive than what we considered for our effort. Combined slowness search and distance ranging are used to identify thunder regions in three dimensions (out to 12 km) and for two overlapping frequency bands (1–10 and 4–40 Hz). Distinct thunder pulses are locatable and used to predict time-of-arrival to neighboring stations and to identify correlated phases across the network. Spatial correlation is also found between the thunder source regions and regions of very high frequency (VHF) radiation as located by the New Mexico Lightning Mapping Array (LMA). Some of the misfit between the LMA and thunder locations is attributable to differences in excitation mechanisms of the respective radiation, which is related to current impulses in lightning channels (for thunder) and incremental ionization of the atmosphere (for VHF emissions).

Thunder Acoustic Specific Research

Acoustic, VHF, and electrostatic measurements throw new light onto the origin and production mechanism of the thunder infrasound signature (<10 Hz) from lightning. This signature, composed of an initial compression followed by a rarefaction pulse, has been the

subject of several unconfirmed theories and models. The observations of two intracloud flashes which each produced multiple infrasound pulses were analyzed for this work. Once the variation of the speed of sound with temperature is taken into account, both the compression and rarefaction portions of the infrasound pulses are found to originate very near lightning channels mapped by the Lightning Mapping Array. It was found that none of the currently proposed models can explain infrasound generation by lightning, and thus propose an alternate theory: The infrasound compression pulse is produced by electrostatic interaction of the charge deposited on the channel and in the streamer zone of the lightning channel.

Acoustic emission from lightning, called thunder, can be classified in terms of the range of frequencies contained in the emission. The origin of the audible portion of the thunder (20–20,000 Hz) is well established and believed to come mostly from shock waves produced by the rapid expansion of a lightning channel as it heats [*Few et al.*, [1967](#); *Few and Dessler*, [1969](#); *Few*, [1969](#)]. In addition to audible acoustic signals, some cloud-to-ground (CG) and most intracloud (IC) lightning flashes produce infrasonic acoustic signals below 20 Hz which are not audible by humans. Numerous observations of the infrasound produced by lightning show discrete pulses characterized by an initial compression followed by a rarefaction with amplitude in the range of 0.05–5 Pa and spectral peak in the range of 0.2–2 Hz [e.g., *Bohannon et al.*, [1977](#); *Bohannon*, [1978](#); *Balachandran*, [1979](#), [1983](#); *Assink et al.*, [2008](#); *Chum et al.*, [2013](#)].

The first theoretical descriptions of infrasound production by lightning were given by *Wilson* [[1920](#)] and *Dessler* [[1973](#)] and predate observations of the phenomena. Both theories predicted that relief of electrostatic pressure in the charge layers in a thunderstorm could produce a rarefaction infrasound signature but did not predict the observed preceding compression. Most recent theories agree that the rarefaction portion of the infrasound signature of lightning is produced by an electrostatic relaxation; however, the mechanism which produces the initial compression has remained a mystery. *Bohannon et al.* [[1977](#)] and *Bohannon* [[1978](#), [1980](#)] suggested a number of different production mechanisms for the compression portion of the signature including rapid intensification of the

electric field just prior to a lightning flash, ohmic heating of the air by charge flowing into the lightning channel, and mutual attraction (and subsequent release) of the positive and negative charge layers. While *Bohannon* [1980] concluded that the most likely mechanism for the infrasound compression was mutual attraction of the positive and negative charge layers, the other mechanisms have been more recently revisited.

Few [1985] revisited the ohmic heating theory and postulated that rapid electrical heating of the air by positive streamer systems during a discharge would produce small but measurable positive pressure perturbations. Modeling results showed that heating by electrical currents would produce a compression pulse which was always smaller than the following rarefaction pulse, but with similar duration. Infrasonic production through the rapid intensification of the electric field was reexamined by *Pasko* [2009]. Originally, Bohannon concluded that the intensification would have to occur very shortly (around 0.5 s) before the lightning flash and would require unrealistic charging rates inside the cloud [*Bohannon*, 1980]. Pasko showed that the time period could be longer, on the order of 2–6 s, and could produce realistic pulses when combined with the electrostatic relaxation mechanism above, at high but reasonable charging rates. Currently, the modeled infrasound pulses produced by Pasko most closely resemble infrasound observed in nature.

3.3 Combined Lightning and Thunder Theory Research

Up until this point, no study had been in a position to validate the existing hypotheses. To do so, required knowledge of the location or shape of the charge layers within the cloud, as well as the location and structure of the parent lightning flash, such measurements were simply not available. In addition, until recently the location algorithms for the infrasound pulses themselves were somewhat limited and unable to locate the pulses precisely. A set of measurements were presented which put the infrasonic pulses produced by lightning in context within the flash and within the storm. Lightning mapping array (LMA) data are used to determine the structure of the lightning flash, as well as the geometry of the charge layers in the thunderstorm. The acoustic localization of the compression and rarefaction portions of infrasound pulses produced by two IC flashes is then compared with their electrical environment. In order to improve the correspondence between the VHF and acoustic localization, a correction is introduced to take into account the variation of the speed of sound with temperature. Included in this comparison are DC electric field measurements on the ground under the storm in the vicinity of the lightning flash. We find that none of the currently proposed models are consistent with these measurements; however, the observations suggest that the infrasound pulses may be produced by electrostatic interaction of charge deposited in the streamer zone of a lightning channel.

Acoustic infrasound pulses were located in lightning with a network of three and four broadband microphone arrays, deployed within 1.5 km from one another in the Magdalena Mountains of central New Mexico. Each array includes microphones located at the center and vertices of an approximate 45 m equilateral triangle. The microphones possess flat response in the 0.01 to 500 Hz band [Marcillo *et al.*, [2012](#)]. At all stations, broadband sound was recorded with data loggers acquiring at a resolution of 24 bits and a sample rate of 1000 Hz. For the study of infrasound, the acoustic signals are then digitally zero-phase

band-pass filtered from 0.2 to 10 Hz. During a survey of the acoustic records obtained during the 2011–2012 storm seasons, approximately 200 flashes were identified with clear infrasound pulses, including virtually every IC flash recorded. Previous studies have indicated infrasound pulses to be somewhat less common than this. The larger fraction found in this study is attributed to quiet recording locations in forested areas where wind noise is reduced. Two IC flashes have been selected from these for detailed study which occurred in storms on 20 July 2011 and 21 August 2012.

For each array, the spatial coordinates (x , y , and z) of the infrasound pulses are found using the Levenberg-Marquardt algorithm [Badillo Ibarra *et al.*, [2013](#)]. The algorithm produces locations with accuracies of approximately 250 m before correcting for the variation of the speed of sound with altitude [Badillo Ibarra *et al.*, [2013](#)]. One of the limitations of acoustic localization is that the position of an acoustic source can be determined, but not precisely when the source was produced. The inputs to the algorithm are the start time of the flash and the times of arrival of the sound to the microphones, using the microphone at the center of the array as a time reference. The start of the flash is obtained by electromagnetic interference pickup on a 30 m unshielded cable and corroborated with the LMA starting time of the flash [Arechiga *et al.*, [2009](#)]. The time difference of arrival between the center microphones and the outer microphones is obtained using cross correlation with a 0.256 s window.

The New Mexico Tech LMA locates VHF sources from lightning using time of arrival differences from six or more stations with an uncertainty of 6–12 m horizontally and 20–30 m vertically for sources located over the array for data processed with 80 μ s windows [Rison *et al.*, [1999](#); Thomas *et al.*, [2004](#)]. The LMA produces detailed 3-D maps showing the entire channel structure of the flash. Lightning flashes can be used as a tracer of the charge regions by identifying the polarity of the breakdown in each channel of each flash [Marshall *et al.*, [2005](#)]. If enough lightning flashes are mapped during a storm by the LMA, the geometry of the charge layers can be determined. The charge layers vary on

the same time scale as the thunderstorm's evolution and remain relatively constant over time periods of 5–20 min.

The electrostatic field is measured by a small network of three custom-designed field mills located around Langmuir Laboratory. The field mills have a dynamic range of ± 20 kV/m and are sampled at 10 Hz.

In previous analyses of acoustic data, the speed of sound was assumed to be constant when determining the location of acoustic sources. This is a good assumption at the ground when using time differences between channels to determine the direction of the source. However, the speed of sound varies significantly with temperature and altitude, making it unclear which constant speed of sound should be used for determining the range to an acoustic source. In practice, previous studies used a constant speed of sound, which was a compromise between the value at ground level and in the cloud. This gives rise to errors which can be estimated as

$$\delta R = v_a \delta \tau + \tau \delta v_a$$

(1)

where τ is the transit time of the sound to the microphone and v_a is the speed of sound. The error in measuring the transit time is due to the duration of the electromagnetic interference pulse, which is approximately 0.2 s for Flash A. Therefore, the first term for the range error contributes between 30 and 34 m of inaccuracy. For nearby sources, both the variation of the speed of sound and the transit time are small, so that the second term is comparable to the first, giving rise to an additional 30–50 m of error. However, for distant sources, both the variation of the speed of sound and the transit time are much larger. The ranging error for distant sources can be as large as 500 m, and this large error can be mitigated by accounting for the variation of the speed of sound with temperature.

In this work, we present a method to account for the variation of the speed of sound with altitude and temperature, based on a linear temperature profile in the storm. The effects of refraction, wind, and other mechanisms are not taken into account, and their significance has not been explored.

The variation of temperature with altitude can be modeled with a linear function:

$$T = T_{\text{ref}} - \Gamma(z - z_{\text{ref}}) \quad (2)$$

where z is the vertical distance above the ground (the source altitude, z_s is a fixed number (e.g., 8 km), while z is a variable ranging from 0 to z_s), T_{ref} and z_{ref} are the reference temperature and height above the ground, and Γ is the temperature lapse rate, which is considered to be a positive constant in K/km. The temperature lapse rate is well defined in dry air undergoing adiabatic expansion, having a value of $\Gamma=9.8$ K/km. In moist air, as a parcel rises, water vapor condenses and releases latent heat, causing the lapse rate to be reduced and become a function of the water mixing ratio. For this reason, the temperature profile needs to be referenced to a temperature near the region of interest. In this case, the temperature profile is referenced to the -40°C level which corresponds to the top of the mixed phase region in the cloud. The environmental temperature lapse rate can then be determined using readily available atmospheric soundings. Calling θ the elevation angle and R the distance from the microphone to the source, $z = R \sin\theta$. Substituting z in equation 2 gives

$$T = T_{\text{ref}} - \Gamma(R \sin \theta - z_{\text{ref}}) . \quad (3)$$

The speed of sound, v_a , in an ideal gas is given by [e.g., *Tipler and Mosca*, 2004, p. 468]

$$\frac{\partial R}{\partial t} = v_a = \sqrt{\frac{\gamma R_g T}{M}} = \sqrt{\frac{\gamma R_g (T_{\text{ref}} - \Gamma (R \sin \theta - z_{\text{ref}}))}{M}}$$

(4)

where $\gamma = 1.4$ is the adiabatic index, $R_g = 8.3145 \text{ J mol}^{-1} \text{ K}^{-1}$ is the ideal gas constant, and M is the average molar mass. For air, $M = 28.95 \text{ g mol}^{-1}$.

Equation 4 defines a separable differential equation with solution

$$\Delta t = \frac{2}{\Gamma \sin(\theta)} \sqrt{\frac{M}{\gamma R_g}} \left(\sqrt{k} - \sqrt{k - \Gamma \sin(\theta) R} \right)$$

(5)

where $k \equiv T_{\text{ref}} + \Gamma z_{\text{ref}}$ and Δt is the transit time of sound from the source to the microphone. Equation 5 can be inverted, giving the range R in terms of the transit time

$$R = \Delta t \sqrt{\frac{\gamma R_g}{M}} \left(\sqrt{k} - \Delta t \sqrt{\frac{\gamma R_g}{M}} \frac{\Gamma \sin(\theta)}{4} \right)$$

(6)

Equations 5 and 6 introduce a method to account for the variation of the speed of sound with temperature. Using this correction improves the agreement between acoustic and VHF localization, especially for high-altitude sources such as those located in the upper positive charge layer. The effects are particularly noticeable in the arrival time histograms shown in Figures 2d and 3d. The histograms are built by calculating the time an acoustic source colocated with an LMA-determined VHF source location would arrive at the center microphone of the acoustic array, and then histogrammed using 0.3 s bins. All VHF sources produced by the lightning flash are included in the histograms. The outlined

histograms uses a constant, 340 m s^{-1} speed of sound, while the solid histograms take into account the variation of the speed of sound with temperature. There is an imperfect but strong correspondence between peaks in the temperature-corrected histogram and the onset of infrasound recorded by the acoustic array. A similar analysis presented in *Arechiga et al.* [2009] found that the locations of acoustic signals recorded during a lightning flash did not directly correspond to acoustic sources originating on lightning channels detected by the LMA. The reason for the different results is that the previous study assumed a constant speed of sound. While the speed of sound correction is only first order, and makes use of a simplistic linear fit to the atmospheric temperature profile, it is particularly easy to implement in practice. In addition, the information required to model the atmospheric temperature profiles is readily acquired from freely available atmospheric soundings.

Flash A occurred 2–6 km southwest of the network of acoustic arrays at Langmuir Laboratory at 20:24:11 UT on 20 July 2011. The flash was a classic bilevel intracloud with negative breakdown traveling upward into the positive charge layer, and positive breakdown traveling downward into the negative charge layer. The acoustic record shows two infrasound pulses separated by approximately 10 s arriving at the arrays (*mg*, *mk*, *ml*, and *mw*). Figure 1 shows the pressure recorded at the *mg* array, averaged over the four receiving microphones. The first pulse (A-19) is somewhat lower in amplitude and noisier, with oscillations during the rarefaction portion of the pulse. The second pulse (A-29) is better defined both for the compression and rarefaction pulses. Rise times, pulse durations, and amplitude fractions for these pulses are given in Table 1.

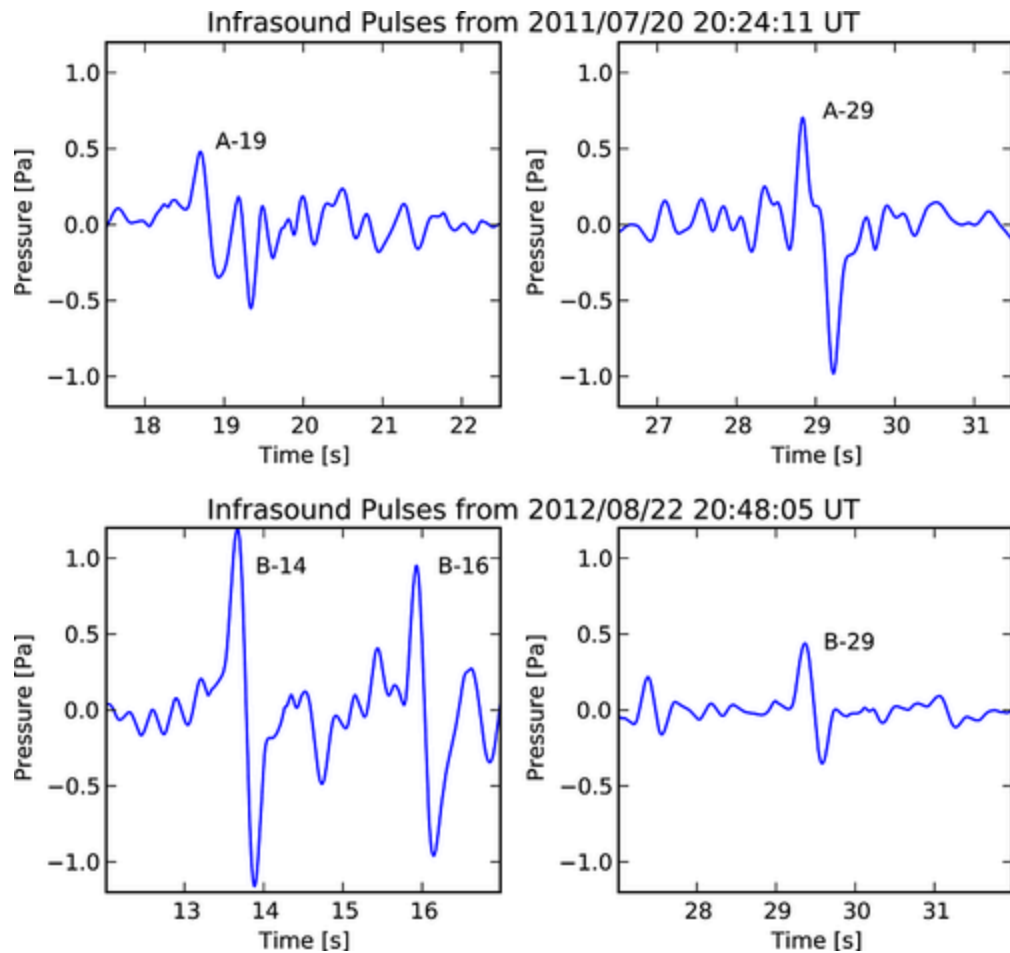


Figure 1

Figure 1 shows the pressure recorded at the *mg* array, averaged over the four receiving microphones. The first pulse (A-19) is somewhat lower in amplitude and noisier with oscillations during the rarefaction portion of the pulse. The second pulse (A-29) is better defined both for the compression and rarefaction pulses. Rise times, pulse durations, and amplitude fractions for these pulses are given in Table 1.

Table 1. Average Parameters for the Infrasound Pulses Observed in the Flashes for Study

Flash	Time	Rise Time	Half Period	Amplitude Ratio
-------	------	-----------	-------------	-----------------

A	19	0.21	0.63	0.87
A	29	0.12	0.37	0.69
B	14	0.23	0.22	1.01
B	16	0.18	0.19	0.98
B	29	0.16	0.23	1.18

^a Time (from start of the flash), rise time, and half period listed in seconds.
Half period measured from compression peak to rarefaction peak.
Amplitude ratio is the ratio of the compression to the rarefaction amplitude.

Figures [2a](#)–[2c](#) show the structure and environment of the flash as determined by the LMA, with the locations of the active acoustic arrays denoted with color-coded triangles. The geometry of the charge layers shown in Figures [2a](#) and [2c](#) was obtained from charge analysis of 10 min of LMA data around the flash. The red regions show the lower and upper positive charge layers, and the blue region shows the main negative charge layer. Infrasound (compression and rarefaction) pulses located by the acoustic arrays are overlaid as circles and squares, respectively, color coded by acoustic array.

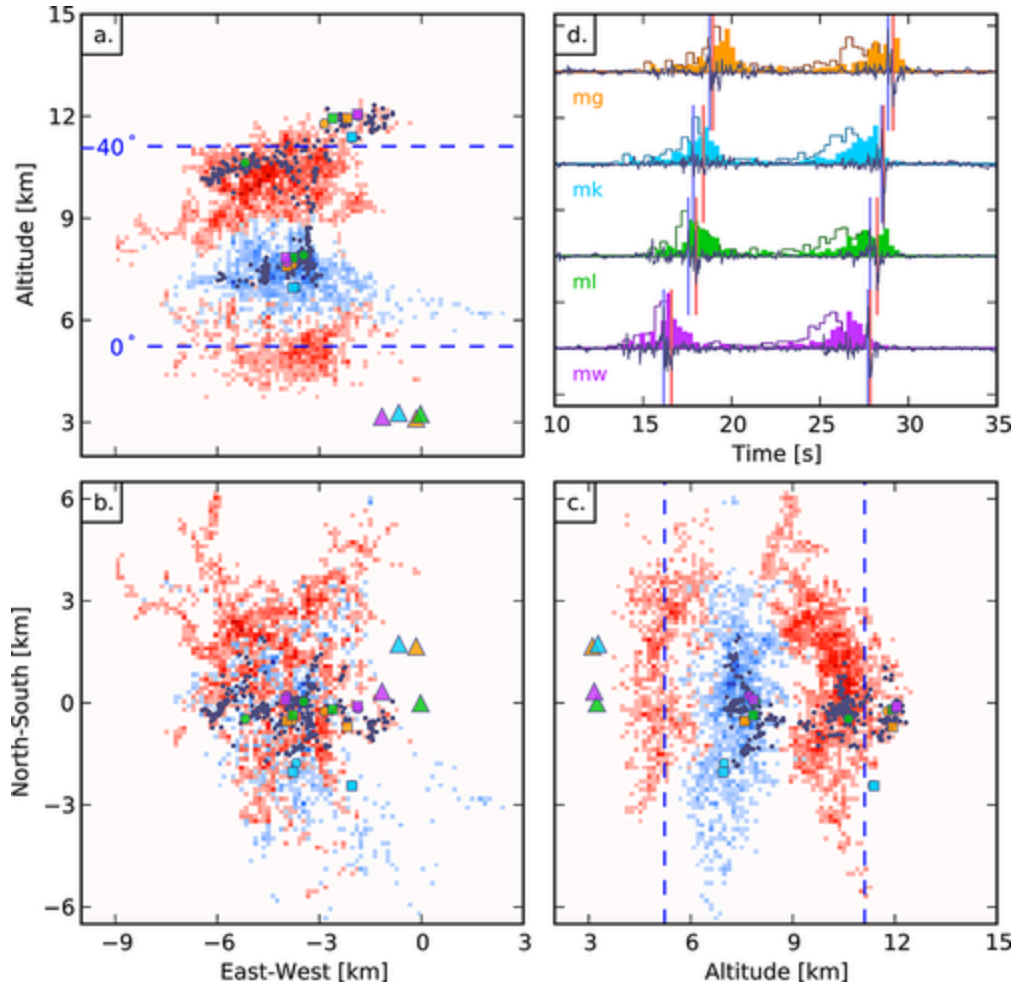


Figure 2

Flash A occurred 2–6 km southwest of the network of acoustic arrays at 20:24:11 UT on 20 July 2011. The flash was a classic bilevel intracloud flash. (a–c) The LMA structure (blue dots) and environment (thunderstorm charge structure inferred from surrounding LMA flashes; red is positive charge, and blue is negative charge) with the locations of the active acoustic arrays denoted with color-coded triangles. Infrasound (compression and rarefaction) pulses located by the acoustic arrays are overlaid as circles and squares, respectively, color coded by acoustic array. (d) A single channel of band pass (0.2–10 Hz) digitally filtered waveforms for each array. Also included are histograms of the LMA source arrivals at each microphone plotted against the measured waveforms.

The dashed lines indicate the altitude of the -40°C and 0°C isotherms, based on temperature soundings from Albuquerque, New Mexico, which is about 100 km from Langmuir Laboratory. The soundings are representative of the background conditions near the storm, but temperatures inside the cloud can deviate from the background values. The -40°C level defines the z_{ref} in equation 2, and the distance from -40°C to 0°C determines the lapse rate Γ .

Figure 2d shows a single channel of band pass (0.2–10 Hz) digitally filtered waveforms for each array. Each waveform exhibits two infrasound pulses with peak amplitudes of ~ 1 Pa and periods between 0.5 and 1 s. The times of the compression and rarefaction portions of the waveforms located above are overlaid with blue and red lines, respectively. Also included are histograms of the LMA source arrivals at each microphone plotted against the measured waveforms, which show very good correlation between the presence of infrasound and peaks in the histogram and are discussed in section 2.1.

Finally, for completeness, Figure 4 shows the electrostatic field measured on the ground below the storm by one of the Langmuir Laboratory field mills. The flash of interest is denoted with a red vertical line at the onset of the flash. Because the field mills switch from a low-field sensitive mode to high-field mode around ± 5 kV/m, the noise seen in the measurement is higher in the ± 5 kV/m range. The field deflection caused by flash A is relatively small compared to the field deflections of the flashes immediately before and after; this is because Flash A is an IC flash during an active storm. The larger field deflections are produced by CG flashes.

The analysis above was repeated for a second IC flash, Flash B, occurring above and south of the acoustic arrays at 20:48:05 UT on 21 August 2012. During this period, a different set of three acoustic arrays were active. Unfortunately, one of the arrays (*mb*) was somewhat noisier than the others, and as a result, acoustic locations from this station are not as accurate. In this flash, the infrasound record

shows three infrasound pulses, average pressure records are shown in Figure [1](#). Two distinct, prominent infrasound pulses (B-14 and B-16) occur in quick succession early in the acoustic record. A third smaller infrasound pulse (B-29) is seen and located at some of the acoustic arrays and is particularly well defined at the *mg* array. Rise times, pulse durations, and amplitude fractions for these pulses are given in Table [1](#).

Figure [3](#) shows the structure and environment of the flash. The charge structure for the parent storm containing Flash B is more complex. There are two cells immediately adjacent to one another, causing the charge layers to show some tilt. As can be seen in Figure [4](#), Flash B was one of the very first flashes to occur during its parent storm, and the flash is very isolated in time. In order to obtain enough flashes to produce a good charge analysis, the 10 min of LMA data after 75,000 sod (20:50:00 UT) were used. As a result, the charge analysis may be somewhat less representative of the charge configuration during the earliest portion of the storm. However, the location and thickness of the charge layers should be accurate.

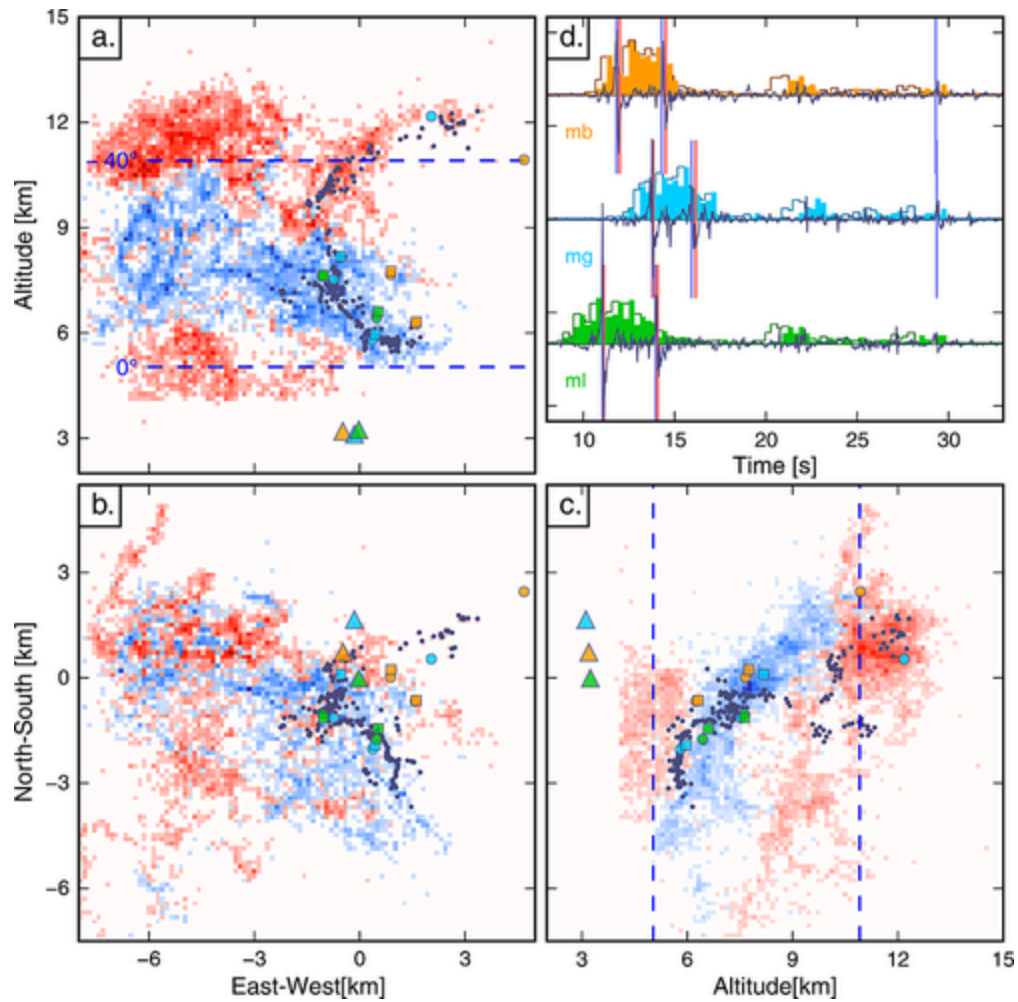


Figure 3

(a–d) Flash B occurred over and south of the acoustic arrays at 20:48:05 UT on 21 August 2012. During this period, a different set of three acoustic arrays were active. Unfortunately, one of the arrays (*mb*) was somewhat noisier than the others, and as a result, acoustic locations from this station are not as accurate. See Figure 2.

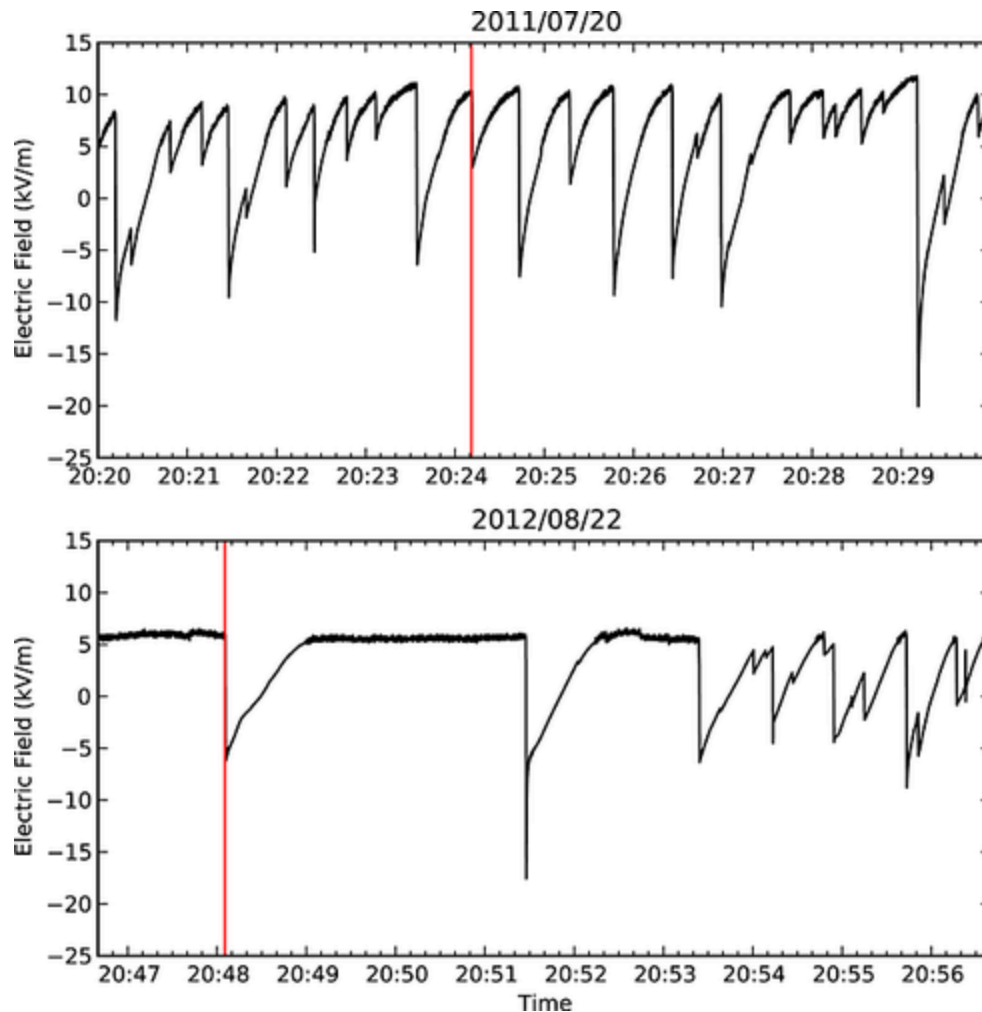


Figure 4

Electrostatic field measured on the ground below the storm by one of the Langmuir Laboratory field mills. The field mills use the physics convention. Positive voltage means electric field is pointing up. The flashes of interest are denoted with a red vertical line at the onset of the flash. The field deflection caused by (top) flash A is relatively small compared to the field deflections of the flashes immediately before and after; this is because Flash A is an IC flash during an active storm. The larger field deflections are produced by CG flashes. (bottom) Flash B was one of the very first flashes to occur during its parent

storm. In both cases the electrostatic measurements are consistent with slow, steady charging inside the cloud.

Based on these observations of two flashes, infrasound pulses produced by in cloud (and likely other) lightning exhibit the following properties: Each IC lightning flash can produce multiple infrasound pulses; typically, one is seen in each of the positive and negative charge layers. Each pulse first has a well-defined compression, followed by a rarefaction which sometimes contains other oscillations; the ratio of the compression amplitude to the rarefaction amplitude is approximately 1. The average half period of the infrasound pulses is 0.33 ± 0.16 s, implying that the pulses were produced by something with a length scale of approximately 200 m. Both the compression and rarefaction portion of the infrasound pulses are locatable and are found to be in close proximity to each other and the lightning channel. It is likely that the compression and rarefaction portions of a pulse are produced in the same location, but the different arrival times give rise to the rarefaction pulse being located slightly farther away. In almost all cases, in cloud lightning will produce at least one infrasound pulse, but frequently, an infrasound pulse is produced in each charge region. Because the main negative and upper positive charge layers are separated vertically by 3–5 km, the infrasound pulses will be separated by 10–17 s. Less frequently, two pulses are produced in a single charge layer, as seen in Flash B. In this case, the pulses are closer together in time.

The DC electric field measurements on the ground are consistent with slow, steady charging in the cloud, and there is no evidence of an accelerated charging rate just prior to a flash. The mechanism which produces the infrasound during the lightning flash should explain all of these properties. Previously suggested mechanisms can be classified into two basic types: ohmic heating in the streamer zones around the lightning channels [e.g., *Few*, [1985](#)] and electrostatic effects in the charge layers of the thunderstorm [e.g., *Pasko*, [2009](#)].

It is unlikely that ohmic heating in the streamer zones of the lightning channel could produce the observed infrasound pulses, although there is evidence that ohmic heating can produce infrasound in sprites [*da Silva and Pasko*, [2014](#)]. While the scale size of the ohmic heating region is correct for producing infrasound pulses with reasonable duration, there are a number of issues with the ohmic heating mechanism. First, the model proposed by *Few* [[1985](#)] described only a monopolar compression pulse being produced by ohmic heating; the rarefaction portion of the pulse was still assumed to be produced by electrostatic contraction of the charge layer. Because the compression and rarefaction portions of the infrasound waves are not produced in the same region of the cloud by this mechanism, it is expected that acoustic localization would identify different origins for the compression and rarefaction in a significant fraction of located pulses. However, observations of infrasound pulses from several lightning flashes and from multiple simultaneous observing locations show that the compression and rarefaction portions of the pulse are located in close proximity to each other. We also note that, consistent with the results presented in *Chum et al.* [[2013](#)], the compression to rarefaction amplitude of approximately 0.4 predicted for this mechanism is also not observed. It may however be possible that a pulse produced by ohmic heating is N shaped, as is seen in acoustic emission from sparks [*Few*, [1969](#)], or may become bipolar after propagating a sufficient distance.

Second, there is not enough heat produced in the streamer zone of the lightning channel to produce the needed pressure change in the volume of air around the channel. To produce an infrasound pulse, the air pressure in the volume around the channel needs to increase by approximately 1 Pa [*Few*, [1985](#)]. If the air inside the streamer zone is assumed to expand adiabatically, and is at a background temperature of -20°C and pressure of 500 mb, then the temperature change needed to increase the pressure by 1 Pa is 1.4×10^{-3} K. To estimate the heat added to the volume, *Few* [[1985](#)] assumed a slab model of the lightning discharge, which (as also observed by *Few*) is not a very realistic model of a

lightning channel. A better model is presented in *Bazelyan and Raizer* [1997, p. 69], where the estimated temperature change in the streamer zone is calculated from the energy stored on a long thin cylindrical capacitor,

$$\Delta T \leq \frac{\epsilon_0 U_0^2 (1 - \epsilon_V)}{r_c^2 c_V \ln(l/r_c)} \quad (7)$$

where ϵ_0 is the permittivity of free space, U_0 is the potential difference between the lightning channel and the cloud, ϵ_V is the fraction of energy transferred into slowly relaxing rotational modes which can be as large as 0.9 for air, r_c is the radius of the streamer zone around the lightning channel, $c_V \sim 500 \text{ J m}^{-3} \text{ K}^{-1}$ is the volumetric heat capacity of air, and l is the length of the channel. For a large range of reasonable l and r_c values, $\ln(l/r_c) \sim 5$. The temperature change in the streamer zone can be estimated by assuming $U_0/r_c \sim 1.5 \times 10^5 \text{ V m}^{-1}$; this is the field needed for positive streamer propagation, and the streamer zone should expand until this criterion is met. Using these criteria implies the radius of the streamer zone, r_c , is very large. If the potential difference between the tip of the leader and the cloud is 50 MV, then $r_c = 330 \text{ m}$. Values of r_c this large are also supported by negative step length measurements inside the cloud [Winn *et al.*, 2011; Edens *et al.*, 2014]. In this case, assuming $\epsilon_V = 0$, the expected temperature change in the positive streamer zone is $8.0 \times 10^{-5} \text{ K}$, far less than that needed to increase the pressure in the volume by 1 Pa. The temperature change is even lower if rotational modes of air are a significant factor.

Electrostatic interactions of the charge regions of a thunderstorm also cannot explain the production of infrasound pulses by lightning. First, if the infrasound pulses were produced by the charge layers, we expect the location of the infrasound pulses to be near the geometric center of the charge layer or perhaps near the charge center of the flash. However, in all cases, the infrasound pulses were located very near the lightning channels as mapped by the LMA. Similar

agreement between acoustic and VHF lightning locations have been observed in studies including higher-frequency acoustic measurements [*Arechiga et al.*, [2009](#); *Johnson et al.*, [2011](#); *Qiu et al.*, [2012](#)]. Second, electrostatic production models of infrasound must make use of charge layers with a thickness of 200–400 m to produce pulses of reasonable duration. However, as can be seen in Figures [2](#) and [3](#), the charge layers in the thunderstorms have a thickness of around 2 km, far exceeding the thickness assumed in electrostatic-based infrasound models. In addition, there is no evidence of a thin layer within the charge layers which have a higher charge density, as proposed by *Dessler* [[1973](#)].

Finally, *Bohannon et al.* [[1977](#)] and *Pasko* [[2009](#)] suggested that the compression portion of the infrasound pulse could be produced by accelerated charging in the cloud prior to the lightning flash. Based on the models presented in *Pasko* [[2009](#)], the rise time of the infrasound compression pulse would require accelerated charging in the cloud 0.2–2 s before the flash. Such an increase in the charging rate may produce a deflection in the DC electric field measurements and should produce a deflection in slow antenna recordings; consistent with the results presented in *Chum et al.* [[2013](#)], such a deflection is not observed in the field mill data of Figure [4](#), or in the slow antenna recordings (not shown).

It is likely that the disagreement between observations and models stem from over simplified geometry and interactions of a lightning flash. We note that a lightning flash does not immediately partially or totally neutralize the charge inside the thunderstorm. Instead, the lightning flash deposits charge of opposite sign along its channels inside the charge layer, then the charge deposited slowly combines with those originally in the charge layer [e.g., *Lu*, [2009](#)]. Based on electric field measurements on the ground, there is little to differentiate a lightning flash discharging a cloud, and a lightning flash depositing charge inside a cloud; both processes produce very similar field changes far from the lightning channel. However, there is a dramatic difference between charge neutralization and charge deposition when the field is measured close to the channel [e.g., *Winn et*

al., [2011](#)]. When modeling the production of infrasound by lightning, these differences should be taken into account. Instead of considering the electrostatic interaction of charge layers in the cloud, one should model the electrostatic interaction of the charge deposited on the lightning channel.

The charge deposited by the lightning channel largely ends up within the streamer zone produced during leader propagation of the lightning channel into virgin air. Electrostatic forces may cause the air in the streamer zone to expand, producing an acoustic compression wave with a period consistent with the size of the streamer zone. The diameter of the streamer zone can be estimated to be between 100 and 500 m based on negative leader step lengths inside the cloud [e.g., *Kitagawa and Brook*, [1960](#); *Winn et al.*, [2011](#); *Edens et al.*, [2014](#)]. Further rapid changes of the potential of the lightning channel, caused by K changes or return strokes, may also cause the charge deposited in the streamer zone to expand or contract, which can cause further acoustic waves. If the wavelength of the acoustic wave is long enough, coherent addition from different portions of the channel can occur at a specific location on the ground, causing complex beaming of sound.

The results presented for these two IC flashes provide a complete picture of the regions in a cloud producing infrasound pulses, as well as the sounding environment of the flash and charge regions. The locations of the infrasound pulses are improved through the use of equations [5](#) and [6](#) to correct for the variation of the speed of sound with temperature. Based on our analysis, none of the currently proposed models can explain the mechanism by which lightning flashes produce infrasound pulses. However, this work provides a number of specific observations which provide constraints for future models and indeed suggests a possible model which may explain the observed infrasound pulses. We note that the work presented here relates to infrasound produced by IC flashes, while most of previous work on infrasound pulses was related to their production by CG flashes which may operate differently. Extending the results presented in this paper to CG flashes is currently underway.

Our approach to thunder detection was based on the knowledge that the frequency of thunder falls into the following categories:

Frequency Breakdown

- **The Crack (1,000 Hz to 2,000 Hz):** Close-up lightning produces sharp, high-pitched snaps or cracks.
- **The Boom (50 Hz to 200 Hz):** Mid-range frequencies create the loud, chest-rattling boom associated with closer strikes.
- **The Rumble (20 Hz to 60 Hz):** Low frequencies and deep vibrations that carry over long distances.
- **Infrasound (below 20 Hz):** Thunder contains significant energy in the infrasonic range. These frequencies are too low for humans to hear but carry immense power over great distances. www.controlbooth.com +4

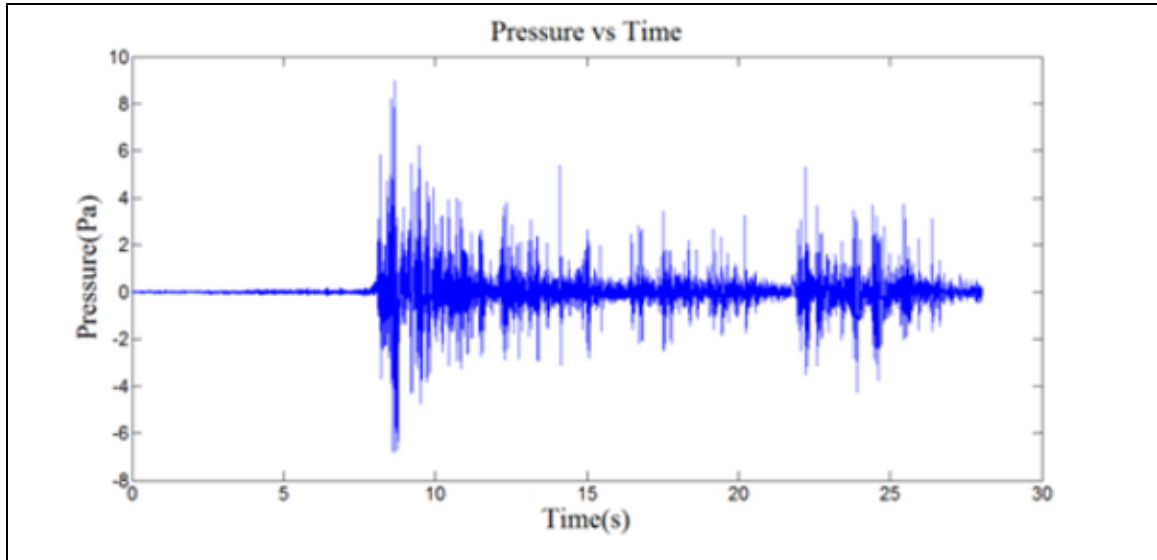
Why the Pitch Changes

- **Close Strikes:** If you are within a few hundred yards, you will hear a sharp, high-pitched blast because high frequencies diminish quickly as they travel through the air. [\[1\]](#)
- **Distant Strikes:** As the sound waves travel miles, the high-pitched "crack" sounds are absorbed by the atmosphere, leaving only the low-frequency "rumbles" and "booms". www.researchgate.net +2

We hypothesize that as the distance one is from thunder can be correlated with both the sound amplitude and frequency of thunder. Therefore, if we could successfully detect and analysis these two parameters in real time, we could detect the distance between the thunder clap and the location of our device.[7]

Pressure over time from thunder can be observed in the below graph.

Figure 3.2 - Pressure vs Time



Our research has confirmed by the work of advanced weather researchers that these acoustic signatures of thunder can be used in understanding some properties of lightning channels. Basic features of thunder signal are qualitatively classified as Peals, Claps and Rumble Characteristics of Peals, Claps and Rumble are directly affected by the acoustic energy, channel tortuosity, the interference of distinct parts of the channel, channel orientation relative to the observer and attenuation of high frequencies during the propagation. The fundamental frequency of audible thunder appears less than 250 Hz. The elapsed time of peals and claps depends on the intensity level and the frequency composition of the fluctuation. The mean fundamental frequency of rumbles is 63 Hz.

The acoustic signatures produced by lightning discharge have been analyzed in time domain on many occasions for quantitative identification of thunder features such as peal, clap and rumbling. The frequency domain analysis as provided by Fourier transformation, short time Fourier transformation with various window functions have contributed considerably in analyzing the acoustic signals. However, the localization and the resolution of the frequency have been degenerated due to mathematical limitations of above methods. Energy per unit length associated with the thunder has been studied using audible fundamental frequency of lightning discharge by introducing short time Fourier transformation. The harmonic nature of audible frequencies has been inferred significantly in distinguishing the ground flashes and cloud flashes.

The audio frequency pressure variation associated with thunder has been estimated by spectral densities and dominant frequency maxima is in the range of 52 to 96 Hz and it is confirmed that the sound pressure is a function of distance and has been initiated by the cylindrical wave-fronts. In general, the terms peal, clap, and rumble are specified qualitatively by listening audible sound pressure variations. A peal defines as a sudden loud burst with repetitive nature and takes a long time to return to the ambient pressure. A sudden isolated sound from main profile lasts for short time duration or impulsive nature refers as a clap and rumble is a weak sound continued over considerable time duration. The distance is also a vital parameter to distinguish the pressure fluctuation of

subjective thunder terms. The occurrence characteristics and the relative amplitudes have been analyzed to enlighten the features of thunder. A level of significance of relative amplitudes has been introduced by using the peak value to effectively distinguish the subjective terms, clap, peal, roll and rumble. The number of occurrence of claps has a certain confidence to identify ground flashes and cloud flashes. The time duration of pressure fluctuation associated with the subjective thunder terms specially peals and claps depends on the fundamental frequency because of their impulsive nature and high intensity level change. In this study, it is intended to analyze the frequency variation of the subjective thunder terms using the time localized spectrograms generated by S-transformation. Relative signal to noise ratio and the decibel level change of signal to noise ratio have been introduced with a certain level of significance to distinguish the characteristic thunder features from the main thunder signature.

3.3 VLF Signal Reception: Antenna Topology

3.3.1 Ferrite Rod vs. Air-Core Loop Antenna

The detector's main sensing element is the VLF antenna, with the primary purpose being to resolve electromagnetic energy in the 3-30 kHz sterics bands radiated by lightning strikes. Two antenna topologies were considered: a ferrite-rod antenna and an air-core loop antenna. Both are magnetic loop designs that respond to H-field signals due to lightning.

A ferrite rod concentrates magnetic flux through a high-permeability core, therefore the same inductance and resonant frequency can be achieved with a much smaller system. This is a significant advantage, considering our project's portability requirement for picnickers and campers (Section 2.1). However, the tradeoff is that the ferrite core's permeability drifts with temperature, and can also saturate under very strong fields. Additionally, the rod's response is directional, with a dead zone along its axis, so the antenna orientation inside the enclosure is important.

The air core loop does not have any problems with core saturation and thermal drift and it is easy to tune, however, the size of the loop that would give comparable sensitivity to the ferrite rod will be tens of centimeters across, which contradicts our objective of having a compact portable device. Commercially available low cost lightning detectors such as AS3935 "Franklin" Lightning Sensor utilize a small ferrite/loop antenna operating at 500 kHz with a Q of about 15 [4]. In view of our objective of developing a portable system, we have decided to use the ferrite rod antennas for both receiver and transmitter portions of the system.

3.3.2 Discrete Receiver Front-End vs. Dedicated Lightning-Sensor IC

The team evaluated two approaches to convert the antenna's induced signal into a usable detection event. We considered a fully discrete front end, consisting of the LC tank, a multi-stage op-amp band-pass amplifier, and a comparator. We also considered a

dedicated lightning-sensor IC, principally the AMS/ScioSense AS3935, which integrates the amplifier, an adaptive noise floor, a disturber-rejection algorithm, and an SPI/I2C host interface and combines it into one [4].

The AS3935 advertises a detection range of up to 40 km, distance-estimation of 1 km, a bank of tuning capacitors for calibrating the antenna to 500 kHz, and firmware rejection of disturber signals such as motor noise. [4]. Notably, ScioSense sells the AS3935 as part of a demonstration kit that ships with a handheld lightning-emulator board.

A fully discrete front end gives our team complete control over gain, bandwidth, and thresholding, but it also means the team must independently implement noise rejection, gain staging, and distance-estimation logic that the AS3935 otherwise provides out of the box. Because the project calls for the team to design and fabricate original analog and firmware content rather than rely on an integrated sensor, our team selected the discrete op-amp/comparator front end for the detector, using the AS3935's as design references for tuning our own antenna network, while implementing our own peak-detection and disturber-rejection logic in firmware (Section 3.4).

3.4 Microcontroller Selection for Detector and Emulator Firmware

Because the detector and emulator have different I/O and connectivity requirements, the team evaluated microcontroller options separately for each subsystem. Simultaneously also validating all firmware logic on Arduino-compatible boards during early bring-up, ahead of receiving the final ESP32, PIC16F1789, and custom PCB hardware.

Table 3.1 Microcontroller Comparison – Detector, Emulator, and Bring-Up Platform

Specification	ESP32 (Detector)	PIC16F1789 (Emulator)	Arduino Uno (Bring-Up)
Clock speed	Up to 240 MHz, dual-core Xtensa LX6 [8]	Up to 32 MHz [5]	16 MHz
Program flash	4 MB [8]	16K x 14-bit words (~28 KB) [5]	32 KB
ADC resolution / channels	12-bit, 18 channels [8]	10-bit, 5 channels (28-pin) [5]	10-bit, 6 channels
Onboard analog comparators / op-amps	None	4 comparators, 2 op-amps [5]	1 comparator

Wireless connectivity	802.11 b/g/n Wi-Fi + Bluetooth 4.2 Classic/BLE (integrated) [8]	None	None
GPIO pin count	Up to 34 [8]	24 (28-pin package)	20 (14 digital + 6 analog)
Approx. unit cost (BOM)	\$10	\$10	~\$5 (bring-up only)

The choice of Microchip for the detector is mainly due to its I/O and easy of development [6].

For the emulator, it does not require any wireless functions and just needs to switch on a driver transistor according to a timer and also read two buttons; hence, the team chose the PIC16F1789 – an 8-bit microcontroller with 32 MHz clock frequency, four built-in op-amps, four analog comparators and 10-bit ADC in 28 pins [5].

In the process of developing the firmware, both roles will be prototyped using an Arduino board. The team decided to do it this way so we are able to test the timer gating, ADC sampling, and the communication mechanism within the Arduino development environment, without having to wait for the ESP32 and PIC16F1789 target boards and the custom printed circuit boards. This will therefore allow for simultaneous development of the detector and emulator firmware.

3.5 Signal Processing for Distance and Direction Estimation

For the advanced objective of measuring relative distance and directionality (2.3.1), the group studied how the lightning detection system translates its measured signal into distance. The AS3935 translates its signal into distance based on its watchdog threshold of energy for the signal from the antenna. This is done using an internal look-up table that gives the corresponding relative distance to the measured energy level, with distance given in discrete steps from 1km up to around 40 km [4]. This is a rough and simple method of measurement based on thresholds, and not necessarily distance calculation.

On this basis, our design method aims to capture the amplified signal from the antenna in the MCU, then employ a peak detection algorithm to detect a strike and categorize the magnitude of the peak in three categories: near, medium and far. These are chosen based on calibration of the signal output of the emulator. The directional trend, approaching and retreating, is estimated in the firmware by evaluating the trend in the peak magnitude using a sliding window of past strike events. Using this technique, an increasing trend

indicates an "approaching" direction while a decreasing trend indicates a "retreating" direction.

3.6 Verbal and Audible Alert Technology

As part of the lightning detection system, an effective method of communicating hazardous conditions to the user is just as important as accurately detecting lightning activity. Once a lightning strike has been detected and classified by the system, the user must be notified in a manner that is both immediate and easily understood. Since users may not always be monitoring a visual display or indicator LEDs, incorporating audible alerts provides an additional layer of notification that can significantly improve overall system effectiveness.

While many portable warning devices rely solely on buzzers or flashing LEDs, these methods provide limited information regarding the nature or severity of an event. Spoken voice messages can convey specific warning information without requiring the user to interpret different tones or flashing patterns. This enables the device to communicate conditions such as lightning detection, storm proximity, or system status using clear, human-readable messages.

Several audio output technologies were investigated during the design process, including traditional piezoelectric buzzers, pulse-width modulation (PWM) audio generated directly by the microcontroller, MP3 playback modules, and dedicated voice playback integrated circuits (ICs). Each approach offers unique advantages and disadvantages with respect to hardware complexity, power consumption, audio quality, memory requirements, and ease of integration. After evaluating these technologies, a dedicated voice playback IC was selected as the most appropriate solution for the project due to its ability to provide high-quality prerecorded speech while minimizing firmware complexity and hardware overhead.

The following sections discuss the motivation for incorporating audible alerts, compare available voice output technologies, evaluate commercially available voice playback ICs, and describe the selected Nuvoton ISD2360 voice playback solution and its planned integration into the lightning detection system.

3.6.1 Importance of Audible Alerts

One of the primary objectives of the lightning detection system is to notify users of nearby lightning activity as quickly and effectively as possible. Although the detection hardware is responsible for identifying potential lightning strikes, the usefulness of the device ultimately depends on how well it communicates that information to the user. Since the system is intended for outdoor environments such as parks, campsites, athletic fields, and recreational areas, users may not always be looking directly at the device when a hazardous condition occurs. As a result, incorporating an audible notification system significantly improves the likelihood that warnings will be noticed immediately.

Traditional warning devices commonly rely on visual indicators such as LEDs or simple audible buzzers. While these methods are inexpensive and easy to implement, they provide little information beyond indicating that an event has occurred. A flashing LED requires the user to maintain visual contact with the device, while a buzzer often relies on different beep patterns that must be learned and interpreted. In emergency situations, this additional interpretation may delay the user's response or lead to confusion regarding the severity of the detected hazard.

Voice alerts provide a more intuitive method of communicating important information. Instead of requiring the user to recognize a particular flashing pattern or audible tone, prerecorded speech can immediately communicate the nature of the warning using clear and concise language. Messages such as "Lightning detected," "Lightning approaching," or "Seek shelter immediately" can be understood without prior training, allowing users to respond more quickly to potentially dangerous weather conditions. Spoken alerts also improve accessibility for users who may have difficulty distinguishing between different audible tones or visual indicators.

Rather than replacing traditional notification methods, voice alerts are intended to complement them as part of a layered warning system. The proposed lightning detector utilizes multiple forms of feedback, including visual LED indicators and spoken voice notifications, to maximize the probability that users recognize hazardous conditions regardless of surrounding environmental factors. For example, bright sunlight may reduce the visibility of LEDs, while high ambient noise may make audible alerts more difficult to hear. By employing multiple notification methods simultaneously, the system provides greater reliability than relying on a single alert mechanism alone.

Based on these considerations, incorporating a dedicated voice playback subsystem was determined to be an effective method of improving the overall usability and safety of the lightning detection system. The following sections investigate several available audio output technologies and evaluate their suitability for implementing reliable verbal alerts within the proposed design.

The notification methods discussed in this section are summarized in Table 3.2, highlighting the advantages and limitations of each approach for use in a portable lightning detection system.

Table 3.2 - Comparison of Common Notification Methods for Portable Lightning Detection Systems

Notification Method	Advantages	Limitations
LED Indicators	Low power, simple, inexpensive	Requires user to be looking at device
Piezoelectric Buzzer	Easy to implement, low cost	Limited information conveyed

Voice Alerts	Clear instructions, intuitive, accessible	Requires dedicated playback hardware
Combined Alerts	High reliability, redundant notification	Increased hardware complexity

3.6.2 Audio Output Technologies

Before selecting a specific hardware component for the lightning detector, several methods of generating audible alerts were investigated. Each technology differs in hardware complexity, audio quality, memory requirements, power consumption, and ease of integration with an embedded system. Since the project requires the ability to deliver clear spoken warning messages while maintaining a compact, low-power design, multiple audio output approaches were evaluated to determine the most suitable solution. The primary technologies considered include piezoelectric buzzers, microcontroller-generated pulse-width modulation (PWM) audio, MP3 playback modules, and dedicated voice playback integrated circuits.

3.6.2.1 Piezoelectric Buzzers

Piezoelectric buzzers are one of the most common methods of generating audible alerts in embedded systems due to their low cost, simple circuitry, and minimal power consumption. A piezoelectric buzzer produces sound by applying an alternating electrical signal to a piezoelectric ceramic element, causing it to rapidly expand and contract. Because the frequency of the applied signal determines the tone that is produced, a microcontroller can easily generate different alert sounds by varying the output frequency.

Although piezoelectric buzzers are well suited for simple notification systems, they are incapable of reproducing intelligible speech or complex audio signals. Users must instead learn to associate different beep patterns or tones with specific system conditions. For a safety-critical application such as a lightning detector, this additional interpretation may delay the user's response or create unnecessary confusion. As a result, while piezoelectric buzzers remain useful as supplemental alert devices, they were not considered sufficient as the primary notification method for this project.

3.6.2.2 PWM Audio Generation

Another possible approach is to generate audio directly from the system microcontroller using pulse-width modulation (PWM). In this method, the microcontroller rapidly switches a digital output at varying duty cycles to approximate an analog audio waveform. After passing through an appropriate filter and amplifier, the resulting signal can drive a small speaker to reproduce prerecorded audio or synthesized speech.

Generating audio through PWM eliminates the need for a dedicated audio playback device, reducing component count and hardware cost. However, this approach places a significant processing burden on the microcontroller, particularly when multiple real-time tasks are already being executed. Because the lightning detector must continuously monitor incoming sensor data, process detection algorithms, and manage communication with other system components, dedicating processor resources to high-quality audio playback could negatively impact overall system performance. Additionally, achieving acceptable audio quality often requires careful timing, filtering, and software optimization.

3.6.2.3 MP3 Playback Modules

Commercial MP3 playback modules, such as the DFPlayer Mini, provide another method of implementing spoken voice alerts. These modules contain onboard storage, an audio decoder, and amplifier circuitry capable of playing compressed audio files directly from removable flash memory. Integration is typically accomplished using a simple serial communication interface, allowing prerecorded voice messages to be triggered through software commands.

While MP3 modules offer excellent audio quality and straightforward implementation, they introduce additional hardware complexity and rely on external storage media such as microSD cards. For applications requiring only a small number of fixed voice prompts, many of the features provided by these modules remain unused. Their larger physical size and dependence on removable memory also make them less attractive for compact embedded systems intended for long-term field deployment.

3.6.2.4 Dedicated Voice Playback Integrated Circuits

Dedicated voice playback integrated circuits are specifically designed for applications requiring prerecorded speech with minimal hardware overhead. These devices integrate nonvolatile flash memory, audio decoding circuitry, and speaker output drivers into a single package, allowing prerecorded messages to be stored and reproduced without placing a significant computational burden on the host microcontroller. Most devices can be controlled using either simple GPIO trigger inputs or standard serial communication protocols such as SPI.

Compared to the previously discussed technologies, dedicated voice playback ICs provide an effective balance between hardware simplicity, audio quality, power consumption, and ease of integration. Because the audio decoding and playback functions are handled internally by the IC, the microcontroller is free to continue executing lightning detection algorithms while spoken warning messages are played independently. For these reasons, dedicated voice playback ICs were identified as the most appropriate

technology for the proposed lightning detection system. The following section compares several commercially available voice playback ICs and discusses the rationale for selecting the Nuvoton ISD2360 for implementation.

The audio output technologies investigated during the design process are summarized in Table 3.3. Based on this comparison, dedicated voice playback integrated circuits were determined to provide the most appropriate balance of audio quality, hardware simplicity, and processing efficiency for the proposed lightning detection system. The following section compares several commercially available voice playback ICs and discusses the rationale for selecting the Nuvoton ISD2360.

Table 3.3 Comparison of Audio Output Technologies Considered for the Lightning Detection System

Technology	Speech Capability	MCU Workload	Hardware Complexity	Overall Suitability
Piezoelectric Buzzer	No	Low	Low	Moderate
PWM Audio Generation	Limited	High	Moderate	Moderate
MP3 Playback Module	Yes	Low	High	Good
Dedicated Voice Playback IC	Yes	Very low	Low	Excellent (Selected)

3.6.3 Dedicated Voice Playback IC Comparison

After determining that a dedicated voice playback integrated circuit provided the most appropriate audio output technology for the lightning detection system, several commercially available devices were evaluated. Important selection criteria included available audio storage capacity, sound quality, interface options, ease of integration with the system microcontroller, and overall suitability for an embedded outdoor warning device. Since the lightning detector requires only a limited number of prerecorded voice messages while maintaining low power consumption and compact hardware, devices intended specifically for embedded voice playback applications were prioritized.

Three representative voice playback ICs were investigated during the component selection process: the Nuvoton ISD1820, the APR33A3 voice recording IC, and the

Nuvoton ISD2360. Each device offers integrated nonvolatile memory and onboard audio playback capabilities, but differs significantly in storage capacity, control interfaces, expandability, and intended application. The following sections summarize the characteristics of each device before presenting the rationale for selecting the ISD2360.

3.6.3.1 Nuvoton ISD1820

The Nuvoton ISD1820 is a low-cost voice recording and playback integrated circuit commonly used in educational projects and simple embedded applications. The device combines onboard flash memory, a microphone input, and speaker driver circuitry into a single package, allowing users to record and replay short voice messages without requiring external memory or complex software development. Playback is typically initiated using dedicated GPIO trigger pins, making the device straightforward to integrate into simple electronic systems.

Although the ISD1820 provides an inexpensive solution for basic voice playback, its limited recording capacity and relatively simple control interface reduce its suitability for more advanced embedded applications. The device offers limited flexibility for managing multiple prerecorded messages and provides fewer interface options than more modern voice playback ICs. For a project requiring several distinct spoken warning messages and future firmware expansion, the ISD1820 was determined to be too limited.

3.6.3.2 APR33A3

The APR33A3 is another dedicated voice recording and playback integrated circuit designed for embedded audio applications. Similar to the ISD1820, the device incorporates onboard nonvolatile memory, microphone support, and speaker output circuitry. Individual voice messages may be assigned to multiple trigger inputs, allowing prerecorded audio clips to be played using external hardware signals without significant software overhead.

Compared to the ISD1820, the APR33A3 provides additional playback flexibility and longer recording durations. However, the device remains primarily intended for standalone voice playback systems rather than applications requiring close integration with a host microcontroller. Limited digital communication capabilities and reduced software configurability make the device less attractive for systems in which voice playback is controlled dynamically through firmware.

3.6.3.3 Nuvoton ISD2360

The Nuvoton ISD2360 is a modern voice playback integrated circuit specifically designed for embedded applications requiring high-quality prerecorded speech and flexible system integration. Unlike the previously discussed devices, the ISD2360 supports both GPIO- and SPI-based control, allowing voice playback to be managed directly through firmware while maintaining compatibility with simpler hardware-triggered operation if desired. The device also provides significantly greater

onboard flash memory, improved audio compression techniques, and multiple playback channels for organizing prerecorded messages.

These features make the ISD2360 particularly well suited for the proposed lightning detection system. Its SPI communication interface enables seamless integration with the primary system microcontroller while allowing future firmware updates and additional voice messages to be implemented without major hardware modifications. Furthermore, the increased storage capacity provides sufficient memory to accommodate multiple warning phrases, system status announcements, and future feature expansion while maintaining high audio quality.

The characteristics of the evaluated voice playback integrated circuits are summarized in Table 3.4. Based on this comparison, the Nuvoton ISD2360 was selected for implementation due to its superior balance of storage capacity, interface flexibility, audio quality, and compatibility with the overall system architecture.

Table 3.4 - IC Comparison

Voice Playback IC	Memory Capacity	Control Interface	Audio Quality	Overall Suitability
Nuvoton ISD1820	Low	GPIO	Moderate	Moderate
APR33A3	Moderate	GPIO	Good	Good
Nuvoton ISD2360	High	SPI / GPIO	High	Excellent (Selected)

3.6.4 Nuvoton ISD2360 Voice Playback IC

The Nuvoton ISD2360 was selected as the dedicated voice playback integrated circuit for the proposed lightning detection system due to its combination of high-quality audio playback, flexible communication interfaces, and compact embedded design. Unlike general-purpose audio playback modules, the ISD2360 is specifically intended for applications requiring reliable playback of prerecorded speech with minimal hardware overhead. The device integrates flash memory, audio processing hardware, communication interfaces, and output drivers into a single integrated circuit, reducing both hardware complexity and processor workload while providing sufficient flexibility for future firmware expansion.

The selected device incorporates an internal 2 Mbit nonvolatile Flash memory capable of storing approximately 64 seconds of prerecorded audio using ADPCM4 compression. This integrated memory architecture eliminates the need for external storage while providing sufficient capacity for multiple spoken warning messages and future firmware

expansion. These features make the ISD2360 well suited for compact embedded applications requiring reliable voice playback.

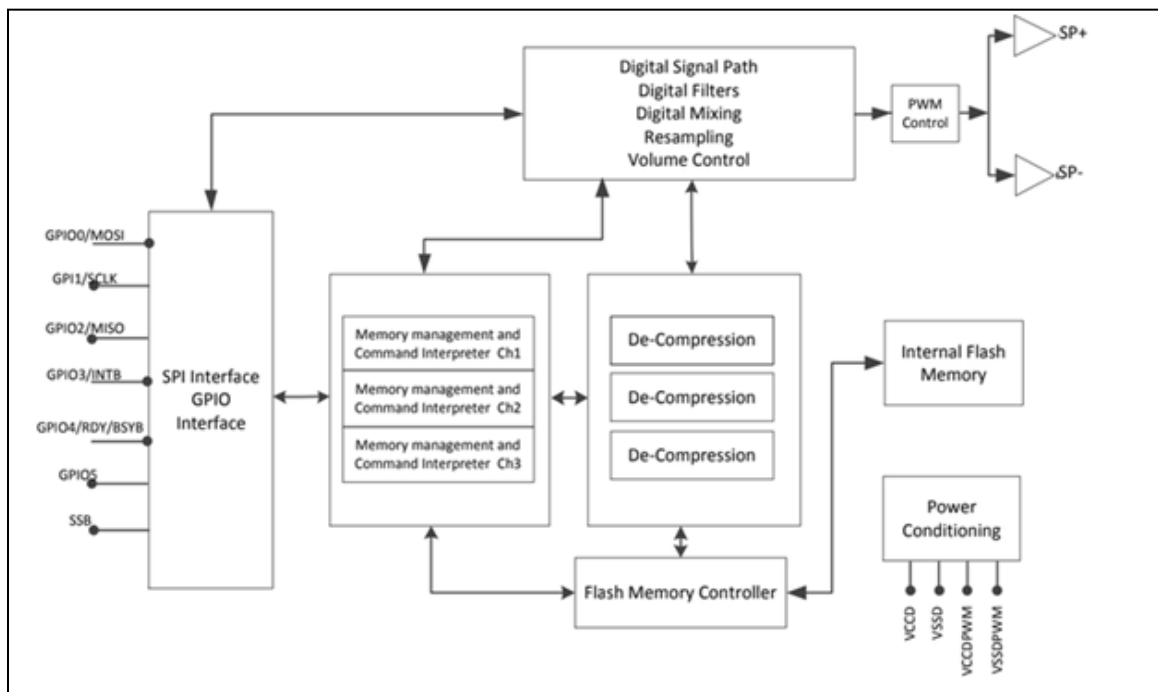
The following sections discuss the major architectural features of the ISD2360 and describe the characteristics that led to its selection for the proposed lightning detection system.

3.6.4.1 Device Architecture

The ISD2360 combines several functional subsystems into a single integrated circuit, including nonvolatile flash memory, an audio processing engine, communication interfaces, and an integrated PWM audio output stage. This highly integrated architecture allows prerecorded voice messages to be reproduced with minimal processing by the host microcontroller. As a result, the primary controller is able to continue executing lightning detection algorithms while voice messages are played independently by the ISD2360.

The functional architecture of the ISD2360 is illustrated in Figure 3.3. As shown, the device incorporates Flash memory, ADPCM decoding hardware, GPIO and SPI communication interfaces, internal control logic, and PWM audio output circuitry within a compact embedded solution.

Figure 3.3 - Functional Block Diagram of the Nuvoton ISD2360 Voice Playback Integrated Circuit.



As shown in Figure 3.3, the ISD2360 combines several key functional blocks required for embedded voice playback into a single device. Audio data is retrieved from the internal Flash memory and processed by the memory management and decompression modules

before entering the digital signal path, where filtering, resampling, and volume adjustments are performed. The processed audio is then routed through the integrated PWM control circuitry before being delivered to the external speaker output. Communication with the host microcontroller is accomplished through either the SPI or GPIO interface, allowing playback functions to be controlled entirely through software or simple hardware triggers depending on the application requirements.

3.6.4.2 Memory Organization and Audio Storage

One of the primary advantages of the ISD2360 is its integrated nonvolatile flash memory, which stores prerecorded audio messages directly within the device. Unlike systems that require external memory cards or removable storage media, the ISD2360 permanently retains programmed voice prompts even after power is removed. This simplifies hardware design while improving overall system reliability.

To maximize the available storage capacity, the ISD2360 utilizes Adaptive Differential Pulse Code Modulation (ADPCM), an audio compression technique that significantly reduces memory requirements while maintaining sufficient speech quality for voice announcements. Since the lightning detector only requires playback of short prerecorded warning messages rather than high-fidelity music, ADPCM compression provides an effective balance between storage efficiency and intelligibility.

The ISD2360 incorporates a 2 Mbit Flash memory organized into 256 sectors, with each sector containing 1024 bytes of storage. This memory organization provides approximately 64 seconds of audio playback when using the device's ADPCM4 compression algorithm. During device initialization, the ISD2360 automatically loads a Memory Header stored in the first Flash sector, which contains the control and indexing information required for proper operation. Because this sector is reserved for device initialization, it should only be programmed using an application image generated by the ISD-VPE2360 software to ensure reliable operation.

This storage capacity provides sufficient space for multiple warning phrases, system status announcements, and future firmware expansion without requiring additional external memory devices.

3.6.4.3 Communication Interfaces

The ISD2360 supports both General-Purpose Input/Output (GPIO) triggering and Serial Peripheral Interface (SPI) communication. GPIO operation allows prerecorded messages to be initiated directly using dedicated hardware inputs, making the device suitable for simple standalone applications requiring little software development.

For the proposed lightning detector, SPI communication was selected as the preferred control method. Through SPI, the system microcontroller can dynamically initiate playback, manage prerecorded voice messages, and coordinate spoken warnings based on system status and detected lightning activity. This approach provides significantly greater

flexibility than hardware-triggered playback while simplifying future firmware modifications and system expansion.

Communication with the ISD2360 is accomplished through a standard four-wire SPI interface, allowing the host microcontroller to program the device, initiate playback commands, and manage stored audio data. During development, prerecorded voice messages are written to the device's internal Flash memory using SPI communication after an application image has been generated by the ISD-VPE2360 software. This firmware-controlled approach provides greater flexibility than hardware-triggered playback and allows voice prompts to be easily modified or expanded as the project evolves.

3.6.4.4 Audio Output Features

In addition to onboard flash memory and audio processing hardware, the ISD2360 incorporates an integrated pulse-width modulation (PWM) audio output stage capable of driving an external audio amplifier or speaker interface with minimal supporting circuitry. Integrating the audio output hardware directly within the device reduces the number of external components required, simplifies printed circuit board (PCB) routing, and lowers overall system complexity.

Because audio decoding and playback are performed internally by the ISD2360, the host microcontroller is not required to continuously generate audio signals during playback. This allows the system processor to remain focused on sensor monitoring, lightning detection algorithms, and other time-critical tasks while voice messages are reproduced independently.

3.6.4.5 Summary of Selected Device

The ISD2360 provides an effective combination of integrated flash memory, compressed audio storage, flexible communication interfaces, and embedded audio playback hardware within a single integrated circuit. Compared to the alternative devices discussed in Section 3.6.3, these features provide greater flexibility for firmware-controlled operation while minimizing additional hardware complexity.

For these reasons, the ISD2360 was selected as the dedicated voice playback solution for the proposed lightning detection system. The following section discusses the software environment used to create, organize, and program prerecorded voice messages into the device.

3.6.5 Voice Programming Software

Unlike traditional audio playback modules that utilize removable storage media, the ISD2360 requires prerecorded voice messages to be programmed directly into its internal Flash memory prior to operation. Nuvoton provides the ISD-VPE2360 software package to simplify this process by allowing developers to organize voice prompts, configure playback parameters, and generate the application image required for programming the

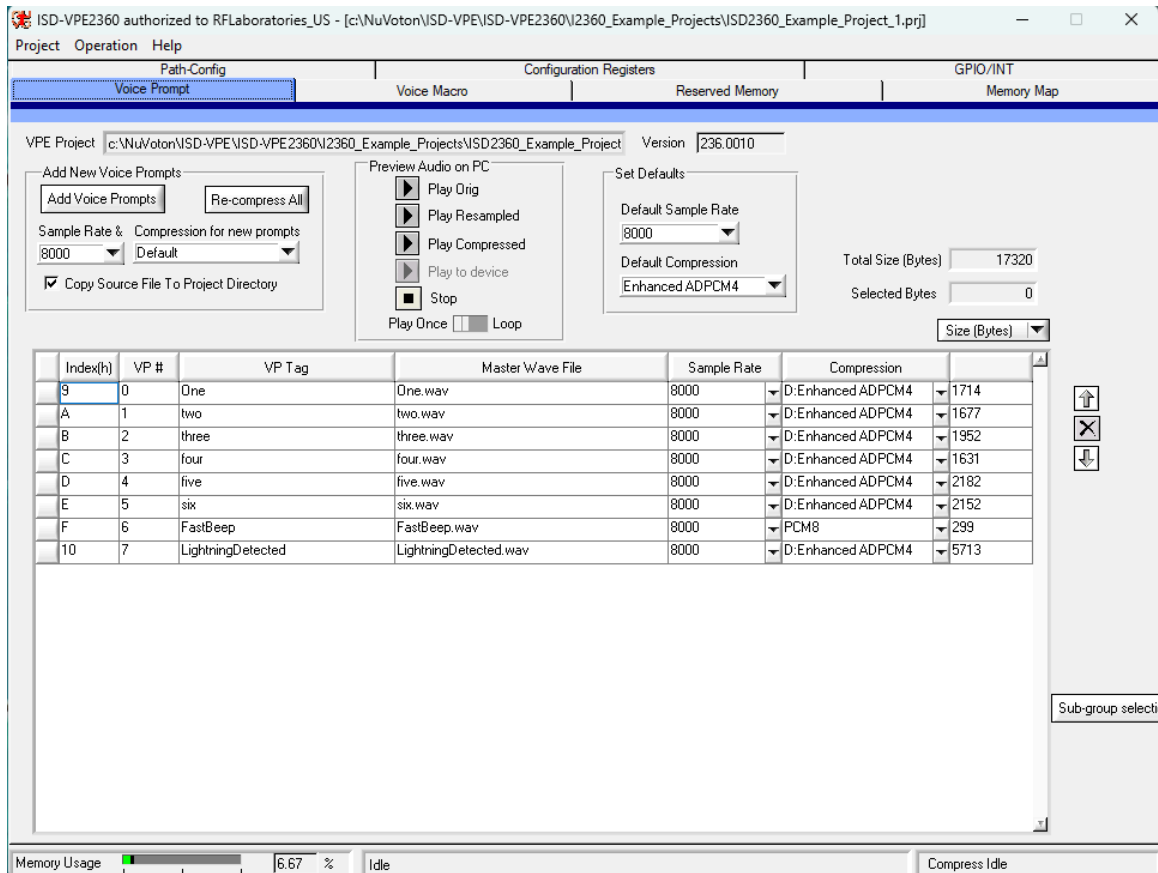
device. The software provides a graphical user interface (GUI), eliminating the need for manually constructing memory maps or programming Flash sectors.

Once individual voice recordings have been prepared, they are imported into the ISD-VPE2360 development environment and assigned to playback locations within the device's internal Flash memory. The software automatically generates the required Memory Header and application image used to initialize the ISD2360 during power-up. This automated process significantly reduces the likelihood of programming errors while simplifying future modifications to stored voice messages.

Programming of the ISD2360 is performed through the Serial Peripheral Interface (SPI), allowing the generated application image to be written directly into the device's onboard Flash memory. During system initialization, the ISD2360 automatically reads the Memory Header located in the first Flash sector to configure playback information and message indexing. Because this sector contains critical initialization data, modifications are performed exclusively through the application image generated by the ISD-VPE2360 software to ensure reliable device operation.

For the proposed lightning detection system, the software will be used to organize prerecorded warning phrases, system status announcements, and future voice prompts prior to deployment. Using a dedicated programming environment simplifies firmware development by separating audio management from the primary lightning detection software, allowing voice messages to be updated without requiring significant modifications to the system architecture.

Figure 3.4 - ISD VPE



3.7 Software

The software portion of the project is divided into two main parts: the mobile application that runs on the user's iPhone and the Bluetooth Low Energy (BLE) firmware running on the Nordic Semiconductor microcontroller. The mobile application is responsible for displaying the lightning detector's status to the user, while the Nordic chip manages the wireless communication between the hardware and the phone. Together, these two components allow the detector to send real-time lightning alerts wirelessly.

For the mobile application, we chose to develop natively for iOS using SwiftUI and Apple's CoreBluetooth framework. We decided against using cross-platform frameworks like React Native or Flutter because CoreBluetooth is the official Bluetooth API for iOS and provides the most reliable way to communicate with BLE devices. Since Bluetooth communication is the primary function of the app, using Apple's native tools helps improve reliability, reduces unnecessary complexity, and gives us direct access to the phone's Bluetooth hardware.

The application is organized into five Swift artifacts, each with a specific responsibility. BLEManager handles all Bluetooth-related operations, including scanning for nearby

devices, connecting to the Nordic BLE device, subscribing to notifications, and receiving data updates. `LightningStatus` is a simple data model that stores whether lightning has been detected along with the timestamp of the latest update. `ContentView` serves as the main controller for the app and decides whether to display the connection screen or the detector status screen based on the Bluetooth connection state. `ScanView` allows the user to search for and connect to the lightning detector, while `StatusView` displays the current detector status. Depending on the data received, it shows either "All Clear" or "Lightning Detected" along with a colored visual indicator and the time the status was last updated.

One of the biggest design decisions was how the application handles changes in data. Instead of constantly checking the Nordic device for updates, the app uses SwiftUI's built-in state management with `@StateObject` and `@ObservedObject`. This allows the user interface to automatically refresh whenever new Bluetooth data is received. When the Nordic microcontroller detects a change in lightning status, it immediately sends a BLE notification to the phone. As soon as the app receives that notification, the display updates without requiring the user to refresh the screen or wait for a timer. This event-driven approach makes the application more responsive while also reducing unnecessary Bluetooth traffic and battery usage. It also helps the system stay well within the project's one-second response time requirement.

To make development easier, we also implemented a demo mode inside the application. During the early stages of development, the hardware was not always available for testing, so demo mode allows us to simulate a connected lightning detector directly on the phone. We can manually switch between the "All Clear" and "Lightning Detected" states to verify that the user interface, animations, and status updates behave correctly. This made it much easier to develop and test the application before the hardware was fully completed.

On the firmware side, the Nordic Semiconductor microcontroller is configured as a Bluetooth Low Energy (BLE) GATT server, which is the standard communication model used by most BLE devices. The chip advertises a custom Bluetooth service that contains a single characteristic used to send lightning status information to the mobile application. When the iPhone connects, it subscribes to notifications from this characteristic. Whenever the lightning detection logic identifies a lightning event or determines that conditions are clear again, the firmware writes a single byte to the characteristic. A value of 0x01 represents "Lightning Detected," while 0x00 represents "All Clear." Because the phone is subscribed to notifications, it receives these updates automatically without continuously requesting new data.

The communication protocol was intentionally kept simple. Using a single-byte message keeps communication fast, reduces processing overhead, and makes debugging much easier during development. It also leaves room for future improvements. Additional information, such as estimated lightning distance, strike count, battery level, signal strength, or sensor diagnostics could be added later without requiring major changes to either the firmware or the mobile application.

3.8 Wireless Notification: BLE Radio Selection

One of the first hardware decisions that affected the software design was the choice of the Nordic chip as the main microcontroller for the detector. The Nordic chip includes an integrated Bluetooth 4.2 radio that supports both Bluetooth Classic and Bluetooth Low Energy. This eliminated the need for a separate BLE module entirely, which simplified both the hardware design and the firmware considerably.

The choice of BLE over Wi-Fi or Bluetooth Classic was straightforward. Wi-Fi requires the device to join a network, which is not practical for outdoor recreational use in a park or campsite where there is no guaranteed network infrastructure. Bluetooth Classic was considered but is generally designed for streaming audio and continuous high-bandwidth connections, and it consumes more power than BLE. For our application, the Nordic Chip only needs to push a small status update to a phone, which is exactly the kind of thing BLE is designed for. The low power consumption also aligns with the battery life requirements we outlined in section 2.3.1.

On the iOS side, Apple's CoreBluetooth framework handles all BLE communication natively and does not require any additional libraries or App Store entitlements beyond a single privacy description in the app's Info.plist file. The app declares itself as a GATT central (the device that initiates connections and receives data) and the Nordic Chip acts as a GATT peripheral (the device that advertises and sends data).

The GATT profile we designed for this system uses a custom 128-bit service UUID and a single characteristic UUID. These identifiers are what the app uses to find the correct device during scanning and to subscribe to the right data once connected. Both UUIDs are defined in the iOS app and will need to be matched exactly in the Nordic Chip firmware. The characteristic is configured for notifications, meaning the Nordic Chip pushes updates to the phone rather than the phone polling the Nordic Chip repeatedly. This is the standard approach for real-time sensor data over BLE and keeps the BLE traffic minimal.

The 100-foot BLE communication range in our engineering specifications is a conservative estimate based on the Nordic Chip's typical output power in an indoor or open environment. In open outdoor conditions, the actual range can be somewhat longer, but 100 feet is a reasonable working assumption for a device being used at a picnic or campsite.

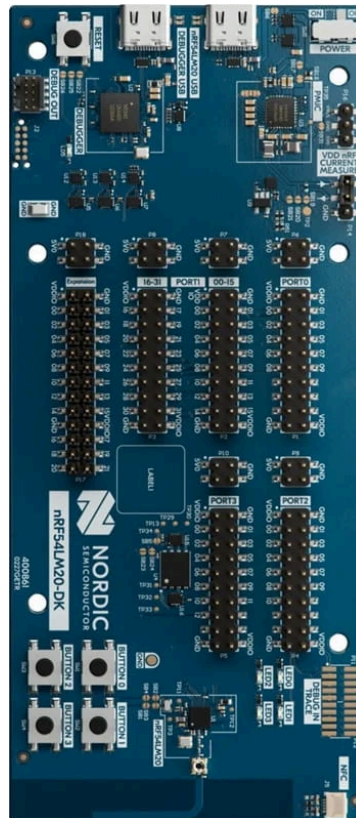
nRF54LM20B wireless SoC BLE eval kit:

After consider several different BLE components, we decided to use the Nordic nRF54LM20. The main reason is that this BLE radio IC had an Arm Cortex-M33 with a full set of digital signal processing (DSP) instructions and a memory protection unit (MPU) for application security. The full-featured, single-precision floating-point unit

(FPU) supports all single-precision instructions. Additionally, the IC has a built in Axon NPU or Neural Processing Unit. We need this to convolve acoustic samples via machine learning for detection of thunder.

This is the IC's development board we procured for preliminary testing:

Figure 3.5 - IC Development Board



The development board has some very important and useful features:

Key features

- 128 MHz Arm Cortex-M33 processor
- 2036 KB NVM and 512 KB RAM
- Multiprotocol 2.4 GHz radio supporting Bluetooth LE, 802.15.4-2020, and 2.4 GHz proprietary modes (up to 4 Mbps)
- Seven serial interfaces (SPI, TWI, UART)
- High-speed USB interface
- Extended set of interfaces, peripherals, and timers including Global RTC available in System OFF, ADC, TDM, PDM, NFC, PWM, and QDEC
- 128 MHz RISC-V coprocessor

- Advanced security including TrustZone® isolation, tamper detection, and cryptographic engine with side-channel leakage protection
- Ultra-compact CSP and QFN packages

Radio

- Bluetooth LE
 - LE 2M, LE 1M, LE Coded
 - Channel Sounding
- IEEE 802.15.4-2020 – 250 kbps
 - Enables Matter, Thread, Zigbee
- 2.4 GHz proprietary GFSK
 - 4 Mbps, 2 Mbps, 1 Mbps
- Single-ended antenna output (on-chip balun)
- 128-bit AES/ECB/CCM/AAR coprocessor
- Configurable TX power with 1 dBm step size from -10 dBm to maximum

Processor

- 128MHz Arm Cortex-M33 CPU
 - FPU, DSP, MPU, TrustZone
 - Debug – SWD, ETM, ITM, DWT, CTI, TPIU

Memory

- 2036 KB NVM (RRAM)
- 512 KB RAM

Edge AI accelerator

- nRF54LM20B only:
 - Axon NPU (AXONS)

Coprocessor/SoftPeripherals

- 128 MHz RISC-V coprocessor (VPR)
- sQSPI available as SoftPeripheral for VPR; see the software documentation for details

Peripherals

- Seven fully featured serial interfaces with EasyDMA, supporting I²C, SPI controller/peripheral, and UART
 - One HS-SPI up to 32 MHz, six SPI up to 8 MHz (SPIM, SPIS)
 - Six TWI up to 400 kHz and I²C compatible (TWIM, TWIS)
 - One HS-UART up to 4 Mbps, six UART up to 1 Mbps (UARTE)
- SAADC with eight programmable gain channels
 - 14-bit at 62.5 ksps (oversampled)

- 12-bit at 125 ksps (oversampled)
 - 10-bit at up to 2 Msps
- High-speed USB interface (USBHS)
- Global RTC can run in System OFF mode and implement a shared system timer (GRTC)
- NFC-A listening device (NFCT)
- Pulse density modulation interface (PDM)
- Time division multiplexed sound interface, also configurable as I²S interface (TDM)
- Three pulse width modulation four-channel units with autonomous waveform generation (PWM)
- Two quadrature decoders (QDEC)
- Two individual watchdog timers for secure and non-secure context (WDT)
- Seven 32-bit timers with counter mode (TIMER)
- Temperature sensor (TEMP)
- Comparator and low-power comparator with wake-up from System OFF mode (COMP, LPCOMP)

GPIO

- Up to 66 GPIO pins
- 64 MHz and 8 MHz ports

Power supply and clock

- Single-inductor DC/DC regulator
- Single 32 MHz crystal operation
- Optional 32.768 kHz crystal

Security

- Arm TrustZone, Root of Trust, secure boot, secure storage
- Security components – Cryptographic engine (CRACEN), Key management (KMU)
- Physical protection – Tamper detectors (TAMPC, GLITCHDET), Side-channel protection

3.9 Summary

In all subsystems, the team's research into technological solutions leads to similar criteria as we outlined in Chapter 2. This includes the aspects of portability, low cost, and use of

well-established, commercially proven technologies (as seen with the AS3935-based antenna tuning, the energy threshold based distance calculation, and the emulator/detector architecture). Additionally, our own implementation will be applied when it comes to detecting, calculating the distance, and generating alerts in firmware code. The selected components, ferrite rod antennas for the detector and emulator, separate op-amp and comparator based receiver circuit, ESP32 for wireless detector, PIC16F1789 for emulator, DFPlayer Mini for audio alerts, and the BLE radio on ESP32 for the phone alerts – have informed the hardware and firmware design in Chapters 6 and 7.

Chapter 4 - Related Standards and Design Constraints

4.1 Applicable Standards

Engineering projects intended for public use must be designed in accordance with applicable industry standards and regulatory requirements to ensure safe, reliable, and legally compliant operation. Although the proposed lightning detection system is being developed as an educational senior design project rather than a commercial product, the design process has considered the standards that would apply if the system were manufactured and deployed in real-world environments. These standards influence hardware selection, wireless communication, electromagnetic compatibility, printed circuit board design, and overall system safety. The following sections discuss the primary standards relevant to the proposed lightning detection system.

4.1.1 FCC Part 15 Radio Frequency Regulations

The proposed lightning detection system incorporates radio-frequency (RF) technology through its Bluetooth Low Energy (BLE) communication subsystem while also utilizing a Very Low Frequency (VLF) receiver to monitor naturally occurring electromagnetic emissions produced by lightning strikes. As a result, the design must consider regulations established by the United States Federal Communications Commission (FCC), particularly those contained within Title 47, Code of Federal Regulations (CFR), Part 15, which governs the operation of unlicensed radio-frequency devices.

FCC Part 15 establishes emission limits intended to minimize harmful electromagnetic interference between electronic devices operating within shared frequency environments. Under these regulations, intentional radiators, such as Bluetooth transmitters, must operate within specified frequency bands and power limits while ensuring that their transmissions do not interfere with licensed communication services. Likewise, devices are required to tolerate interference received from other electronic equipment without causing unsafe or unintended operation.

Although the VLF receiver used within the proposed lightning detector functions only as a passive receiving device and does not intentionally transmit radio-frequency energy, the BLE subsystem operates within the 2.4 GHz Industrial, Scientific, and Medical (ISM) band and would therefore be subject to FCC certification requirements if the product were commercially manufactured. Selecting a commercially certified BLE module simplifies regulatory compliance by utilizing hardware that has already been evaluated for RF performance and emissions.

Compliance with FCC regulations also influenced several design decisions throughout the project. The use of low-power BLE communication minimizes unnecessary radio-frequency emissions while providing sufficient wireless range for the intended application. Additionally, careful printed circuit board layout, proper grounding practices, and separation between analog signal conditioning circuitry and digital communication circuitry help reduce electromagnetic interference that could degrade the sensitivity of the VLF receiver. These design practices improve both regulatory compliance and overall system performance.

Although FCC certification is outside the scope of this academic prototype, consideration of FCC Part 15 requirements during the design process provides a foundation for future commercial development and demonstrates adherence to accepted engineering practices for wireless embedded systems.

4.1.2 Bluetooth Low Energy (BLE) Communication Standard

Bluetooth Low Energy (BLE) is the wireless communication standard selected for the proposed lightning detection system to provide communication between the detector and a companion mobile application. BLE was developed as part of the Bluetooth 4.0 specification and has since been enhanced through subsequent revisions to provide low-power wireless communication for embedded devices and Internet of Things (IoT) applications. Unlike Classic Bluetooth, BLE is specifically optimized for applications that exchange relatively small amounts of data while minimizing power consumption, making it well suited for battery-powered systems.

The BLE subsystem within the proposed lightning detection system is intended to transmit lightning detection status, warning notifications, and system information to nearby mobile devices. Communication is accomplished through the Bluetooth Generic Attribute Profile (GATT), which organizes data into services and characteristics that can be read, written, or subscribed to by the mobile application. This standardized communication architecture simplifies application development while ensuring compatibility with modern smartphones and tablets that support the Bluetooth Low Energy standard.

Compliance with the Bluetooth standard also improves interoperability and long-term maintainability of the system. Rather than implementing a proprietary wireless protocol, the use of BLE allows the lightning detector to communicate using an internationally recognized standard supported by virtually all modern mobile operating systems. This

reduces software development complexity while providing users with a familiar and reliable wireless interface.

Another significant advantage of BLE is its emphasis on energy efficiency. Because the lightning detector is intended to operate as a portable, battery-powered device, minimizing wireless power consumption is an important design objective. BLE accomplishes this by transmitting only small packets of data at scheduled intervals and allowing both the detector and connected mobile device to remain in low-power operating states between communication events. This reduces overall energy consumption while maintaining reliable wireless connectivity throughout normal system operation.

The use of Bluetooth Low Energy supports the project's goals of portability, low power consumption, and ease of use while providing a standardized communication interface that can be expanded in future software revisions if additional system features are implemented.

4.1.3 IPC-2221 Printed Circuit Board Design Standard

The proposed lightning detection system will be implemented using custom printed circuit boards (PCBs) to integrate the receiver circuitry, microcontroller, communication modules, and supporting electronic components into a compact and reliable package. During the PCB design process, the team considered the recommendations provided by IPC-2221, the Generic Standard on Printed Board Design. Although compliance with IPC standards is not required for an academic prototype, following these design guidelines improves manufacturability, reliability, and long-term maintainability.

IPC-2221 establishes general design recommendations for printed circuit boards, including conductor spacing, trace width, component placement, thermal management, and electrical isolation. Adhering to these guidelines helps reduce manufacturing defects while improving electrical performance and mechanical durability. Proper trace routing and component spacing also minimize the possibility of unintended short circuits, signal degradation, and electromagnetic interference between sensitive analog circuitry and high-speed digital communication lines.

These considerations are particularly important for the proposed lightning detector because the VLF receiver processes relatively small electromagnetic signals that may be susceptible to electrical noise generated by nearby digital components. Separating analog signal conditioning circuitry from the microcontroller, Bluetooth module, and other digital electronics reduces the likelihood of interference and improves overall detection reliability. Proper grounding techniques and careful PCB layout further enhance signal integrity while simplifying debugging and future revisions.

Following IPC-2221 design practices also supports efficient manufacturing and assembly of the prototype. Standardized component spacing, appropriate pad dimensions, and proper routing practices improve solderability and reduce assembly complexity during prototype fabrication. As the project progresses into hardware implementation, these

design principles will help ensure that the completed PCB operates reliably while providing a foundation for future design iterations.

4.2 Design Constraints

Throughout the design process, several engineering constraints influenced the selection of hardware components, communication methods, and overall system architecture. Because the lightning detection system is intended to operate as a portable outdoor safety device, factors such as power consumption, physical size, and environmental operating conditions were considered alongside system performance. These constraints guided many of the design decisions discussed throughout Chapters 2 and 3 and continue to influence the hardware and software implementation of the project.

4.2.1 Power Constraints

Power consumption is a primary design constraint for the proposed lightning detection system because the device is intended to operate as a portable, battery-powered unit. Unlike permanently installed lightning monitoring systems, the proposed detector must function reliably without access to continuous external power, requiring careful consideration of the energy requirements of each subsystem. The VLF receiver, signal conditioning circuitry, microcontroller, Bluetooth Low Energy (BLE) module, and voice playback subsystem all contribute to the overall power budget.

To minimize energy consumption, the design incorporates low-power embedded components wherever practical. Bluetooth Low Energy was selected instead of traditional Bluetooth due to its significantly lower communication power requirements, while the dedicated ISD2360 voice playback integrated circuit performs audio playback independently of the primary microcontroller, reducing processor workload during operation. Future firmware development will also investigate the use of sleep modes and power management techniques to further extend battery life when the system is idle.

The power subsystem must also provide stable operating voltages to both analog and digital circuitry. Proper voltage regulation and power distribution are essential to ensure reliable operation of the sensitive VLF receiver while minimizing electrical noise generated by digital components. Balancing battery life, system performance, and hardware complexity remains an important consideration throughout the design process.

4.2.2 Portability Constraints

Portability is another major design constraint because the proposed lightning detector is intended for outdoor recreational use, including parks, sporting events, campsites, and similar environments. Users should be able to transport and operate the device without requiring specialized equipment or permanent installation. Consequently, the overall size, weight, and physical configuration of the system were considered during component selection and mechanical design.

The system architecture favors compact, highly integrated components to reduce the overall footprint of the device while simplifying assembly. The use of integrated communication modules, dedicated voice playback hardware, and a custom printed circuit board minimizes the number of discrete components required and allows the electronics to be packaged within a portable enclosure. The enclosure itself must provide sufficient protection for the internal electronics while remaining lightweight and convenient to carry.

Portability also influenced the user interface design. Audible voice announcements, visual indicators, and Bluetooth communication provide multiple methods of notifying users without requiring continuous interaction with the device. This enables the detector to operate independently while remaining accessible to users in a variety of outdoor situations.

4.2.3 Environmental Constraints

The lightning detection system is intended for operation in outdoor environments where temperature, humidity, precipitation, and electromagnetic interference may affect system performance. Unlike laboratory equipment operating under controlled conditions, the proposed detector must continue functioning reliably while exposed to changing environmental conditions commonly encountered during thunderstorms and other outdoor activities.

Moisture protection is an important consideration because the system may be exposed to rain or elevated humidity during normal operation. Although complete waterproofing is beyond the scope of the initial prototype, the enclosure design will seek to reduce direct exposure of electronic components to environmental contaminants while allowing adequate ventilation where necessary.

Electromagnetic interference also represents a significant environmental constraint. Because the detector relies on receiving naturally occurring VLF electromagnetic emissions produced by lightning, nearby electronic devices and switching power supplies have the potential to introduce unwanted noise into the receiver circuitry. Careful printed circuit board layout, grounding practices, and separation of analog and digital circuitry help reduce the effects of electrical interference and improve measurement reliability.

Finally, the detector must remain visible and audible under a variety of outdoor conditions. Bright sunlight, background noise, and varying user distances influenced the decision to incorporate both visual indicators and spoken voice alerts, ensuring that warning notifications remain effective across a wide range of operating environments.

4.3 Practical Constraints

In addition to technical requirements and industry standards, the development of the proposed lightning detection system is influenced by several practical constraints. As a university senior design project, limitations involving budget, manufacturing capability,

and project schedule significantly affect component selection and system implementation. These practical considerations require engineering tradeoffs to ensure that the final prototype remains both technically feasible and achievable within the available resources.

4.3.1 Cost Constraints

Project cost represents one of the primary practical constraints affecting the design of the lightning detection system. Since the project is being completed within the scope of an undergraduate senior design course, component selection must balance system performance with affordability. Wherever possible, commercially available off-the-shelf (COTS) components were selected to reduce development costs while maintaining reliable operation.

During the component selection process, multiple hardware alternatives were evaluated based not only on technical performance but also on availability and price. Selecting widely available components reduces procurement delays and simplifies replacement should hardware failures occur during testing. This approach also improves the potential for future replication of the project by other students or researchers without requiring specialized or difficult-to-source components.

Cost considerations also influenced the decision to develop a custom prototype rather than relying on expensive commercial lightning detection equipment. By integrating low-cost embedded hardware with custom firmware, the project demonstrates that a portable lightning detection system can be developed using readily available electronic components while remaining financially practical for an educational environment.

4.3.2 Manufacturability Constraints

The proposed lightning detection system must be designed so that it can be assembled, tested, and modified using the manufacturing resources available to the project team. Unlike commercial products produced using automated manufacturing processes, the prototype will be fabricated primarily through student assembly, requiring design decisions that simplify construction and troubleshooting.

Custom printed circuit boards will be designed to consolidate the system electronics while reducing wiring complexity and improving overall reliability. Component placement, connector selection, and PCB layout will be developed with assembly and debugging in mind to facilitate testing throughout the development process. Where practical, modular hardware interfaces will be used to allow individual subsystems to be tested independently before full system integration.

Mechanical manufacturability also influences enclosure development. The enclosure is expected to be fabricated using additive manufacturing techniques, allowing rapid iteration as hardware changes occur during the design process. Three-dimensional printing provides a cost-effective method of producing custom mechanical components

while minimizing fabrication time and enabling modifications without specialized tooling.

4.3.3 Schedule Constraints

Project development is governed by the timeline established for the Senior Design sequence, requiring the system to progress through research, design, implementation, testing, and documentation within the allotted academic schedule. This limited development period requires careful planning to ensure that critical project milestones are completed on time while maintaining sufficient opportunity for subsystem integration and testing.

To reduce project risk, the overall system has been divided into multiple subsystems that can be developed independently by individual team members before final integration. This parallel development approach allows hardware, firmware, mobile application development, and voice playback implementation to progress simultaneously while reducing scheduling conflicts between project tasks.

The inclusion of a lightning emulator also addresses schedule constraints by allowing subsystem testing to occur independently of natural weather conditions. Without the emulator, verification of the detector would depend upon naturally occurring lightning activity, significantly increasing project uncertainty and delaying system validation. The emulator therefore provides a controlled and repeatable testing environment that supports efficient development throughout both Senior Design I and Senior Design II.

4.4 Other Design Considerations

In addition to the technical and practical constraints discussed previously, several broader engineering considerations influenced the overall system design. Although these considerations did not directly determine individual hardware or software selections, they affected the long-term usability, reliability, and future development potential of the proposed lightning detection system.

4.4.1 Safety Considerations

Because the proposed system is intended to warn users of potentially hazardous weather conditions, system reliability is an important safety consideration. Although the detector is designed to provide early notification of nearby lightning activity, it should not be considered a substitute for official weather forecasts or established lightning safety procedures. Users should continue to follow guidance issued by the National Weather Service and local emergency management agencies regardless of the detector's reported status.

The hardware must also be designed to operate safely under normal operating conditions. Proper voltage regulation, protected power distribution, and appropriate electrical

isolation reduce the likelihood of hardware failures that could compromise system operation. The enclosure should protect users from accidental contact with energized circuitry while shielding sensitive electronic components from physical damage during transportation and normal outdoor use.

4.4.2 Scalability

Although the current prototype is intended primarily as a portable standalone lightning detector, the system architecture has been designed with future expansion in mind. The modular organization of the hardware and firmware allows additional sensors, communication interfaces, or software features to be incorporated with minimal modification to the existing system.

Future revisions of the project may include enhanced lightning localization algorithms, GPS functionality, cloud-based data logging, expanded mobile application features, or additional environmental sensors. Selecting standardized communication interfaces such as SPI and Bluetooth Low Energy provides flexibility for future development while minimizing compatibility concerns between new hardware and the existing system architecture.

4.4.3 Reliability and Maintainability

Long-term reliability and ease of maintenance were considered throughout the design process to simplify both prototype development and future improvements. Wherever practical, commercially available components with extensive documentation and community support were selected to reduce development risk and simplify troubleshooting. Modular subsystem design also allows individual hardware components to be repaired, replaced, or upgraded without requiring complete redesign of the system.

Maintainability extends beyond hardware into firmware development. Separating the lightning detection algorithms, Bluetooth communication, and voice playback functionality into independent software modules improves code readability and simplifies debugging. This modular approach also allows future developers to modify individual subsystems without introducing unnecessary changes to unrelated portions of the project.

4.4.4 Future Development Considerations

As an academic prototype, the proposed lightning detection system represents the first stage of a design that could be further refined for commercial or recreational use. Additional development beyond the scope of this project may include weather-resistant enclosure improvements, extended battery life through advanced power management techniques, expanded wireless communication capabilities, and additional user interface features.

These future enhancements were considered during the current design process by selecting components and communication protocols that support system expansion without requiring substantial hardware redesign. This forward-looking approach improves the long-term adaptability of the project while providing a strong foundation for continued development during Senior Design II.

Chapter 5 - Artificial Intelligence Use

In both the design process and research phase of this project, our group used ChatGPT and related generative AI technologies as research and documentation tools. In this chapter, we will provide an analysis of this process, as is expected from our program, and document the prompt and result which provided our insight into the capabilities and risks associated with these technologies. Our goal is not to present AI as a replacement for engineering judgment, but to be honest about the areas where it helped us and where it caused risk.

5.1 Platforms Used

The main AI tool used by the team was ChatGPT (GPT-4-class model by OpenAI), accessed via its regular chat-based interface, mainly used for research. The other members utilized the code completion tool in GitHub Copilot within the Arduino IDE and VS Code during the firmware bring-up process of our Arduino testbeds (Section 3.3). At times, we would confirm information with a second chatbot in case they were unsure about the answer.

5.1 Benefits and Disadvantages

Since these tools were used in a targeted way, they provided our team with a number of practical benefits:

Immediate orientation on new hardware. Only two members of our team had prior experience working with 8-bit PICs. For the rest of the team, this represented a significant learning curve, since low-level peripheral behavior is rarely intuitive from documentation alone. By asking ChatGPT to explain in simple terms how a peripheral like Timer2/CCP works, we were able to build a mental model of unfamiliar concepts such as prescalers, PWM duty cycle registers, and interrupt flags before diving into the dense, register-level detail of the datasheet. This meant that when we did turn to the official documentation, we already had a rough map of what we were looking for and

why it mattered, rather than reading it cold. In effect, ChatGPT acted as a bridge between having zero context and being able to productively parse manufacturer documentation.

Debugging assistance. Firmware development on unfamiliar boards inevitably produced compile errors and unexpected runtime behavior, and diagnosing these issues manually is often slow, particularly when the root cause is a subtle configuration mismatch rather than a logic error. When our firmware failed to compile or behaved unexpectedly on the Arduino boards, pasting the exact compiler error message, or a plain-language description of the observed behavior, into ChatGPT typically returned a shortlist of likely causes, such as missing library includes, incorrect baud rate configuration, or mismatched pin definitions. This was consistently faster than manually searching through forums or issue trackers, where relevant answers are often buried among unrelated threads or specific to slightly different hardware setups. In several cases, this let us isolate a bug within minutes rather than the better part of an hour.

Research acceleration. During the technology research step detailed in Chapter 3, our team needed to quickly get a lay of the land across a range of candidate components, protocols, and approaches before making design decisions. ChatGPT proved useful here not as a source of authoritative facts, but as a way to quickly surface relevant search terms, comparison criteria, and general background information that we could then verify against primary sources such as datasheets, official documentation, and peer-reviewed material. This meant our research time was spent evaluating credible sources rather than figuring out what to search for in the first place.

5.3 Limitations and Drawbacks

These same platforms presented several challenges and restrictions that needed to be dealt with by the team:

- **Hallucinations.** ChatGPT mentioned register names and bit field names of the ESP32 that were not present in their actual datasheets. Since these mistakes were explained using the same confidence level as accurate information, it was possible to detect them only after the team started checking all technical facts in the manufacturer's datasheet before implementing them anywhere.
- **Knowledge cut-off and outdated information.** The data used by these platforms is outdated, so sometimes we were provided with advice from the AI console that was generic or outdated.
- **Risk of diminished comprehension.** The purpose of Senior Design is to develop engineering intuition by tackling problems directly, specially through challenging problem solving efforts. Sometimes AI is unable to aid individuals to do this, for instance, when the team took the output from AI as a solution, it would sometimes lead to solutions that were easy to implement but with important fail-points that were not acknowledged previously.
- **Lack of specific knowledge about the project.** ChatGPT did not have any built-in knowledge about our diagrams, our firmware state, or our BOM. The

responses given by it were generic unless one of the team members provided it with the relevant information in the form of prompts, therefore the quality of the prompt determined the relevance of the answer.

5.4 Examples of Impact on Learning Experience

The following case highlights the advantages and disadvantages mentioned above through the experiences of the team and the team's firmware. The complete prompt and answers of the case can be found in Appendix B.

5.4.1 Case 1 (Positive, with a detected hallucination): PIC16F1789 Timer2/CCP Configuration

Before implementing the firmware in the emulator (Section 3.2.2), the team requested ChatGPT to explain how the Timer2 and CCP peripherals on the PIC16F1789 microcontroller work together. This was explained well and helped the team become familiar with the architecture quite fast. But when we questioned ChatGPT about the exact bit positions in the control registers, the answer pointed to a bit field which is not present in this chip. Overall result: very positive regarding concept comprehension, but another example of why everything had to be independently checked.

5.4.3 Case 3 (Mixed, with a detected hallucination): Literature Review on PIC-Based Embedded Control Systems

In the beginning stages of researching the project, the team turned to ChatGPT and requested to give a conceptual summary of common methods of creating real-time control loops on PIC microcontrollers along with mentioning any related academic literature on the subject. This summary itself proved to be useful by providing an understanding of common practices, such as using interrupts to sample signals, using fixed-point arithmetic for control loops, and common trade-offs between pollings and interrupts. In order to add credibility to the conceptual summary, the team then requested ChatGPT to mention some references that would include authors, dates of publication, and names of conferences and journals where these papers were presented. However, after searching for them in the university library and Google Scholar, none of the suggested papers could be found, and one of the journals didn't even seem to exist.

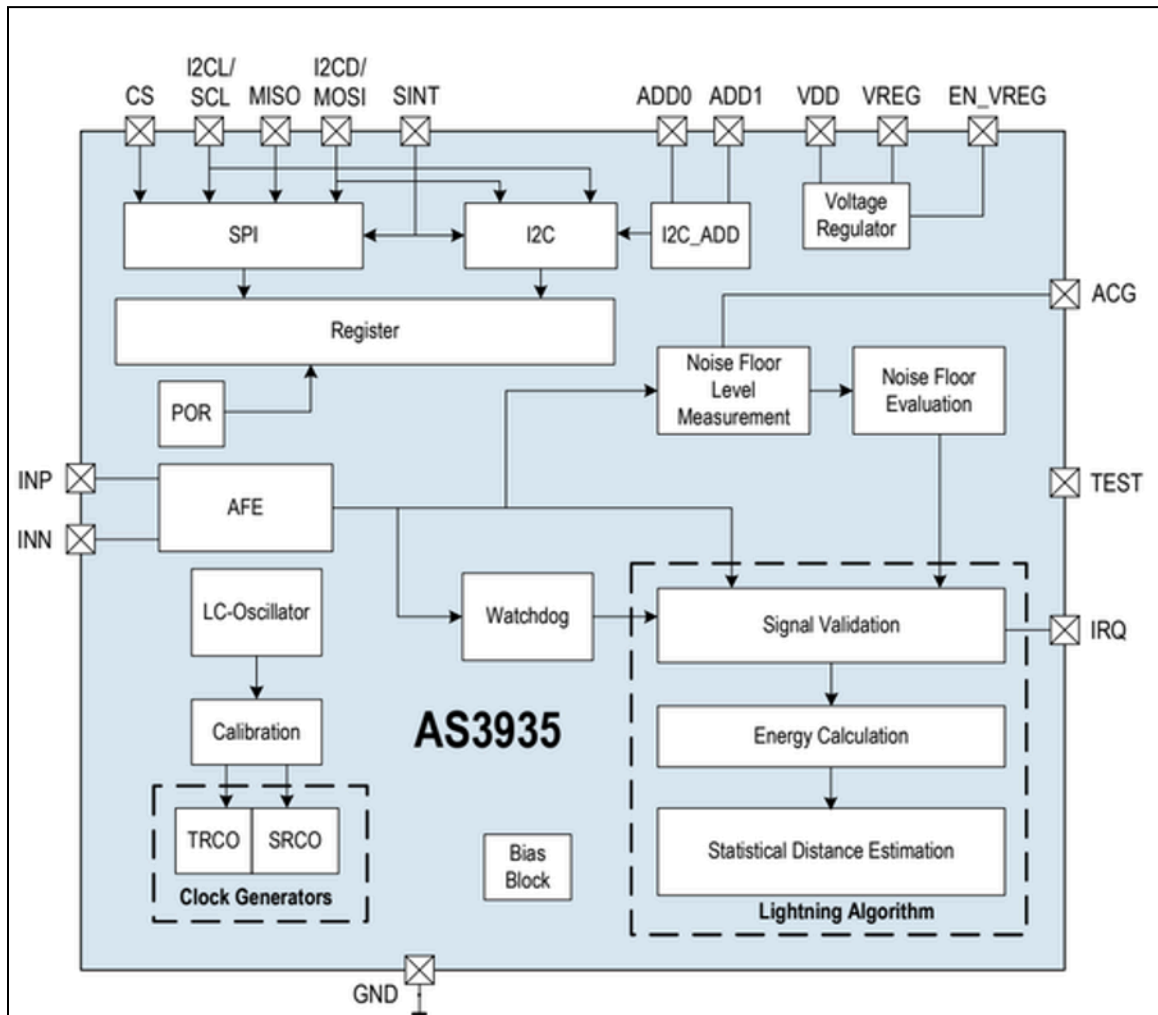
Chapter 6 - System Hardware Design

6.1 System Hardware Overview

This chapter presents the hardware implementation of the lightning detector and lightning emulator developed during the project. The following sections include system schematics, printed circuit board (PCB) layouts, and supporting hardware design files that illustrate the electrical implementation of the proposed system.

The radio integrated solution we found is made by ScioSense previously Austrian Microsystems.

Figure 6.1 - IC Schematic



The radio IC is a programmable fully integrated VLF radios IC with Lightning Sensor capability. The embedded algorithm it contains checks the incoming signal pattern to reject the potential man-made disturbers.

The radio IC can also provide information on the noise level and inform the external unit (e.g. microcontroller) in case of high noise conditions, with the noise floor generator and noise floor evaluation blocks.

The radio IC can be programmed via a 4-wire standard SPI and is also compatible with I2C. Also, in case the latter is chosen, it is possible to choose from three different addresses. Two clocks are internally generated by two different RC-Oscillators:

TRCO and SRCO. An automatic calibration procedure can increase the precision of those oscillators. The radio IC can be either supplied by an internal voltage regulator or directly by VDD.

6.2 System Schematics

The complete electrical schematics for both the lightning emulator and lightning detector are presented below. These schematics illustrate the electrical connections between the microcontrollers, analog front-end circuitry, power regulation, communication interfaces, and peripheral devices used throughout the project.

6.2.1 Emulator Schematic

Figure 6.2 - Emulator Schematic

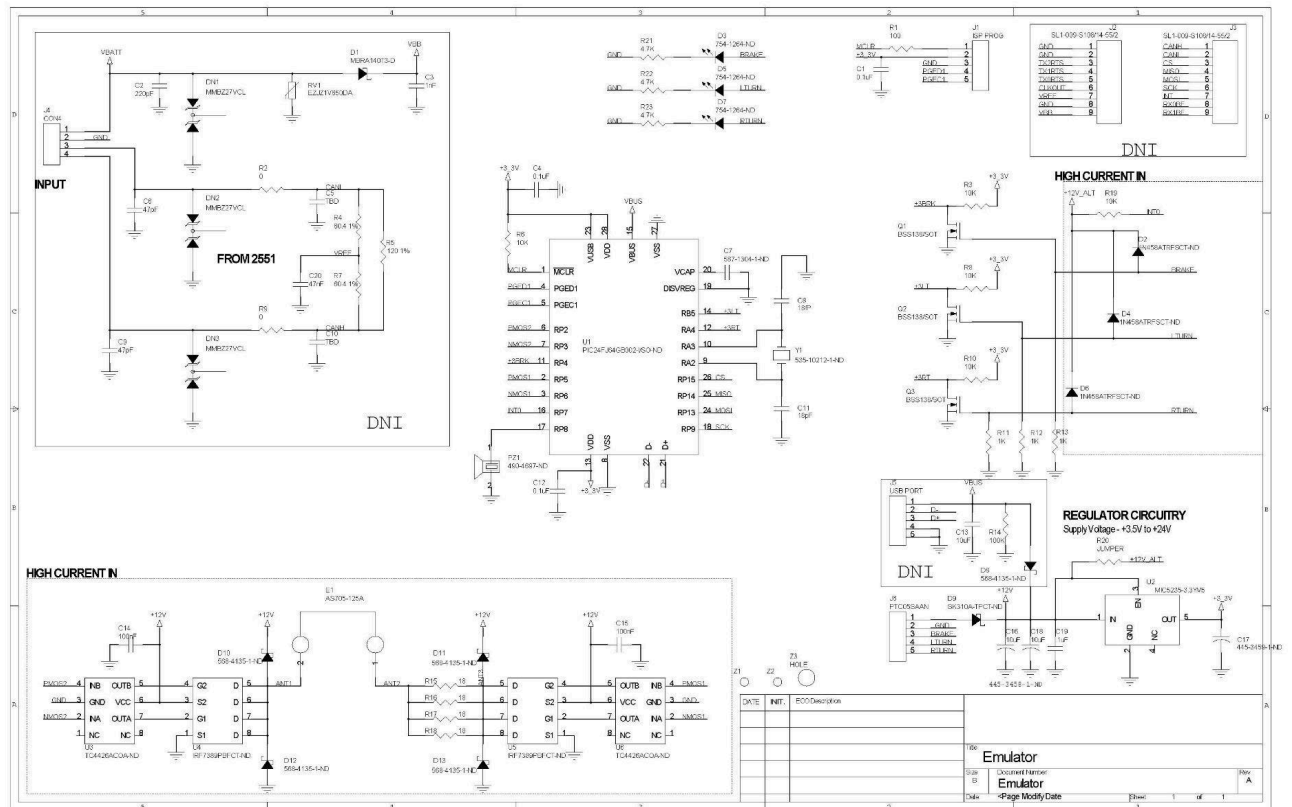


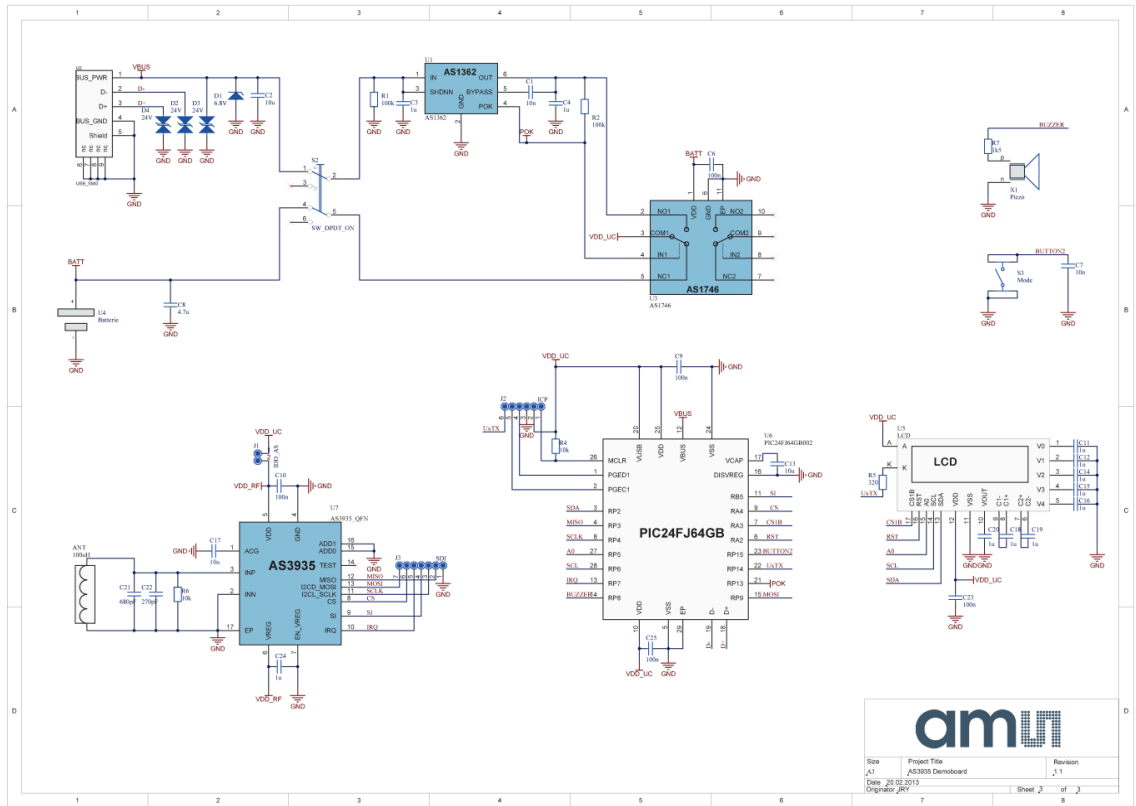
Figure 6.3 - Detector Schematic



6.3 Alternative Design Approaches

Alternative approach designs:

Figure 6.4 - Alternative Design Schematic

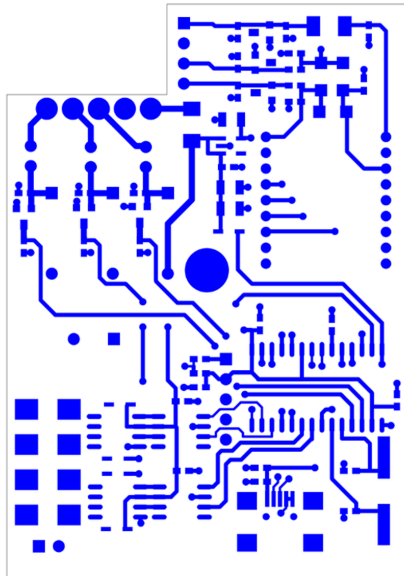


6.4 PCB Layouts

6.4.1 Emulator

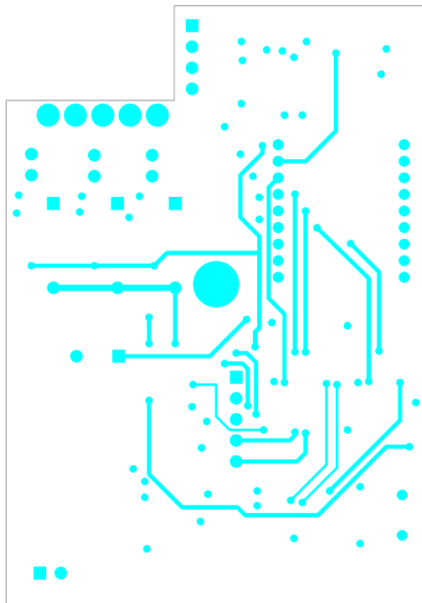
6.4.1.1 Emulator PCB Top Layer:

Figure 6.5 - Emulator PCB Top Layer



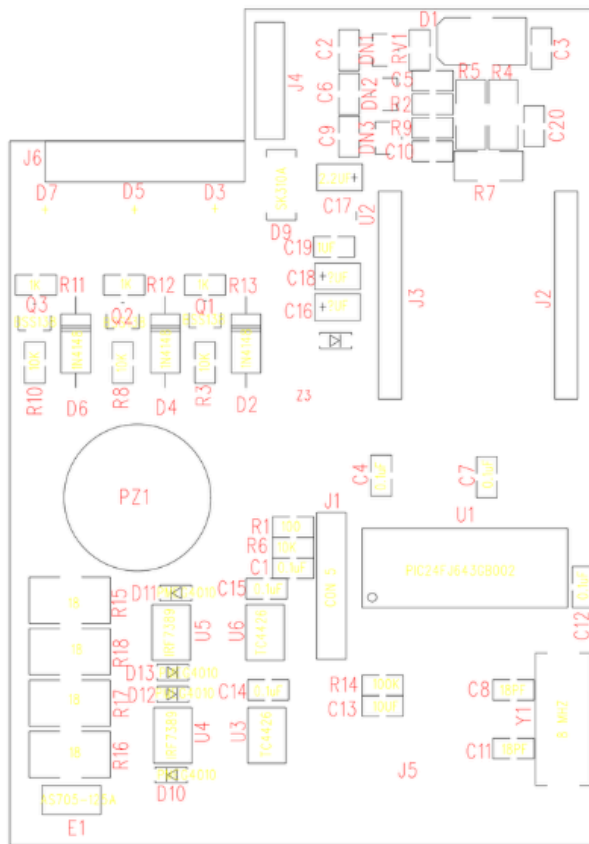
6.4.1.2 Emulator PCB Bottom Layer

Figure 6.6 - Emulator PCB Bottom Layer



6.4.1.3 Emulator PCB Silk Screen Layer:

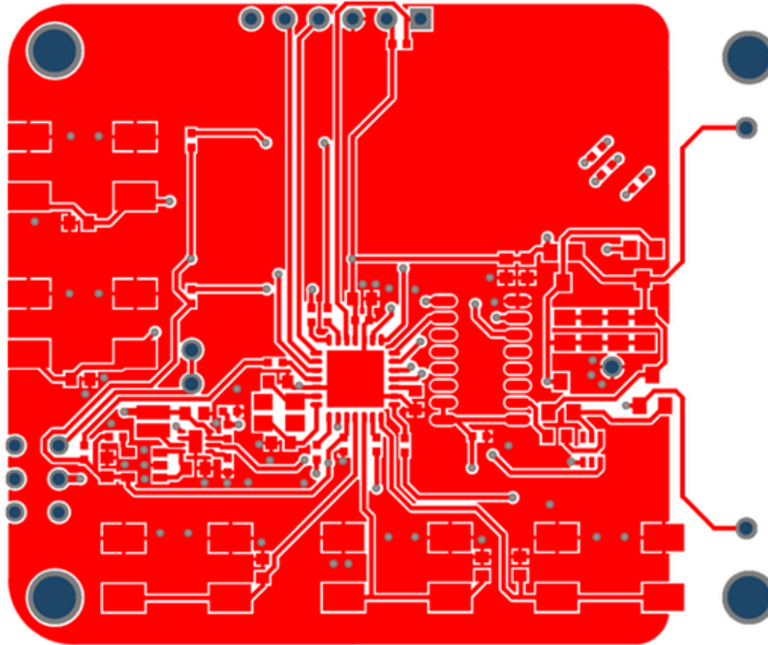
Figure 6.7 - Emulator PCB Silkscreen Layer



6.4.2 Emulator Test Board

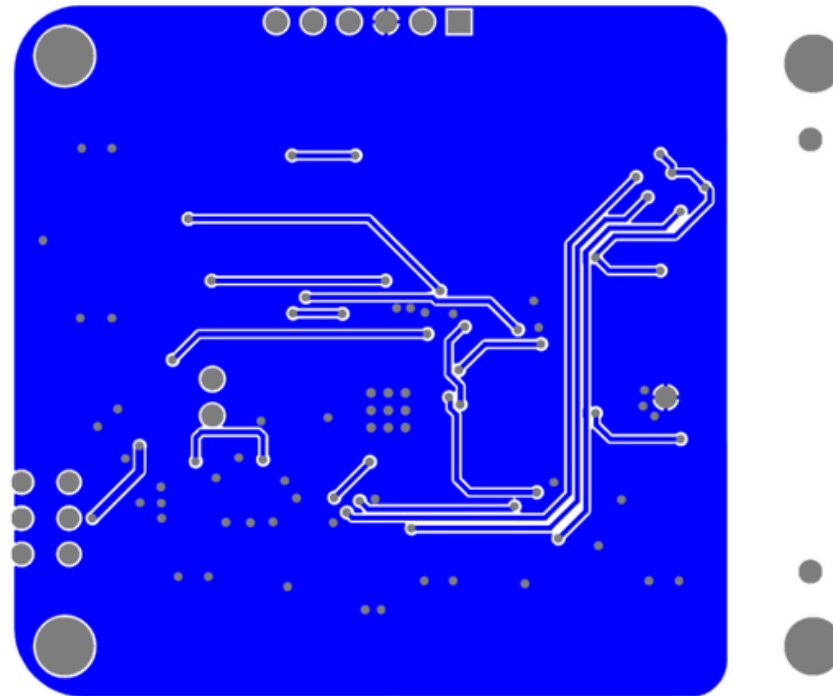
6.4.2.1 Emulator Test Board PCB Top Layer:

Figure 6.8 - Emulator Test Board PCB Top Layer



6.4.2.1 Emulator Test Board PCB Bottom Layer

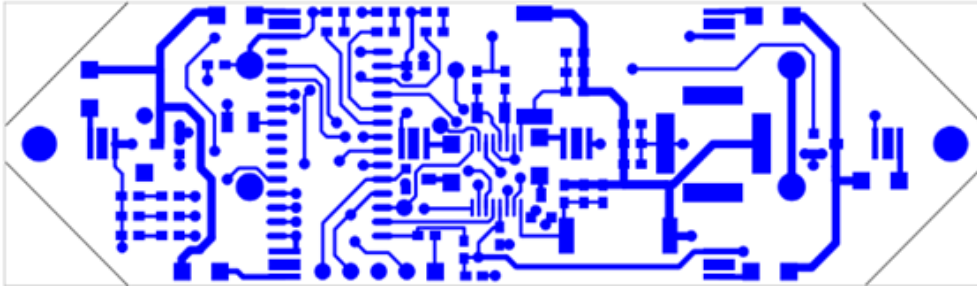
Figure 6.9 - Emulator Test Board PCB Bottom Layer



6.4.3 Detector PCB

6.4.3.1 Detector PCB Top Layer

Figure 6.10 - Detector PCB Top Layer



6.4.3.2 Detector PCB Bottom Layer

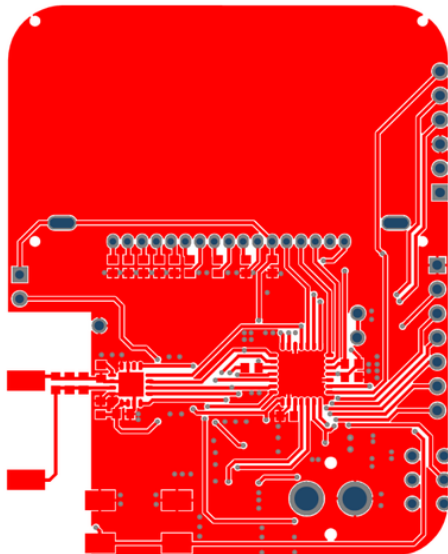
Figure 6.11 - Detector Test Board PCB Top Layer



6.4.4 Detector Test Board

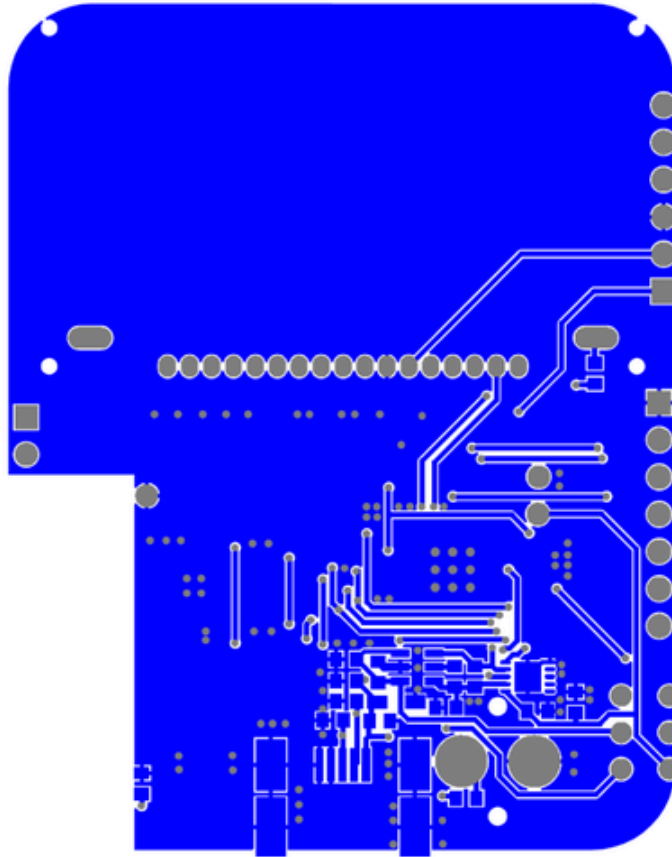
6.4.4.1 Detector Test Board PCB Top Layer

Figure 6.12 - Detector Test Board PCB Top Layer



6.4.4.2 Detector Test Board PCB Bottom Layer

Figure 6.13 - Detector Test Board PCB Bottom Layer



Detector Secondary Test Board:

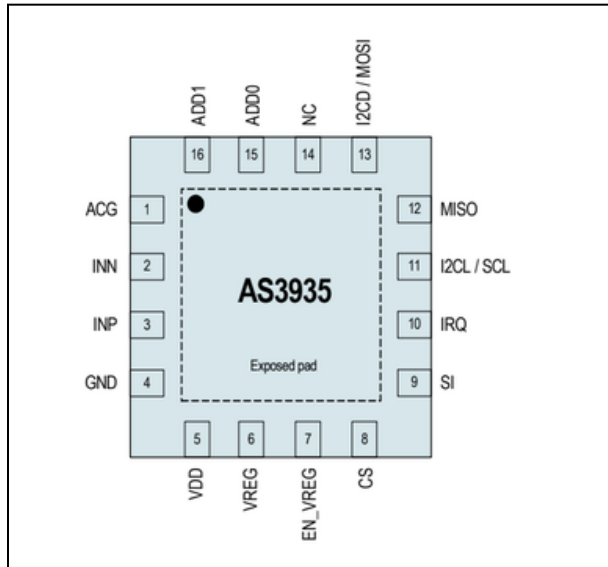
Figure 6.14 - Secondary Test Board



6.5 IC

IC Pinout and Overview:

Figure 6.15 - IC Pins



Radio IC Pin Assignments:

Pin Number	Pin Name	Pin Type	Description
1	ACG	Analog I/O	AC-Ground
2	INN		Antenna ground
3	INP		Antenna positive input
4	GND	Supply pad	Ground
5	VDD		Positive supply voltage
6	VREG		Positive supply voltage / Regulated voltage
7	EN_VREG	Digital input	Voltage Regulator Enable
8	CS		Chip Select (active low)
9	SI		Select Interface (GND → SPI or VDD → I ² C)
10	IRQ	Digital output	Interrupt
11	I2CL/SCL	Digital input	I ² C clock bus or SPI clock bus (according to SI setting)
12	MISO	Digital output	SPI data output bus

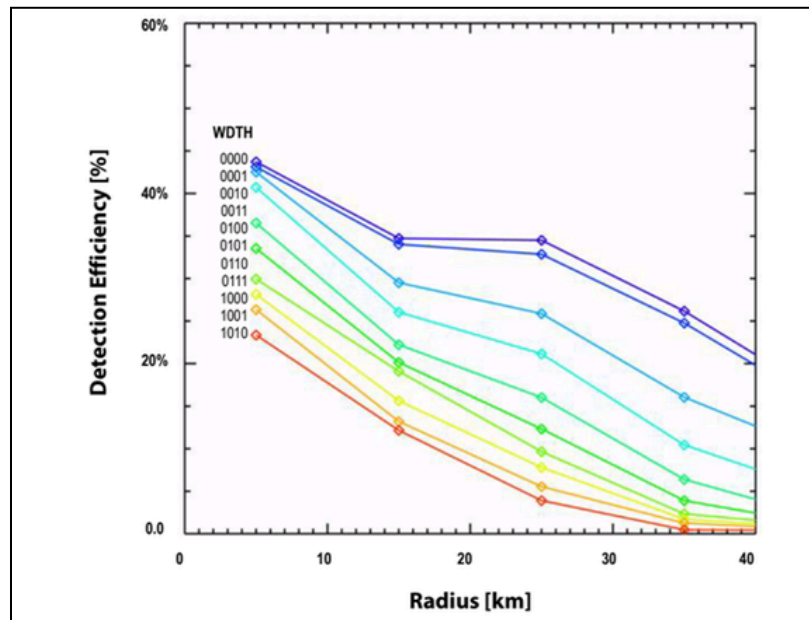
Signal Verification

The signal verification mode is based on the Algorithm block. Every time the watchdog threshold is passed the IC enters the signal verification mode. The watchdog threshold can be set. If the signal is classified as disturber the chip immediately aborts the signal processing and goes back into the listening mode. Otherwise, the energy calculation is performed and the distance estimate provided.

Overview of Radio IC Registers:

Register #	7	6	5	4	3	2	1	0
0x00	Reserved		AFE_GB					PWD
0x01	Reserved	NF_LEV				WDTH		
0x02	Reserved	CL_STAT	MIN_NUM_LIGH		SREJ			
0x03	LCO_FDIV		MASK_DIST	Reserved	INT			
0x04	S_LIG_L							
0x05	S_LIG_M							
0x06	Reserved			S_LIG_MM				
0x07	Reserved		DISTANCE					
0x08	DISP_LCO	DISP_SRCO	DISP_TRCO	Reserved	TUN_CAP			
0x3A	TRCO_CALIB_DONE	TRCO_CALIB_NOK	Reserved					
0x3B	SRCO_CALIB_DONE	SRCO_CALIB_NOK	Reserved					

Figure 6.17 - Radio IC Response Frequency vs. Range



Noise Floor Threshold Setting for Radio IC:

Continuous Input Noise Level [μVrms] (Outdoor)	Continuous Input Noise Level [μVrms] (Indoor)	REG0x01[6]	REG0x01[5]	REG0x01[4]
390	28	0	0	0
630	45	0	0	1
860	62	0	1	0
1100	78	0	1	1
1140	95	1	0	0
1570	112	1	0	1
1800	130	1	1	0
2000	146	1	1	1

Chapter 7 - System Software Design

7.1 Overview

The software system for the lightning detector is divided into three layers: the firmware running on the microcontroller, the BLE communication layer that bridges the hardware and the mobile app, and the iOS mobile application itself. Benjamin Titus is responsible for the mobile app, the BLE GATT profile design, and the BLE firmware.

7.2 Mobile Application

7.2.1 Mobile Application Architecture

The mobile application is a native iOS app built with Swift and SwiftUI, targeting iOS 17 and later. SwiftUI was chosen because it is Apple's current UI framework and handles reactive state updates cleanly when the BLE connection state or the lightning status changes, the UI reflects that change instantly without requiring any manual update logic.

The app is structured around five source artifacts:

BLEManager.swift is the core of the application. It is a class that manages the entire Bluetooth lifecycle. When the app launches, BLEManager creates a CoreBluetooth central manager and waits for the phone's Bluetooth radio to be ready. When the user taps

Scan, it begins scanning specifically for devices advertising our custom GATT service UUID, so only our lightning detector shows up in the list rather than every Bluetooth device nearby. Once the user selects a device and the connection is established, BLEManager discovers the detector's service, finds the status characteristic, and subscribes to notifications. From that point on, any time the chip sends an update, BLEManager receives it, decodes the single byte, creates a new LightningStatus value with the current timestamp, and publishes it. SwiftUI automatically re-renders the status screen when that published value changes.

LightningStatus.swift is the data model. It holds two things: a boolean indicating whether lightning is currently detected, and a timestamp for when that update was received. The timestamp lets the status screen show something like "Updated 2 minutes ago," which helps users know the device is still actively communicating and the status is not stale.

ContentView.swift is the top-level router. It checks the connection state of BLEManager and either shows the scan screen or the status screen. This keeps the navigation logic simple and all in one place.

ScanView.swift is the first screen the user sees. It shows a list of nearby BLE devices that match our service UUID, a scan/stop button, and a small "Try Demo Mode" button at the bottom for testing without hardware.

StatusView.swift is the main experience. It shows a large circular indicator that pulses when lightning is detected, the status text, a subtitle, and a relative timestamp. The icon transitions between a checkmark (safe) and a bolt (detected) using SwiftUI's symbol effect transition. In demo mode, a toggle button appears below the timestamp so the developer can simulate state changes.

7.2.2 BLE Communication Protocol

The GATT profile is designed to be as simple as possible at this stage while leaving room to expand later.

Service UUID: 4FAFC201-1FB5-459E-8FCC-C5C9C331914B

Status Characteristic UUID: BEB5483E-36E1-4688-B7F5-EA07361B26A8

Characteristic properties: Notify

Data format: Single byte (0x00 = clear, 0x01 = lightning detected)

The firmware will need to advertise this service UUID and write to the characteristic whenever the detection state changes. The iOS app will automatically receive that notification and update the UI. Because the characteristic is notify-only at this stage, the phone receives updates when the pushes them rather than the phone requesting them on a schedule. This is more efficient and keeps the BLE traffic low.

The single-byte format is intentional. When the distance estimation and directional trend firmware from Beatriz's side is ready, we can extend the protocol to include additional bytes for distance and trend data without changing the UUID or the subscription setup on the iOS side.

7.2.3 Version Control and Repository Setup

The project repository is hosted on GitHub and is organized to support work across multiple machines. The repository contains the full Xcode project at the root level along with a README that documents the file structure, setup steps, the BLE UUIDs, and the data protocol.

Link: <https://github.com/Benjamintitus03/LightningDetector/>

The branching strategy uses two branches: master, which holds stable working code, and develop, which is the active development branch used for day-to-day changes. When a feature is ready and tested, it gets merged into master. This keeps the master branch clean in case we need to demo or hand something to the review committee.

The repository is synced through OneDrive, which allows the project to be accessed from both a Windows machine and a Mac without needing to push and pull manually for every small change. The .xcodeproj file and all Swift source files live inside a LightningDetector subdirectory matching the standard Xcode project structure. The README is at the root of the repository.

7.2.4 iOS Deployment Plan

Current state (development): The app is signed with a personal Apple Developer account (free tier) and is deployed directly to a physical iPhone over USB via Xcode. This is sufficient for development and testing. The free account does not allow App Store distribution but allows running on up to a handful of personal devices, which is all we need for the demo and for testing with the hardware.

Short-term (class demo and grading): We will continue using direct USB deployment for the demo. If we want to share the app with other team members' phones for testing, we can add their device UDIDs to the provisioning profile under the free account, or upgrade to the paid Apple Developer Program if needed.

Production iOS deployment: For public distribution on the App Store, the process is as follows. First, enroll in the Apple Developer Program (\$99/year). Then, in Xcode, set the build configuration to Release, create an Archive of the app, and submit it through App Store Connect. Apple reviews the app (typically a few days for a first submission) and

then it becomes publicly available. Before submitting to the App Store, we would want to use TestFlight, Apple's beta testing platform, to distribute the app to a small group of testers without going through the full App Store review.

Future Android support:

The current app is SwiftUI, which only runs on Apple platforms. When Android support becomes a priority, the recommended path is to rebuild the app using Flutter, Google's cross-platform framework that supports both iOS and Android from a single codebase. The BLE protocol we have designed is not platform-specific — the same service UUID, characteristic UUID, and byte protocol will work with Android's Bluetooth LE API as well. The main work would be rewriting the UI and BLE logic in Dart (Flutter's language), but the overall structure would be very similar to what we have now. React Native is another option, but Flutter has stronger native BLE support and is generally preferred for hardware-adjacent applications.

7.2.5 Demo Mode

A demo mode was added to the app to support development and testing before the hardware is complete. Tapping "Try Demo Mode" on the scan screen bypasses the BLE scan and sets the app into a simulated connected state. In this mode, the status screen shows "Demo Mode" in the connection bar instead of a device name, and a toggle button appears that lets the developer switch between the safe and detected states manually. Disconnecting from demo mode returns the app to the scan screen as normal. This was essential for iterating on the UI and animations without needing to have the Nordic connection up and running detection firmware every time a UI change was made.

Below are attached photos of our current application progress. You can see portrait and landscape orientations depending on the device being used(Laptop,Ipad, Cell Phone):

10:42



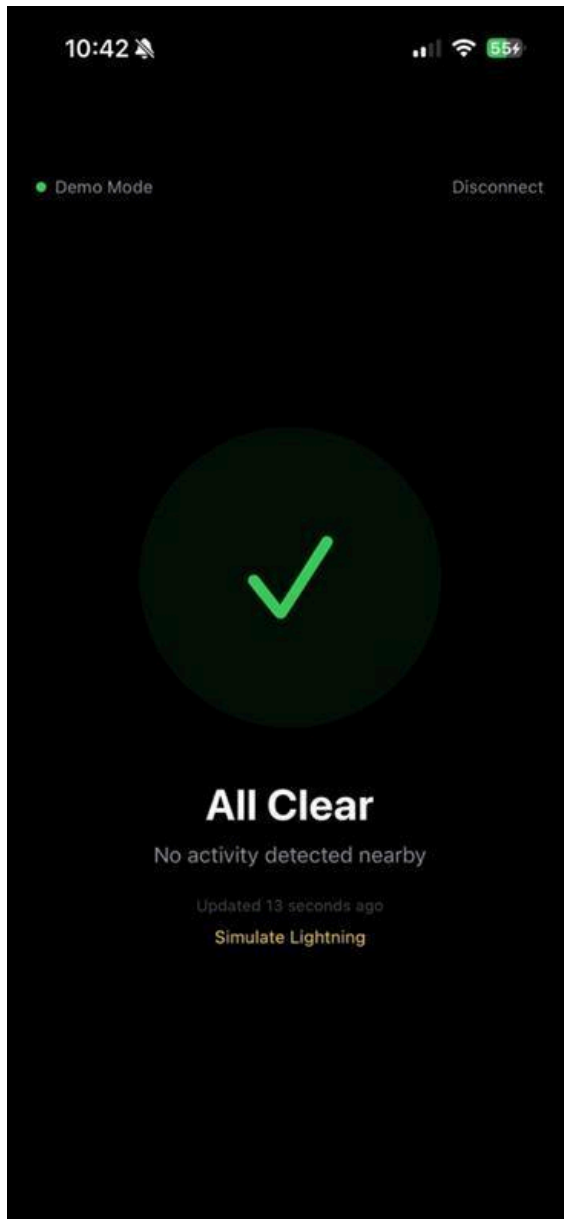
Lightning Detector

Connect to your device to begin

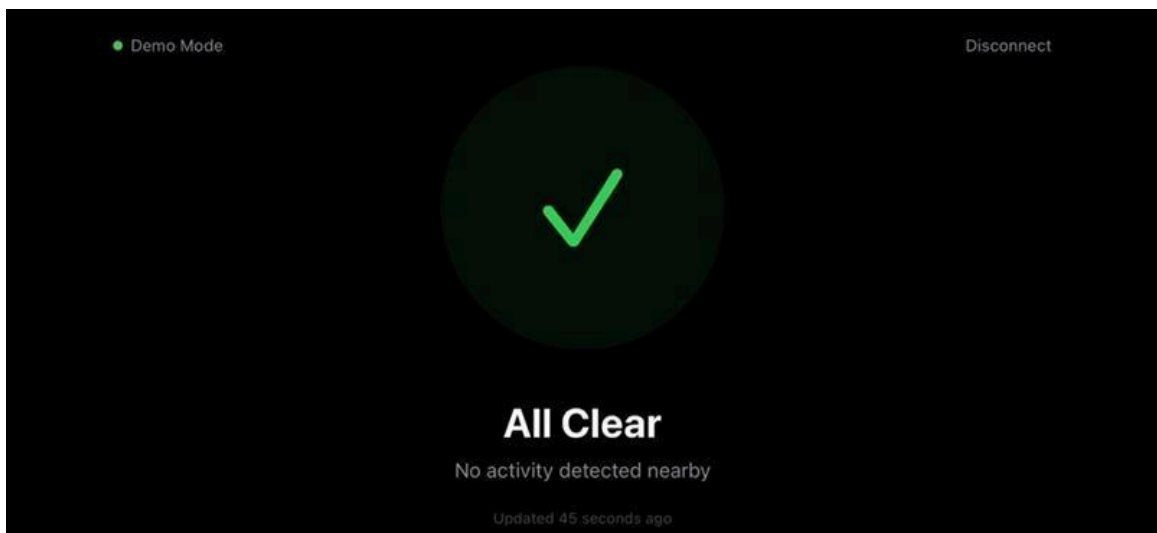
No devices found

Scan for Devices

Try Demo Mode



Screenshots taken from a Laptop



7.2.6 Software Appendix

Why not React Native or Flutter from the start? Both are reasonable cross-platform options, but for a project where BLE is the entire communication mechanism, going native on iOS with CoreBluetooth removes one layer of abstraction. Cross-platform BLE libraries add complexity and sometimes have platform-specific quirks. Since we knew iOS was the initial target and Android was a stretch goal, native iOS was the cleaner choice. If Android becomes a concrete requirement before the project is done, it is worth revisiting Flutter at that point.

Why a single byte for the data protocol? Keeping the initial protocol at one byte makes the early firmware and app integration as simple as possible. The format is easy to parse on both ends and leaves no ambiguity. Extending it later is straightforward and we can add a second byte for distance category (near/medium/far) and a third for directional trend (approaching/retreating) when that firmware is ready, without changing anything in the existing connection or subscription setup. That way the hardware team and the software side are not blocked on each other.

On the Git repo and collaboration: The repository tracks all source code for the iOS app and will eventually include firmware as well. Having everything in one place makes it easier for the review committee to see the full scope of what was built and for the team to coordinate. The OneDrive sync means the Mac running Xcode and the Windows development machine stay in sync without extra steps.

7.3 Lightning Algorithm

The lightning algorithm we intend to employ is as follows. False events (man-made disturbers) which might trigger the Radios IC are rejected, while lightning events initiate calculations to estimate the distance to the head of the storm.

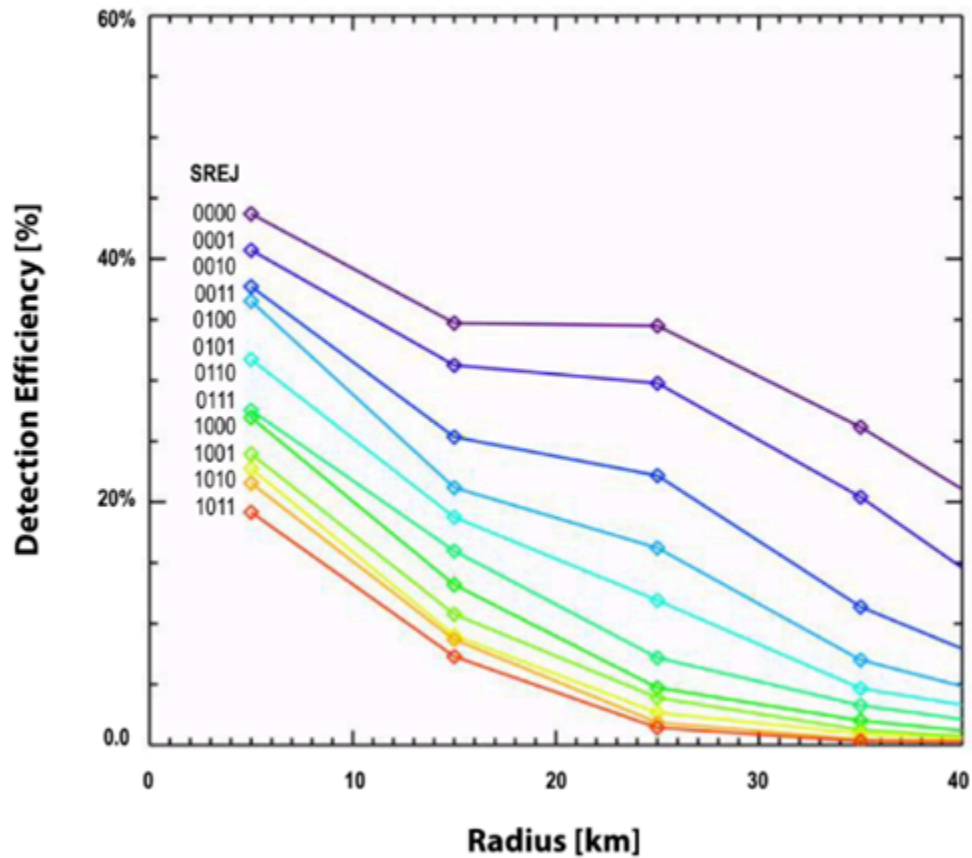
The Lightning algorithm is broken into three sub blocks:

1. **Signal Validation:** Verification that the incoming signal can be classified as lightning.
2. **Energy Calculation:** Calculation of the energy of the single event.
3. **Statistical Distance Estimation:** According to the number of stored events (lightning), a distance estimate is calculated. In case the incoming signal does not have the shape characteristic to lightning, the signal validation fails and the event is classified as disturber. In that case the energy calculation and statistical distance estimation are not performed and the device automatically goes back to listening mode. The shortest time span between two lightning strikes that the Radios IC can resolve is approximately one second. Once a signal is classified as disturber the sensor is deactivated for a further 1.5s period. As the duration of disturber signals can vary, this sensor down time will prevent the sensor from triggering repeatedly due to longer disturber events.

Signal Validation

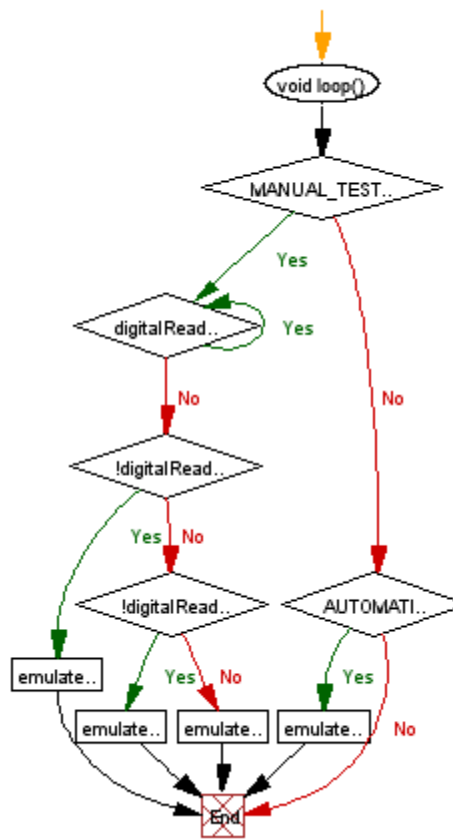
During the signal validation phase the shape of the incoming signal is analyzed. The device can differentiate between signals that show the pattern characteristic of lightning strikes and man-made disturbers such as random impulses.

Below is a graph of our estimated Detection Efficiency vs Radius of Strike:

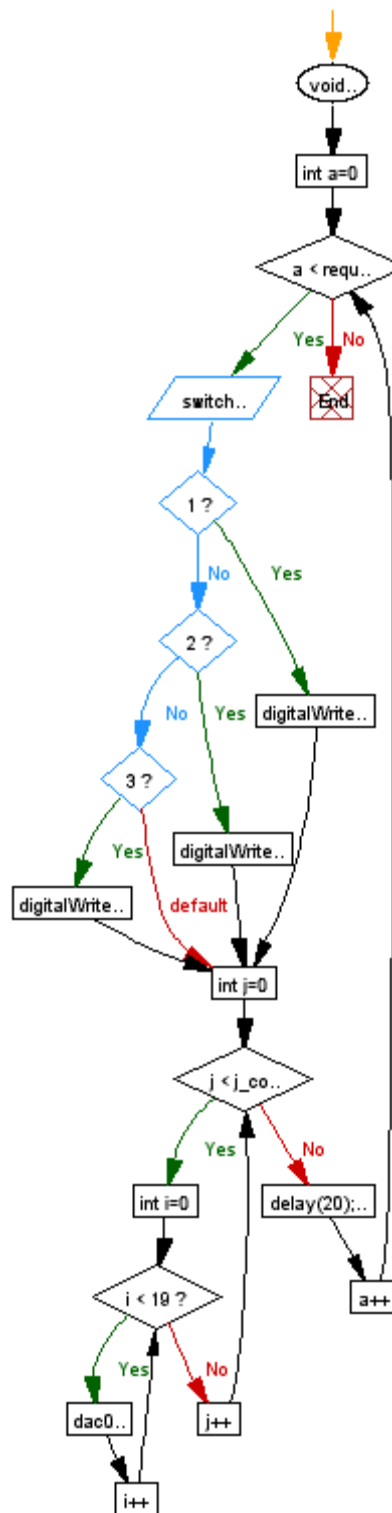


Emulator and Detector Test Firmware Functional Diagram:

Emulator and Detector Test Firmware:

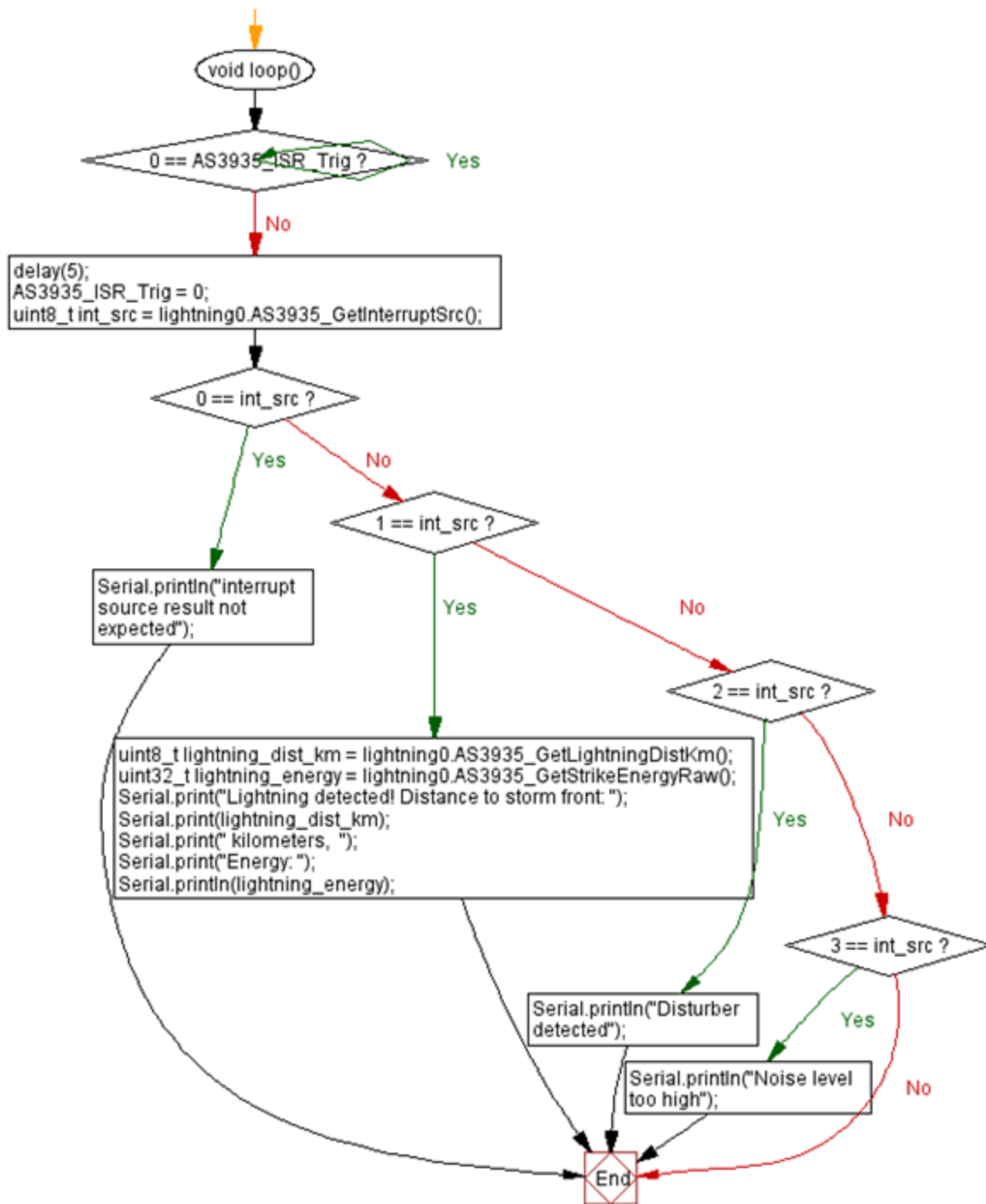


Emulation:



Detector:





7.4 Voice Playback Firmware Integration

The voice playback subsystem provides spoken notifications to inform users of lightning activity and system status. Rather than generating audio signals directly with the primary microcontroller, the firmware communicates with the Nuvoton ISD2360 voice playback integrated circuit using the Serial Peripheral Interface (SPI). This approach minimizes processor workload while allowing prerecorded voice messages to be triggered dynamically based on system conditions.

The following sections describe the firmware architecture used to control voice playback within the proposed lightning detection system.

7.4.1 Firmware Operation

The system firmware continuously monitors data collected from the lightning detection circuitry while executing the primary detection algorithm. When the measured signal exceeds predetermined thresholds indicating nearby lightning activity, the firmware determines the appropriate warning level and selects the corresponding prerecorded voice message stored within the ISD2360.

Playback requests are transmitted to the ISD2360 through SPI commands, allowing the dedicated voice playback IC to reproduce the selected audio message independently of the host microcontroller. Because the ISD2360 performs all audio decoding and playback internally, the primary microcontroller is free to continue monitoring sensor data and evaluating additional lightning events while spoken alerts are being reproduced.

7.4.2 Firmware Workflow

The overall firmware sequence for voice playback is illustrated in Figure 7.X. After the lightning detection algorithm identifies a valid event, the firmware determines the appropriate alert level, selects the corresponding prerecorded voice message, and issues the necessary SPI command to initiate playback. Once playback begins, the ISD2360 independently reproduces the selected message while the microcontroller resumes execution of the lightning detection firmware.

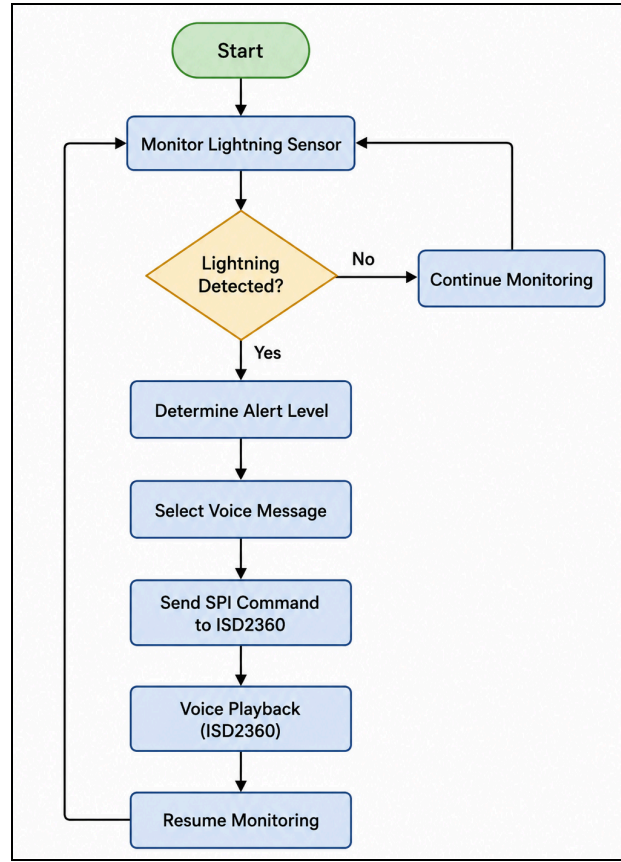


Figure 7.. Voice Playback Firmware Workflow.

7.4.3 Future Firmware Development

At the time of this report, firmware development for the voice playback subsystem is in the early stages. Future work will focus on implementing SPI communication routines, integrating playback commands into the lightning detection firmware, organizing prerecorded voice messages within the ISD2360 memory, and validating reliable operation under normal system conditions. Additional testing will ensure that voice alerts are generated with minimal latency while allowing the lightning detection subsystem to continue operating without interruption.

Chapter 8 - System Fabrication & Prototype Construction

8.1 System Fabrication and Prototype Construction

At the time of this Midterm Milestone Report, fabrication and construction of the final lightning detector and lightning emulator prototypes have not yet been completed.

Current project efforts have focused on system research, component selection, hardware and software design, schematic development, printed circuit board (PCB) layout, and preliminary firmware implementation. As hardware fabrication and subsystem integration occur during the remainder of the Senior Design sequence, this chapter will be expanded in the Final Report to document the manufacturing process, prototype assembly, integration procedures, and any design revisions made during construction and testing.

8.2 Firmware Prototype

Before the final ESP32, PIC16F1789, and custom PCB hardware became available, preliminary versions of the firmware code for both the detector and emulator modules were prepared and verified. This was done by utilizing Arduino-compatible hardware modules. This method, as we described previously in Section 3.4, made it possible for the group to write, test, and verify the core logic of the system early during the development process, without having to wait for the completion of the hardware production process and availability of components. The reason behind this lies in the fact that, since the Arduino Uno follows the same basic protocol (SPI, GPIO interrupts, timed digital I/O) of the target platforms ESP32 and PIC16F1789, the firmware should port easily to them.

8.2.1 Detector Firmware Prototype

When it comes to the detector subsystem, an Arduino Uno was interfaced to the SEN-39003 breakout board, which includes the AS3935 IC used in the lightning sensor, via SPI connection (consisting of four wires; you can see Chapter 9 for details of hardware configuration and breakout board setup). There are three main stages of operation of the firmware that have been implemented as a prototype: initialization, event detection using interrupts and event classifications.

Initialization stage involved the configuration of AS3935 IC using the SPI interface in terms of operating indoors/outdoors, tuning the internal antenna capacitance to resonate at 500 kHz, and setting the watchdog and noise floor thresholds described in Section 6.5. This was all done with research on the ideal settings for our use case, as well as developing using test driven development skills.

After this configuration was completed, the ISR (Interrupt Service Routine) is attached to the IRQ pin of the AS3935. This makes the Arduino wait idle until the interrupt is triggered by the AS3935 and not poll the SPI interface all the time. This is done to be consistent with the low-power usage of the final detector, when the MCU will wait in a low-power monitor mode instead of constantly being triggered (Section 4.2.1).

When the IRQ pin is triggered, the firmware then enters its main event-handling loop, illustrated in the flowcharts of Section 7.3. A short delay is first introduced to allow the AS3935's internal state to settle, after which the firmware reads the Interrupt Source register to determine the cause of the interrupt. These possible three outcomes are handled differently, depending on the cause of the interrupt:

- **Noise level too high:** the ambient noise exceeds the configured threshold, and the event is discarded without further processing.
- **Disturber detected:** the incoming signal shape does not match the pattern characteristic of lightning (Section 3.3.2) and is rejected without triggering a distance calculation.
- **Lightning validated:** the signal passes shape validation tests, and the firmware queries the sensor for the estimated distance to storm front and the raw strike energy value, printing both to the serial monitor.

This differentiating technique allowed the team to directly observe, in real time, how reliably the AS3935 rejected common sources of electrical noise (such as fluorescent lighting and nearby switching power supplies) in contrast to how it responded to real, actual simulated strikes from the emulator. Logging both the classified outcome and the raw energy and distance values also gave our team an early dataset to do sanity checks. This in return allowed us to test our distance estimation approach that we described in Section 3.5 before implementing the team's own peak detection and trend-analysis firmware algorithms.

8.2.2 Emulator Firmware Prototype

For the emulator system, another Arduino Uno was combined with a breakout shield and an output circuit designed for generating test VLF signals in the 3-30 kHz frequency range (the photo of the emulator shield can be seen in Chapter 9). In contrast to the detector firmware which is largely interrupt driven, the emulator firmware is built around a polling routine that constantly polls the state of the two pushbuttons on the board using the function `digitalRead()`.

There are two separate, distinct operating modes available in the emulator:

1. **Manual Test Mode:** a single button press immediately triggers an emulated lightning signal. This allows our team to generate on demand test events while observing the detector's response live.
2. **Automatic Test Mode:** the firmware cycles through a pre- configured sequence of strike patterns of varying intensity and timing. These are selected using a switch statement that maps a strike index to specific sets of waveform values, including amplitude and pulse width.

Once a strike pattern is selected, the firmware writes a sequence of samples to the DAC in a tightly timed loop. This reproduces the waveform shape at the appropriate frequency to fall within our AS3935's detection band. Additionally, a visual flash is also toggled during transmission time, which gives the team an unambiguous, visual confirmation that a test event was actually produced. This proved useful for distinguishing, during testing, whether the emulator didn't fire or if the detector didn't detect during early debugging and testing. This logic is shown in the "Emulation" flowchart of Section 7.3.

8.2.3 Integration and Preliminary Results

With both prototypes operating successfully and independently, the team then focused on connecting the emulator and the detector boards over a distance range. This helped our team prove that our detector's firmware could reliably determine that the waveform signals generated by the emulator were valid lightning events; While at the same time still being able to ignore the sources of noise that existed in the lab as we were testing. As we explained in detail in Chapter 9, this well-rounded test provided our team with an early proof of concept for the entire detect, reject, and estimate distance pipeline before any custom PCB hardware was even available.

Aside from testing the actual functionality of the sensor and the waveform signals themselves, this implementation with the help of an Arduino board allowed the firmware and the hardware development to proceed in parallel. Eventhough the detector and the emulator hardware was still undergoing the development process, their firmware logic was for the most part developed and tested and ready to be moved onto ESP32 and PIC16F1789, respectively. The next steps for the firmware work, outlined in Section 7.4 and Chapter 8, include the implementation process as well as testing the hardware and firmware performance after implementation. Additionally, we still had to test the porting of the existing logic to the target microcontrollers, in addition to the integration with BLE and the voice playback subsystems, as well as the addition of the LED and buzzer alerts of the final design.

8.2.4 BLE Firmware Prototype

In addition to the detector and emulator firmware as we presented in Section 8.2.1 to 8.2.3, a preliminary prototype of the firmware for the Bluetooth Low Energy (BLE) communication module was designed and tested. As the customized printed circuit board equipped with the Nordic nRF54LM20B chip was still under fabrication when this report was written, this prototyping process was done using the Nordic nRF54LM20B wireless SoC BLE evaluation kit. This enabled our team to confirm the GATT design and iOS communication protocol before the fabrication process was done.

Development Board Set Up

The development board nRF54LM20-DK (Figure 3.5) was used as the main platform for early BLE firmware development. In contrast to our development process via Arduino board for detector and emulator, this development board included the actual radio IC that is supposed to be used in the final production version. This meant that the firmware we have developed and tested on this development board will have very small changes when we finally flash it to the final custom board. With a few minimal changes expected for the reassignment of the pins, as necessary. The initial steps for the firmware development included bringing up the development board by making sure that the development environment could actually build and flash the firmware onto the board. We also tested logging by using UART with the board's USB interface and also using the on-chip radio.

GATT Server Firmware

Once the basic set up process was completed, we implemented the GATT server firmware of Section 3.7 and Section 7.2.2. The nRF54LM20B firmware sets up the chip to be a BLE peripheral that shows cases a unique 128-bit service UUID (4FAFC201-1FB5-459E-8FCC-C5C9C331914B) with one notify-only characteristic. It is worth noting that this setup was made to be deliberately minimalistic and simple during our prototyping phase. Our current firmware offers just enough functionality to show that the main device can discover the service, subscribe to the characteristic and receive notifications about its values changes.

The prototype firmware has a simple testing procedure that toggles the characteristic value between the All Clear (0x00) and Lightning Detected (0x01) state. This is done either periodically, or after receiving a specific serial command from the debug terminal. It was designed to allow testing the entire notification process, including the characteristic value write, dispatching a notification and receiving it by the central device. More importantly, this testing was able to be done without needing the whole lightning detection algorithm to be ready. Once the detection firmware (Section 8.1.1) is implemented on the ESP32, this implementation will be replaced with a call from the detection algorithm, which will write a value to the characteristic only when the actual state of lightning changes, following the lightning detection protocol.

8.2.5 iOS Application Integration Testing

Since the GATT server firmware was operational with our development kit, we used it to directly test the BLE stack for the iOS app instead of relying on the Demo Mode feature of the app for testing (Section 7.2.5). The BLEManager Class that we explained in Section 7.2.1 allowed the iOS app to scan for and filter based on the custom service, connect with the development board, discover the status characteristic, and then finally post the notification. Once the value of the status characteristic of the development board changed during our testing, the iOS app successfully received the change and updated the Status View Interface from “All Clear” to “Lightning Detected,” as we expected.

This integration test was an essential validation test in contrast to Demo Mode. This is because Demo Mode involves just the application's own state management and user interface logic while this test involved the CoreBluetooth stack, as well as the wireless communication channel between the phone and the developer kit and the real-time GATT subscription and notification system. The connection and notification time periods we recorded in our testing were comfortably well within the one second mobile application response time that we previously mentioned in Table 2.1.

Remaining BLE Firmware Work

As we were writing this report, we are still in the testing process for the prototype of the BLE firmware. We are only testing the communication channel and not using any actual live detection data yet. The future development of the firmware will include the following: implementing actual detection event reporting (instead of the test toggle

routine, which will be done through the implementation of real detection with the use of the lightning detection algorithm described in Section 7.3; expanding the one-byte protocol with more information such as the estimated category of distance and the direction in which the lightning bolt was headed, as predicted in Section 7.2.2; implementing at least basic power management capabilities for entering the idle state of the radio, taking into account the power limitations described in Section 4.2.1; as well as ensuring that the final custom PCB's antenna design preserves the range of the communication channel we have established in development kit tests.

8.2.6 Voice Synthesizer Firmware Prototype

Development of the voice synthesizer firmware prototype focused on evaluating the functionality of the Nuvoton ISD2360 voice playback integrated circuit and establishing a foundation for future integration with the lightning detection system. Initial efforts involved familiarization with the ISD-VPE360 development environment, which is used to organize, program, and manage prerecorded audio messages stored within the device's internal flash memory.

To support prototype development, a collection of prerecorded voice prompts was created for use by the lightning detector. These audio files include system status messages, warning announcements, and numerical distance values that will allow the system to communicate estimated lightning distance to the user. Prior to programming the device, each audio file was converted to the format required by the ISD2360 and verified for proper playback within the development software.

Early testing concentrated on validating the basic operation of the voice playback hardware using the development board provided by Nuvoton. Individual audio messages were successfully imported into the development environment and organized for future playback. The ISD-VPE360 software also provided the ability to test playback functionality and verify audio quality before integration with the system microcontroller.

The final implementation will utilize Serial Peripheral Interface (SPI) communication between the primary microcontroller and the ISD2360. Under normal operation, the lightning detection firmware will determine the appropriate warning message based on detected lightning activity and transmit the corresponding playback command to the voice playback IC. Once playback begins, the ISD2360 independently reproduces the selected audio message while allowing the microcontroller to continue executing the lightning detection algorithm without interruption.

Future firmware development will focus on implementing the complete SPI communication routines, integrating playback commands into the main application firmware, and validating reliable operation through subsystem and system-level testing. Additional work will also include optimizing message organization within the ISD2360 memory to reduce playback latency and simplify future firmware updates.

Voice Message Programming

Before beginning to program any of the firmware to use the ISD2360, we built an initial collection of prerecorded voice messages. We also sorted them and compiled them with the help of Nuvoton's ISD-VPE2360 software (Section 3.6.5, Figure 3.4). Within the prototype system, the messages we have set were deliberately made very small in number and were made up of a few critical voice messages like "Lightning Detected" and "All Clear." We also included a tone solely for testing the correct operation of audio playback and amplifier connection. We assigned an index called Voice-Prompt (VP) to each message in the memory map of the software (Figure 3.4). This was done so the firmware could ultimately, once development is done, call a specific message using its index without worrying about the different memory addresses. Finally, the Application Image, was programmed into the Flash memory of ISD2360 over SPI according to Section 3.6.4.3.

SPI Communication Firmware

Using voice messages embedded in the device, firmware was created to interact with the ISD2360 via its four-wire SPI interface. This early prototype of the firmware was initially set up on an Arduino Uno. This allowed the SPI command sequence verification on hardware, which was more familiar to the team members before we moved the implementation to the actual microcontroller.

The firmware has very simple functionality, which consists of several essential SPI messages: checking the status/ready bit of the device (this is done to make sure that ISD2360 is not in a playback mode before sending it a command) and sending a play command by referencing the required VP index and issuing a stop command if needed. After the development, a simple test suite was implemented in order to sequentially play all programmed voice messages by sending commands to the device from the serial-monitor. This way it made it possible for our team to verify that all messages were being played back properly and that the ISD2360 reported that the playback was completed.

Trigger Logic Prototype

Along with testing the of the basic operations of the ISD2360 and the integration with the microcontroller's interrupt code, our team also developed a prototyped trigger routine that will be used to integrate the voice subsystem into the lightning detection logic. Instead of feeding this routine actual, real detection data, the prototype firmware mimics a lightning event via a pushbutton signal. Which is very similar to the method used in the manual test mode of the emulator (Section 8.1.1). After receiving a pressed button signal, the code picks a message, sends an appropriate SPI play command to the ISD2360, and polls the ready flag on the device, and later goes back to idle mode. This allowed the team to be able to validate the entire path from "event detected" to "message selected" to "audio played" with the same sequence of operations the Voice Playback Firmware. Advantageously without needing the detection algorithm implemented.

During the initial testing period, an important integration issue was actually discovered. Since the playback happens automatically after the device is triggered, we found it was necessary for the trigger routine on the microcontroller to not send any other play commands until the message had finished playing. To solve this, we began using the ready flag of the device before sending a play command. This effectively prevented the problem, and this same pattern will also be used in the final firmware of the detector.

Remaining Voice Firmware Work

The voice synthesizer firmware prototype as we were writing this report was proven to work with the SPI communication protocol, voice message programming process, and manual trigger function. However, we still have to link it to the real lightning detection messages or implement it in the ESP32 hardware architecture. Future work will mainly focus on linking the SPI trigger functionality to the alerts from the detection algorithm, implementing the complete set of voice messages for status notifications and also ensuring that there is no delay in the triggering of these messages beyond the 100 ms threshold.

8.3 Enclosure Design Considerations

The enclosure for the proposed lightning detection system was designed around several key engineering objectives intended to support portability, usability, and future manufacturability. Although the mechanical design remained under development throughout this phase of the project, consideration was given to the physical requirements imposed by the electronic hardware, user interface, and intended operating environment. These design considerations served as the basis for future enclosure development and prototype fabrication.

One of the primary design objectives was portability. Because the device is intended for use during outdoor recreational activities, the enclosure must be compact enough to be easily transported while remaining large enough to accommodate the printed circuit board, battery, speaker, and supporting electronics. A compact enclosure also improves user convenience and encourages the detector to be carried during activities where lightning hazards may be present.

Ease of assembly and maintenance was also considered during the mechanical design process. The enclosure should allow access to the internal electronics for programming, debugging, battery replacement, and future hardware revisions without requiring destructive disassembly. A multi-piece enclosure secured with standard fasteners was identified as the preferred approach because it simplifies assembly while allowing the internal components to remain securely mounted during normal operation.

Placement of external components was another important design consideration. The speaker opening should be positioned to maximize the clarity of spoken warning messages while protecting the speaker from accidental damage. Similarly, the status LEDs and user push button should remain clearly visible and easily accessible without

interfering with normal handling of the device. Any charging or programming connectors should also be located to permit convenient access while minimizing cable strain during firmware updates.

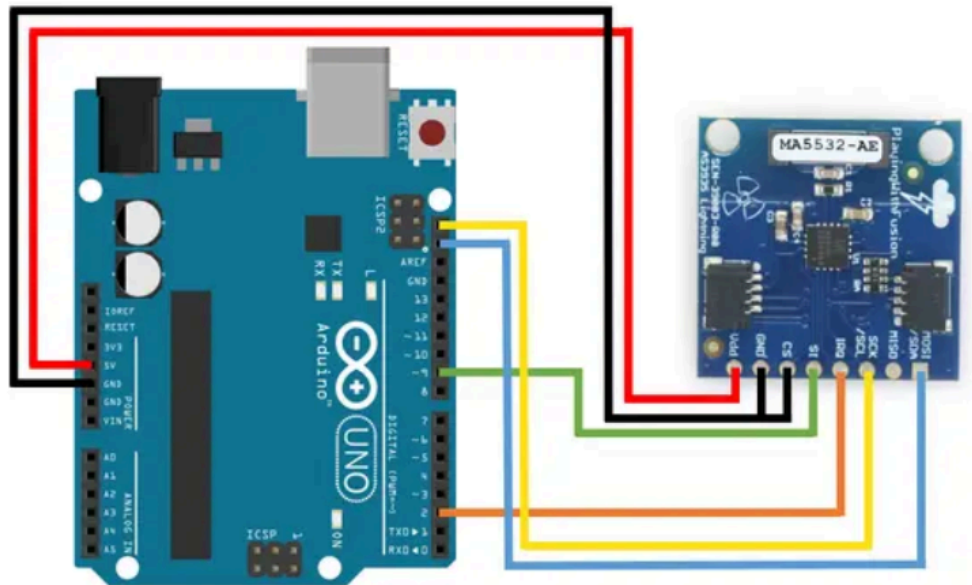
Finally, the enclosure was designed with future improvements in mind. Additional revisions may incorporate weather-resistant seals, improved mounting features for the printed circuit board, and alternative manufacturing methods better suited for higher production volumes. By considering these design objectives early in the project, the enclosure can evolve alongside the electrical and firmware development without requiring significant redesign of the overall system.

Chapter 9 - Testing and Evaluation

9.1 Testing Materials

We identified and procured a test/breakout board for some initial testing for the Radio IC and lightning frequency test and verification. The proof of concept was successful. The board we used was a SEN-39003. It had the Radio IC that we needed to verify. We used an Arduino Uno test together with the breakout board to successfully test our proof of concept.

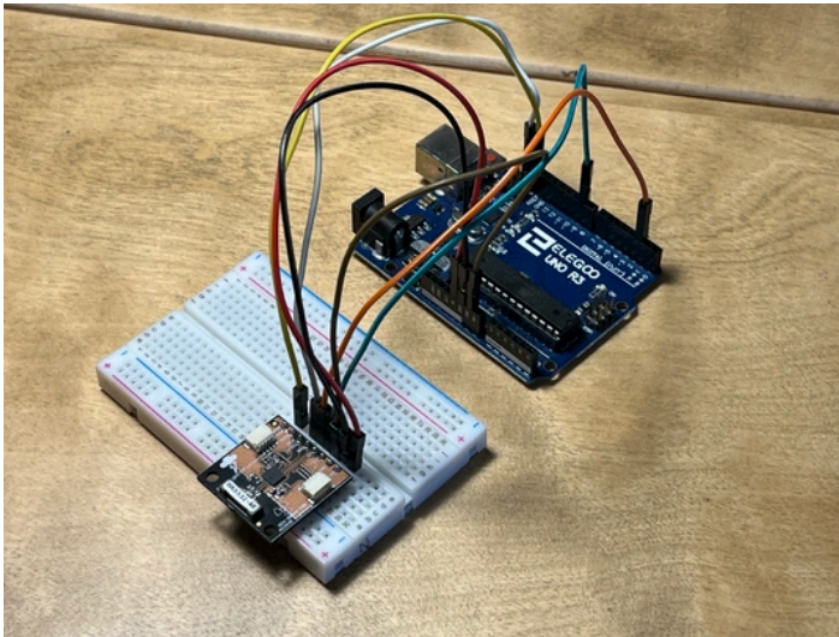
Arduino Uno Schematic



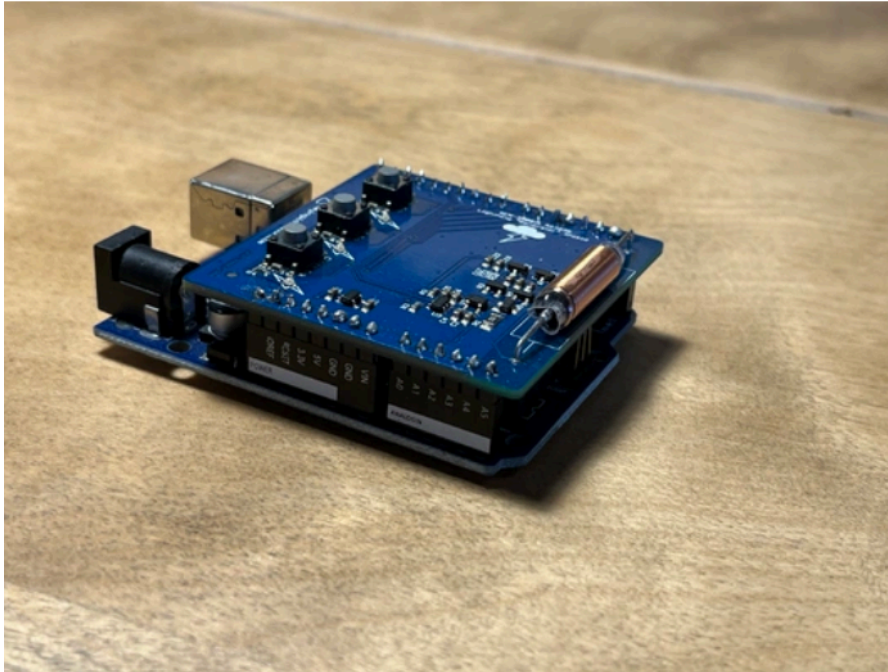
SEN-39003 is a Qwiic-compatible breakout board for the AMS / Franklin AS3935 Digital Lightning (that's right, lightning, not "lighting") sensor IC. This low-cost sensor "listens" to the EM spectrum in the 500 kHz range to identify lightning strikes up to 40

km away. This is about 4 times farther away than a typical human can discern lightning proximity using the “30-30 rule”, in which lightning is within 10 km if thunder is heard within 30 seconds of a visible lightning flash. With lots of settings and calibrations, the sensor can be tuned for a wide variety of applications and locations. These settings and readings are accessed by either SPI (using the breakout header) or I2C, either with the header or Qwiic- compatible cable. Unlike other suppliers, Playing With Fusion calibrates and provides this cal value (in pF) for each sensor’s antenna frequency. We also store the value in case you happen to lose it.

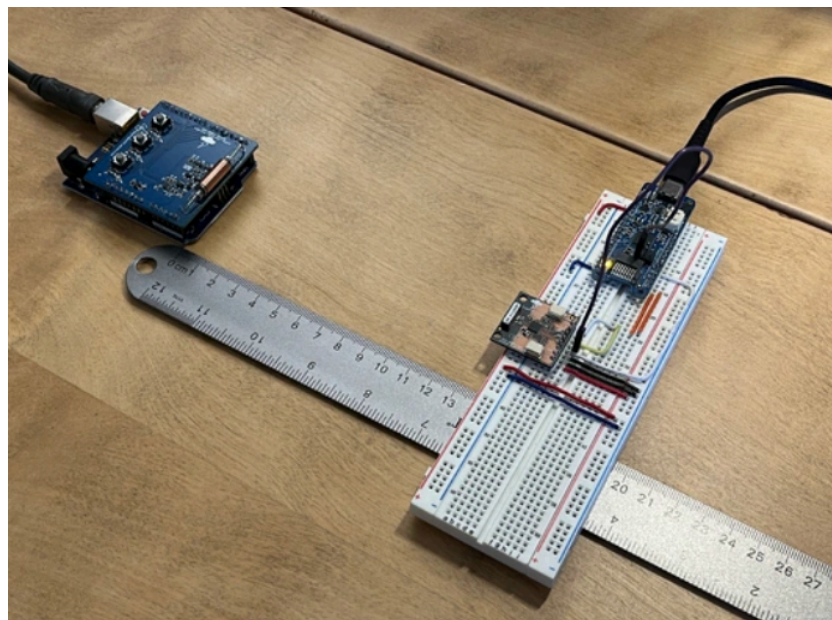
Below the interface between the Radio IC breakout board and the arduino can be seen.



In a like fashion, we used an Arduino Uno together with a breakout emulator shield to test transmission of our emulator.



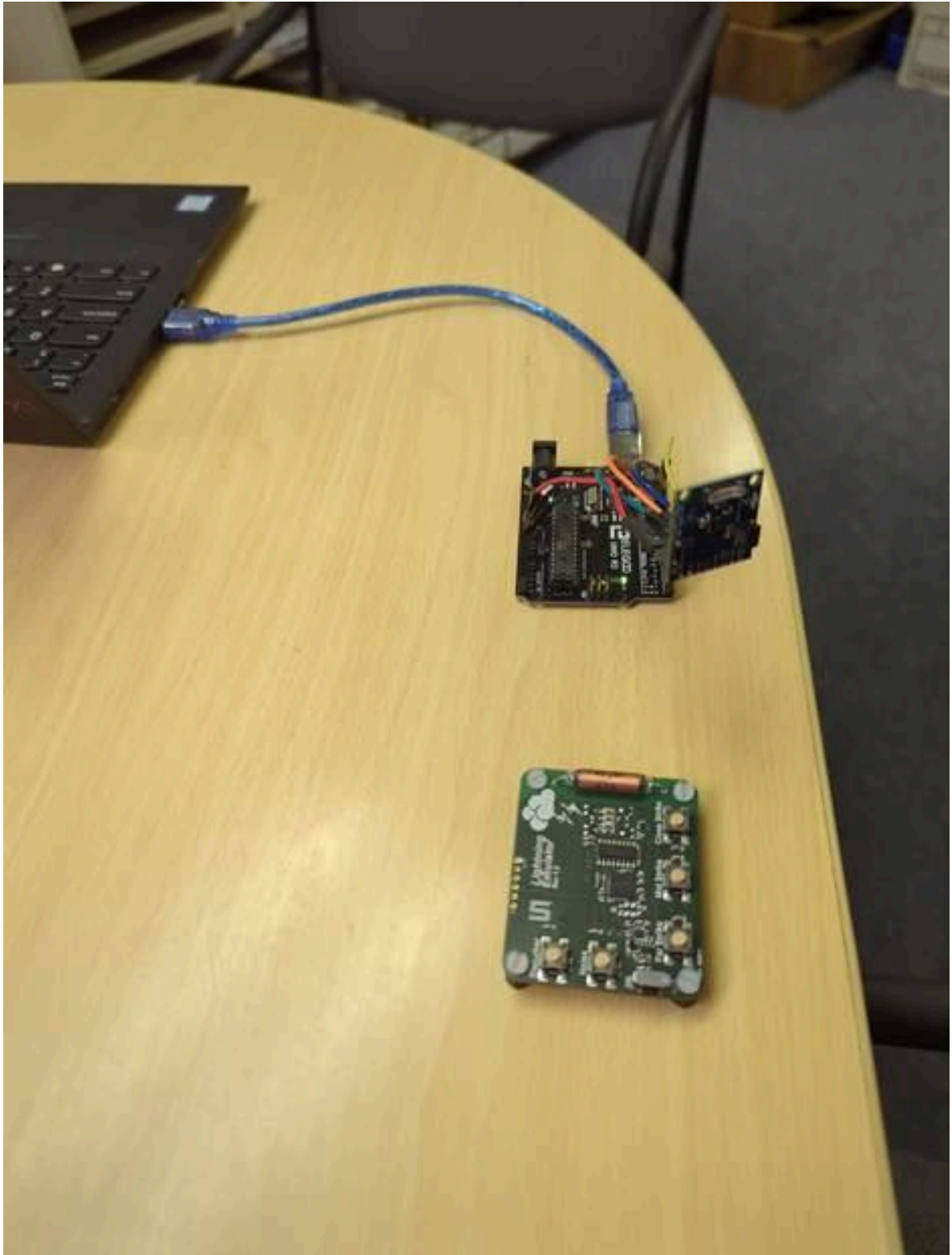
We conducted some initial testing of lightning strike range estimates with interpolation of kilometers to inches here.



A Tesla coil was employed to emulate lightning. It successfully activated our lightning detector.



A test was also conducted with a test VLF radio to emulate EMF in the same spectrum that lightning is found. Here, we attempted to detect the VLF radio emissions with our radio. It was successful.



Additionally, we attempted to detect VLF, EMF energy from a greater distance from the Tesla coil. This was also successful.

9.1 Firmware Testing

Because the final custom PCB hardware was not available for testing at the time this report was written, we conducted all firmware tests in this section using the Arduino test platforms and the manufacturer development boards and not the final ESP32 and PIC16F1789. This testing phase is not intended to verify the final system performance, instead we verified that the basic firmware functionality, communications protocols, and subsystem interfacing worked correctly. Ultimately in order to implement these systems to their final hardware. Each subsystem was tested independently based on a set of objectives and, if required, then subsystems were tested in combination with each other.

9.1.1 Lightning Detector Firmware Testing

The objectives of the detector firmware testing were the following:

- To verify that the AS3935 radio IC could be properly configured over the SPI protocol
- That the event handling using the interrupts was able to distinguish properly between the cases of different disturbers, high noise levels, and actual lightning events
- The distance and energy values returned by the sensor are consistent and reliable.

For testing the detector firmware, an Arduino Uno board interfacing to the breakout board, SEN-39003 (see Section 9, Figuring showing wiring schematic) containing the AS3935 lightning sensor IC, was used. The communication between the Arduino and the breakout board was done over the SPI bus. The signal from the IC connected to the digital input pin of the Arduino board programmed to run the interrupt service routine.

Some test events were simulated using the Arduino emulator prototype we mentioned before. The location of the emulator with respect to the detector breakout board was at a fixed test distance that was within short range (5 inches). This made it possible for the team to create test events without the need to wait for lightning activity.

Table 9.1 Firmware Testing

Test Case	Description	Expected Result	Result
1	Power-on initialization	IC responds to configuration writes, no SPI communication errors	Pass

2	Ambient noise floor exceeds threshold (TV playing loudly)	Firmware reports "Noise level too high"; no distance and energy was calculated	Pass
3	Single emulated lightning strike at close range	Firmware reports valid lightning event with distance and energy values	Pass
4	Reiterated, same single emulated lightning strike at close range	Firmware reports similar valid lightning event with distance and energy values, similar to last run	Pass
5	Repeated emulated strikes in rapid succession, less than 1 second apart	Firmware respects AS3935's one second minimum resolvable interval between strikes	Pass

Results and Discussion

The set up of the AS3935 using the detector firmware through the SPI interface was achieved consistently without errors in all of the tests conducted. The interrupt driven method functioned as planned, where the microcontroller is idle except when an event is generated by the chip.

Additionally, the disturber rejection and the noise floor detection tests worked well during testing, offering our team some confidence about the internal signal validation mechanism of the AS3935. During our 5th test, we determined that the AS3935 has a minimum one second resolvable interval for separate lightning strikes. Meaning that whenever the module was triggered within a very short time, the individual strikes are grouped as a single lightning event. This is within normal sensor operation and not a firmware error. For future implementation, the firmware has to take this constraint into consideration when we begin implementing the direction estimation.

Some inconsistencies with the recorded distances and energies between repetitions of identical test hits were seen. However, these inconsistencies are not due to any errors or mistakes with the sensor or the firmware. Instead, these are more likely due to the somewhat experimental nature of the breadboard test setup, which includes the inconsistency of the position and orientation of the antenna in comparison to the transmitting antenna of the emulator.

Chapter 10 - Administrative Content

10.1 Budget

Our budget is estimated at around \$1000 USD. This limit is set by our sponsor, RF Laboratories, Inc.

A goal is to limit out-of-pocket expenditures from team members. We are being supported by a company which has inventory which we can pull from for components.

10.2 Bill of Materials

Table 10.1 Bill of Materials

Sub-System	Description	Part Name	Price	Quantity
Lightning Detector	MCU	ESP32 DevKit V1	\$10	1
	VLF Receiver	Ferrite Rod Antenna + Receiver Circuit	\$20	1
	Voice Synthesizer	DFPlayer Mini MP3 Module	\$6	1
	LED Indicator	High Brightness LED	\$1	3
	Speaker	8 Ω Mini Speaker	\$3	1
	Buzzer	Active Piezo Buzzer	\$2	1
	BLE Transceiver	Integrated ESP32 BLE Radio	\$0	1
	Battery	3.7V Li-Ion Battery Pack	\$12	1
	Battery Charger	TP4056 Charging Module	\$2	1
	Voltage Regulator	AMS1117-3.3V Regulator	\$2	1

	Misc. Components	PCB, Resistors, Capacitors, Connectors, Wiring	\$25	1
Lightning Emulator	MCU	MC PIC16X Dev Brd	\$10	1
	VLF Transmitter	Ferrite Rod Antenna + Driver Circuit	\$20	1
	Push Button Interface	Tactile Push Buttons	\$2	2
	Misc. Components	PCB, Resistors, Capacitors, Connectors, Wiring	\$20	1
Enclosure	3D Printer Filament	1kg PLA	\$15	1
	Fasteners	Assorted Screws, Nuts, Standoffs, and Mounting Hardware	\$10	1
		TOTAL	\$164	

10.3 Distribution of Work

Below is the distribution of the project in between all team members. The table specifies each team member's contribution to the project. However, each member's contribution relies on the other member's contribution as well. Collaboration will be present in all aspects of the project throughout the design and development process. This table will ensure each member is aware of their duties to the project as well as their expectations.

Table 10.2 - Distribution of Work

Electrical Engineering	Responsibilities
David Zima	Design/provision of VLF radio receiver, antenna and transmitter.
	Design/provision of 2.4 GHz, BLE radio transceiver, antenna, and Z matching.

	Design/provision of VLF radio transmitter.
	Design of power profile and circuitry for device to function off battery source and minimal duty cycle.
Electrical Engineering	Responsibilities
Will Zabin	Firmware design for MCU to interface with voice synthesizer
	Design/provision of voice synthesizer and associated circuitry
	Design of electrical button interface.
	Design of enclosure for device.
Computer Engineering	Responsibilities
Benjamin Titus	Design of frontend, backend and database of our mobile app for users.
	Mobile app deployment for public use, and repository maintenance
	BLE infrastructure to communicate with aforementioned app for user push notifications
	Firmware design to run and control BLE functions and interfaces.
Computer Engineering	Responsibilities
Beatriz Navas	Firmware design for MCU to interface with VLF receiver.
	Firmware design for MCU to interface with VLF transmitter.
	Firmware design to process detection of lightning and its movement.

	Website design and management.
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10.4 Milestones

For a project this size, it is imperative to keep track of the team's progress and maintain due-dates in order to stay on track. This is why we are using a milestones table to keep our team on track. It will keep each team member in track with specific goals and dates for progress goals, as well as keeping the team informed on what each team member is doing. Our project's milestones table can be found below, with

Table 10.3 - Project Milestones

Milestone Number	Milestone Description	Start Date	Planned End Date	Dependencies & Requirements
1	Finalize Project Idea	5/18/26	5/26/26	Set mentor and finalize idea for project
2	D&C Finalization	5/18/26	5/29/26	Finalize D&C document, make appointment with mentor, finalize reviewer committee
3	D&C Meeting	6/3/26 3:00-3:30pm	6/3/26	Complete D&C document, review document with mentor
4	Website	5/25/26	6/6/26	Finish website set up
5	Midterm Report Submission	7/6/26	7/6/26	Finalize midterm report
6	Midterm Report Submission	7/8/26	7/10/26	Review midterm report with mentor
7	Final Report Submission	7/10/26	7/24/26	Upload draft of final report, complete demo video and upload to YouTube
8	Final Report Feedback Meeting	7/24/26	7/28/26	Meet with mentor to review final report
9	Finalized Final Report	7/24/26	7/28/26	Submit final submission

				of report and upload finalized demo video
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Scope:

The scope of this proposal includes design and support of the following aspects of the application:

WIRELESS AND ANALOG DESIGN:

1. Design/provision of VLF radio receiver.
2. Design/provision of VLF radio receiver antenna.
3. Design/provision of VLF radio transmitter.
4. Design/provision of VLF radio transmitter antenna.
5. Design/provision of 2.4 GHz, BLE radio transceiver.
6. Design/provision of 2.4 GHz, BLE radio transceiver antenna.
7. Design/provision of 2.4 GHz BLE Z matching.
8. Design/provision of voice synthesizer and associated audio circuitry.
9. Design of power profile for minimal duty cycle.
10. Design of electrical button interface.
11. Design of power circuitry for device to function off battery source.

DIGITAL DESIGN:

1. Design/provision of voice synthesizer and associated circuitry.

FIRMWARE DESIGN:

1. Firmware design for MCU to interface with VLF receiver.
2. Firmware design for MCU to interface with voice synthesizer.
3. Firmware design for MCU to interface with VLF transmitter.
4. Firmware design to process detection of lightning and its movement.

5. Firmware design to run and control BLE functions.

SOFTWARE DESIGN:

1. Design of app to communicate with BLE notifications on lightning status.
2. Design and implementation of the iOS mobile application frontend, including the scan screen for BLE device discovery, the real-time status screen displaying safe and detected states, connection state management, and demo mode for hardware-independent testing.
3. Design of the BLE GATT communication profile, including the custom service and characteristic UUIDs, the single-byte data protocol for lightning status updates, and the CoreBluetooth central manager implementation on the iOS side.
4. Version control infrastructure including GitHub repository setup, branching strategy (master/develop), cross-platform OneDrive sync, and documentation of the BLE protocol and app setup in the project README. iOS deployment pipeline using Xcode and the Apple Developer Program, with a migration path to Android via Flutter as a future deliverable.

MECHANICAL DESIGN:

1. Design of enclosure for device.

SCHEMATIC, LAYOUT AND MECHANICAL

1. Schematic capture of radio receiver.
2. Schematic capture radio transmitter.
3. Schematic capture associated circuitry.
4. Layout of radio receiver.
5. Layout of radio transmitter.
6. Layout associated circuitry.
7. Cam and Gerber generation for fabrication of device.
8. PCB fabrication and panel stencil of receiver device.
9. Assembly of receiver unit.
10. PCB fabrication and panel stencil of transmitter device.

11. Assembly of transmitter unit.
12. Cam and Gerber generation for transmitter unit.
13. Assembly of transmitter unit

HARDWARE TEST, VERIFICATION AND OPTIMIZATION:

14. Verification, modification and optimization of receiver unit.
15. Verification, modification and optimization of transmitter unit.
16. Impedance matching, tuning and optimization of Bluetooth antenna.
17. Impedance matching, tuning and optimization of VLF RX antenna.
18. Impedance matching, tuning and optimization of VLF TX antenna.
19. Impedance matching, tuning and optimization of NTN antenna.

ENCLOSURE FABRICATION:

1. 3D printing of enclosure.

FIRMWARE MODIFICATION AND OPTIMIZATION

20. Embedded firmware design, test and optimization for radio functions.
21. Embedded firmware design, test and optimization for voice synthesizer.
22. Embedded firmware design, test and optimization for BLE

TEST, VERIFICATION AND PROJECT SUPPORT:

23. Test, debug, analysis, verification and optimization of system function and performance.
24. Part research and procurement for device.
25. Mechanical support
26. General project support and maintenance.

REGULATORY COMPLIANCE AND VERIFICATION

27. FCC unintentional radiation part 15.109 prescreening

Appendices

Appendix A - Reference

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